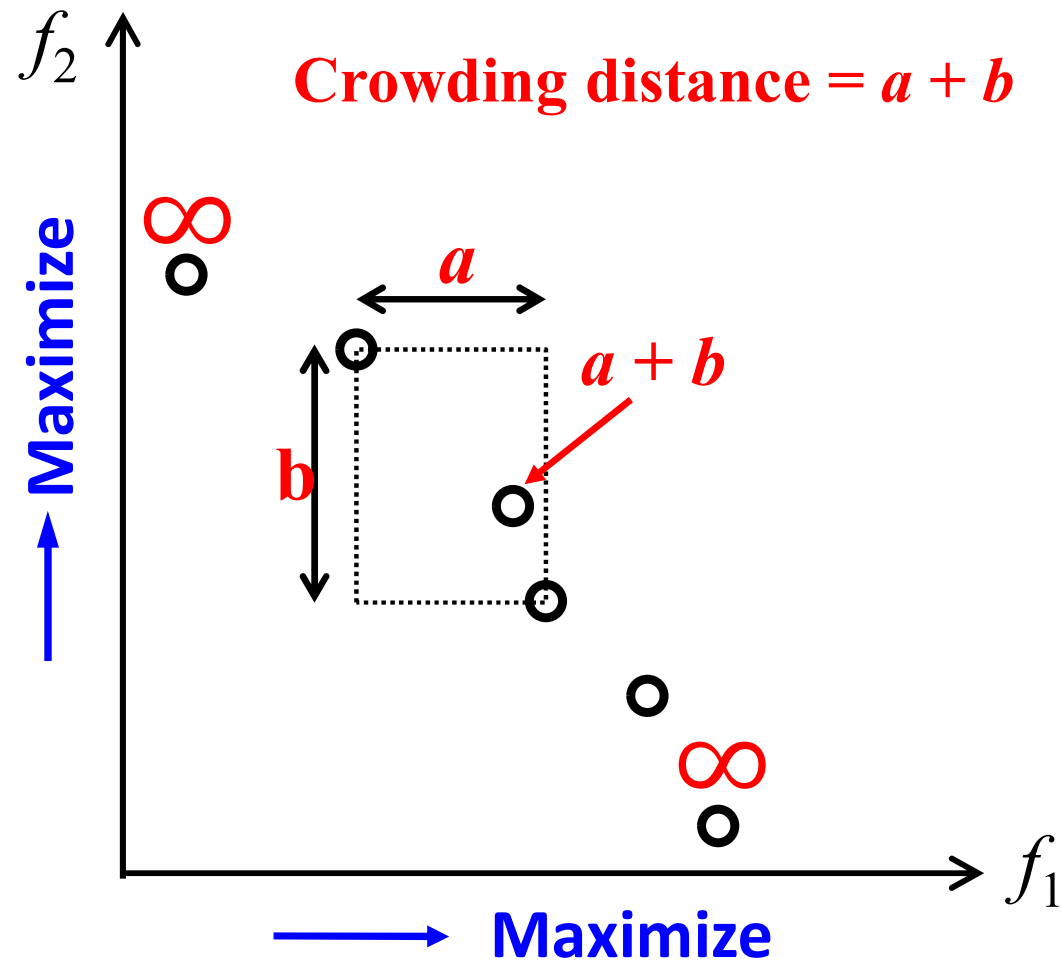


One Comment about Crowding Distance:

Exactly speaking, the normalization of each objective is needed in crowding distance calculation.

Crowding Distance: $a + b \implies a / (f_{1\max} - f_{1\min}) + b / (f_{2\max} - f_{2\min})$



Optimization Methods

1. Introduction.
2. Greedy algorithms for combinatorial optimization.
3. LS and neighborhood structures for combinatorial optimization.
4. Variable neighborhood search, neighborhood descent, SA, TS, EC.
5. Branch and bound algorithms, and subset selection algorithms.
6. Linear programming problem formulations and applications.
7. Linear programming algorithms.
8. Integer linear programming algorithms.
9. Unconstrained nonlinear optimization and gradient descent.
10. Newton's methods and Levenberg-Marquardt modification.
11. Quasi-Newton methods and conjugate direction methods.
12. Nonlinear optimization with equality constraints.
13. Nonlinear optimization with inequality constraints.
14. Problem formulation and concepts in multi-objective optimization.
15. Search for single final solution in multi-objective optimization.
- 16: Search for multiple solutions in multi-objective optimization.**

How to evaluate the obtained solution set ?

Performance Evaluation and Performance Indicators

E. Zitzler, L. Thiele, M. Laumanns, C. M. Fonseca, and V. G. da Fonseca, “Performance assessment of multiobjective optimizers: An analysis and review,” *IEEE Trans. on Evolutionary Computation* 7 (2), 117-132, 2003.

M. Li and X. Yao, Quality evaluation of solution sets in multiobjective optimisation: A survey, *ACM Computing Surveys*, 52(2), 2019.

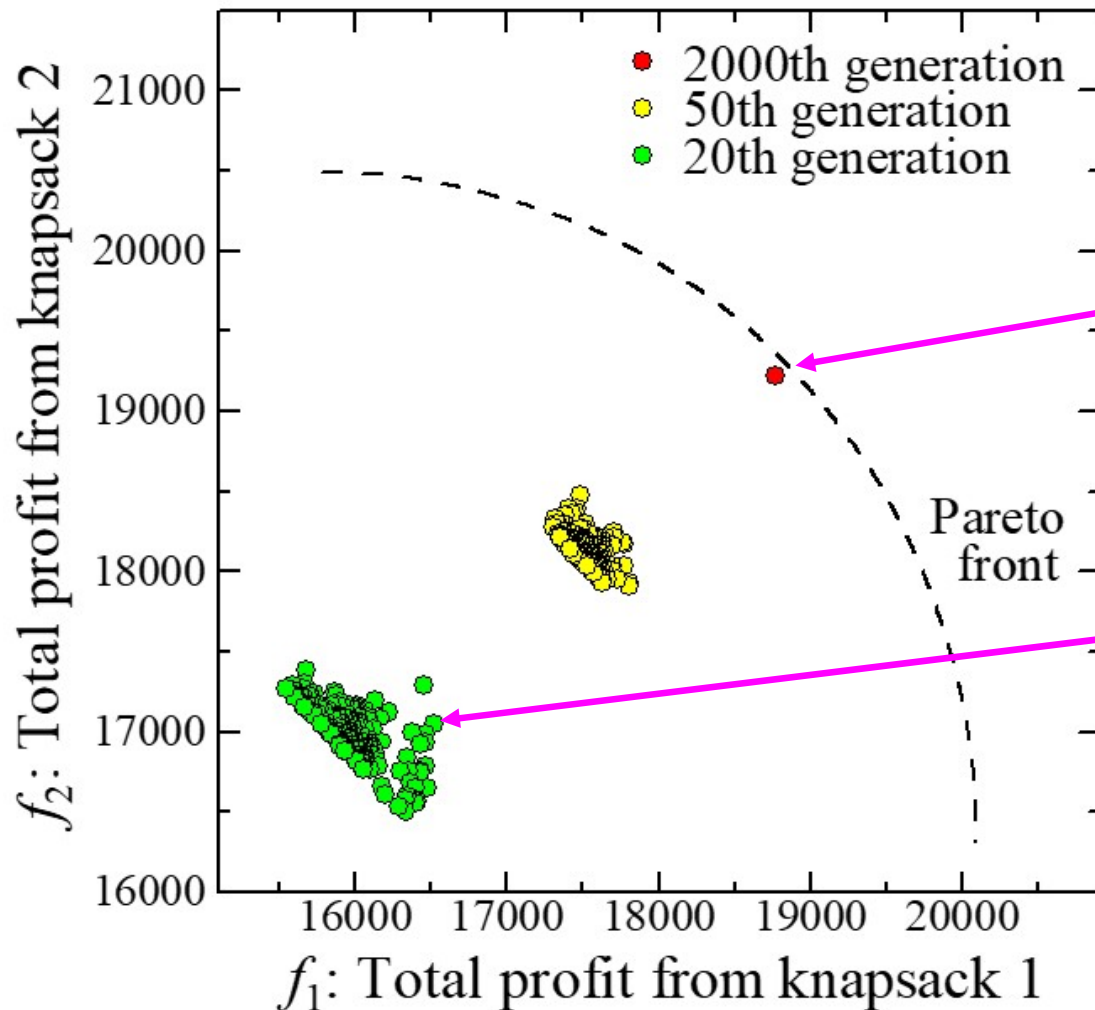
H. Ishibuchi, L. M. Pang, and K. Shang, Difficulties in fair performance comparison of multi-objective evolutionary algorithms, *IEEE Computational Intelligence Magazine*, 17 (1), 86-101, 2022.

In standard single-objective optimization

The performance of an algorithm

= The performance of the finally obtained solution

Algorithm performance = Solution performance



Maximize $f_1(\mathbf{x}) + f_2(\mathbf{x})$

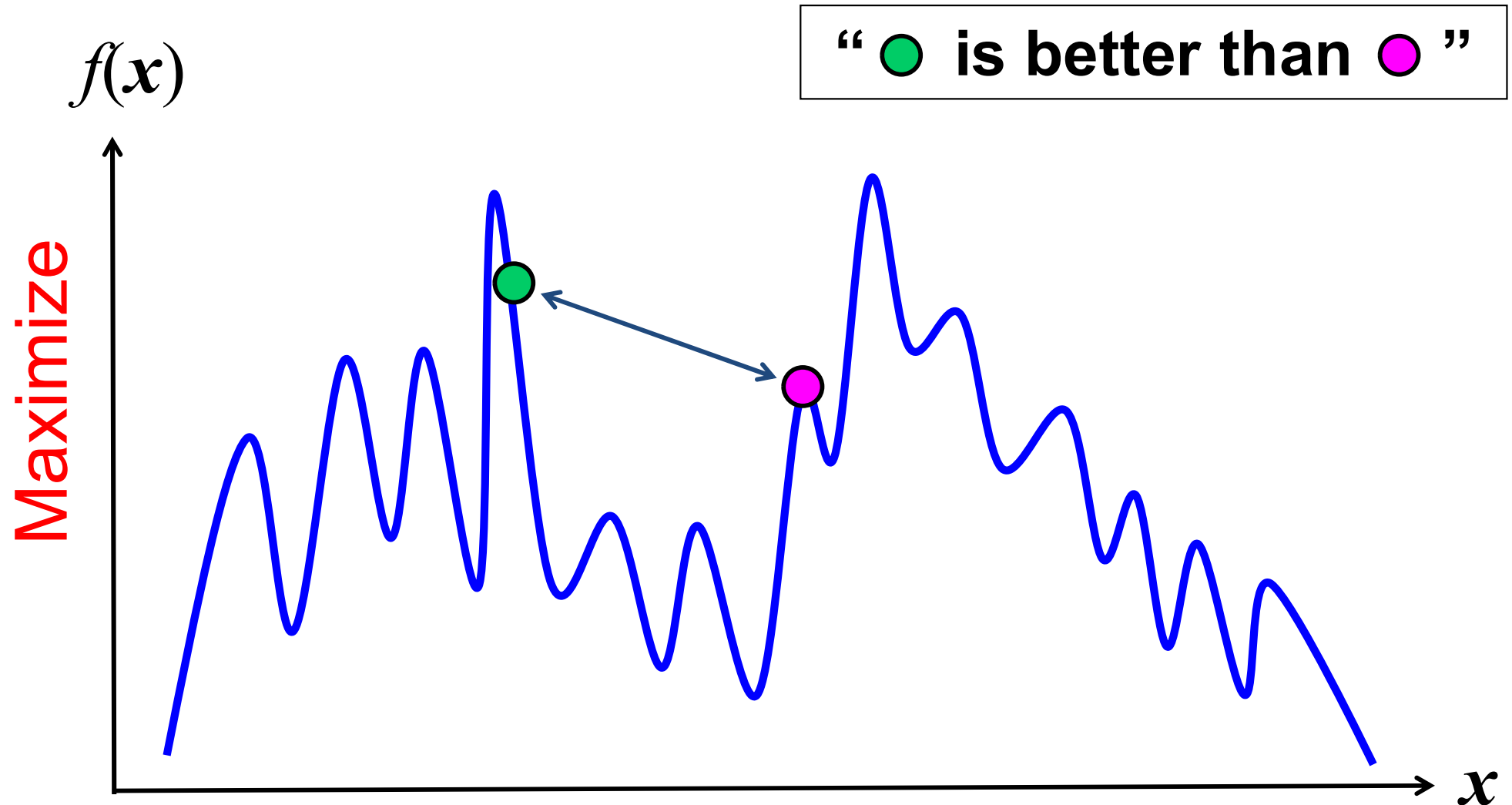
Algorithm performance =
This solution's performance

Each solution can be evaluated
in the same (or similar) manner
(i.e., based on the value of
 $f_1(\mathbf{x}) + f_2(\mathbf{x})$).

In standard single-objective optimization

The final result of optimization is a single solution.

Comparison of solutions is easy.

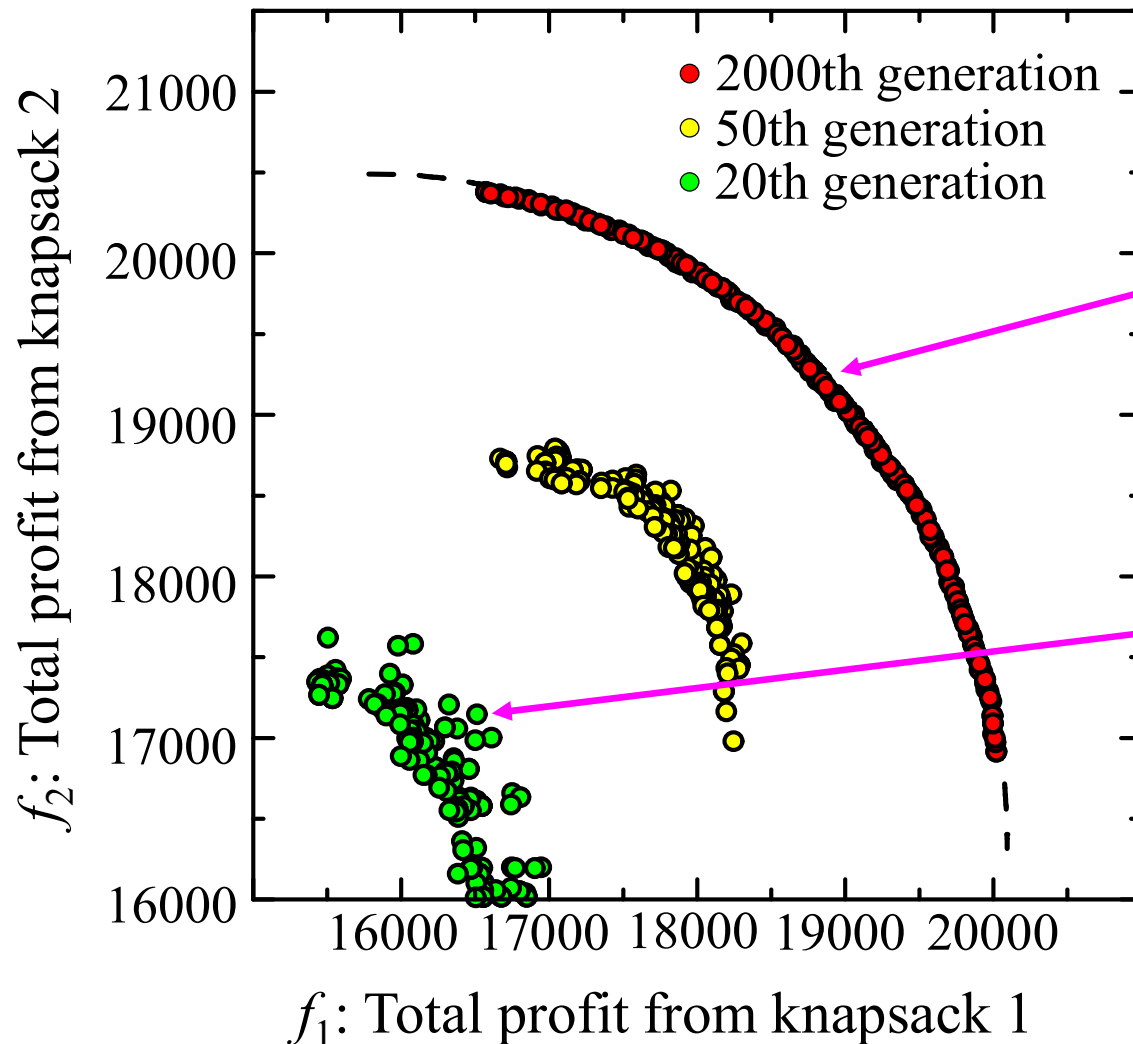


In evolutionary multi-objective optimization (EMO)

The performance of an algorithm

= The performance of the finally obtained solution set

Algorithm performance = Population performance



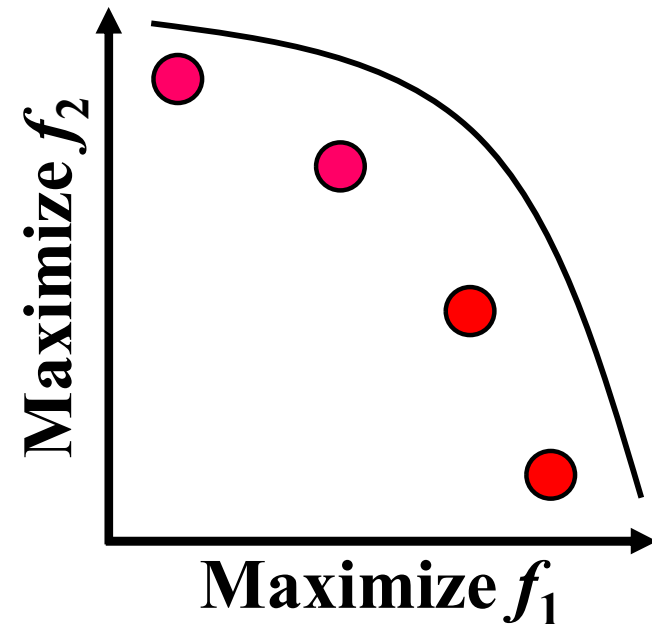
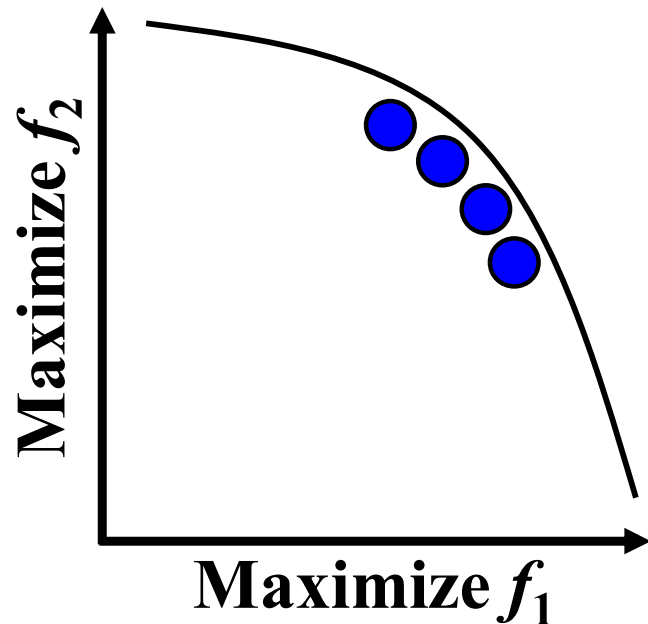
Algorithm performance =
This population's performance

Each solution is evaluated in a
totally different manner.

In evolutionary multi-objective optimization (EMO)

The final result of optimization is a solution set.
Comparison of solution sets is not easy.

Which is a better solution set?



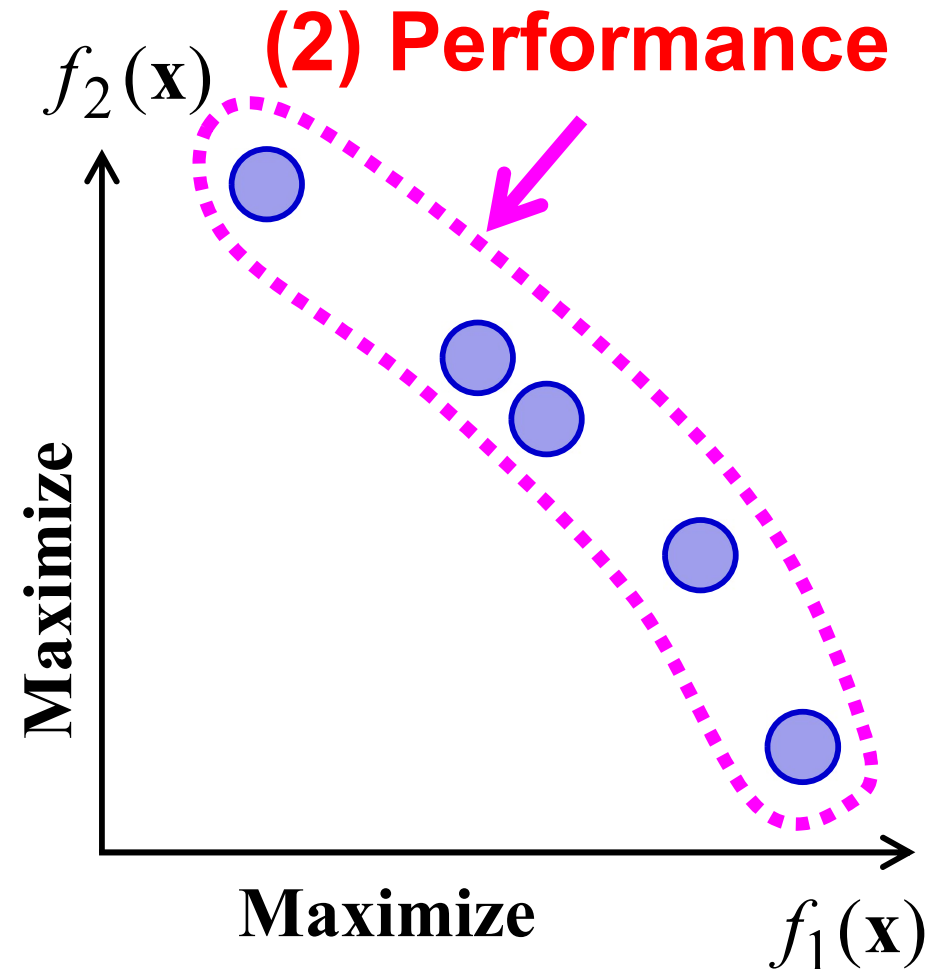
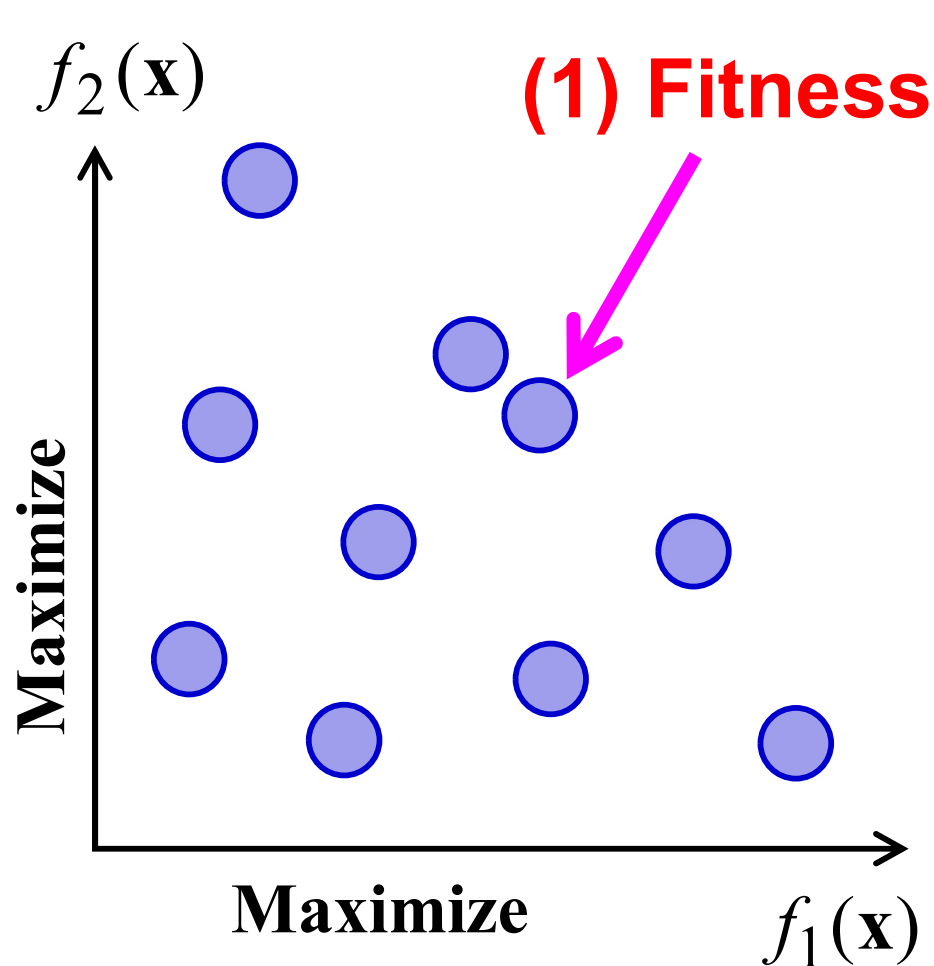
Two Levels of Evaluation

(1) Solution level:

for solution evaluation within an algorithm

(2) Solution set level:

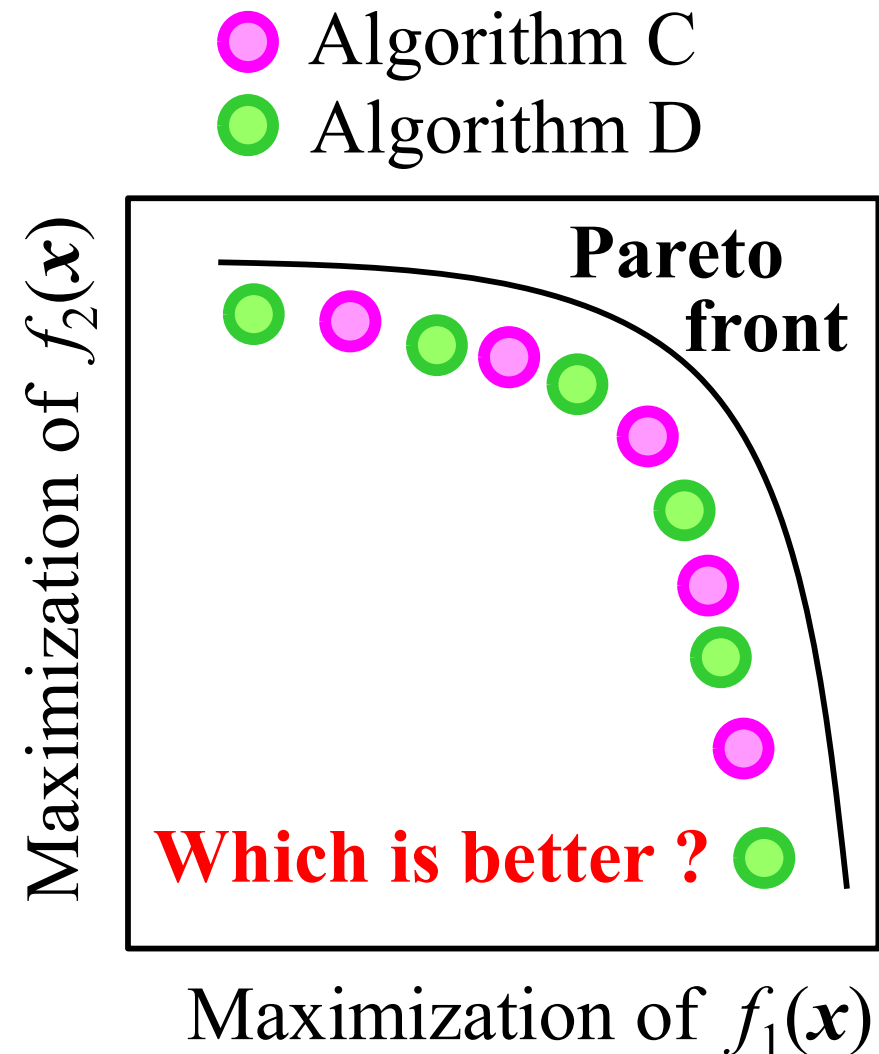
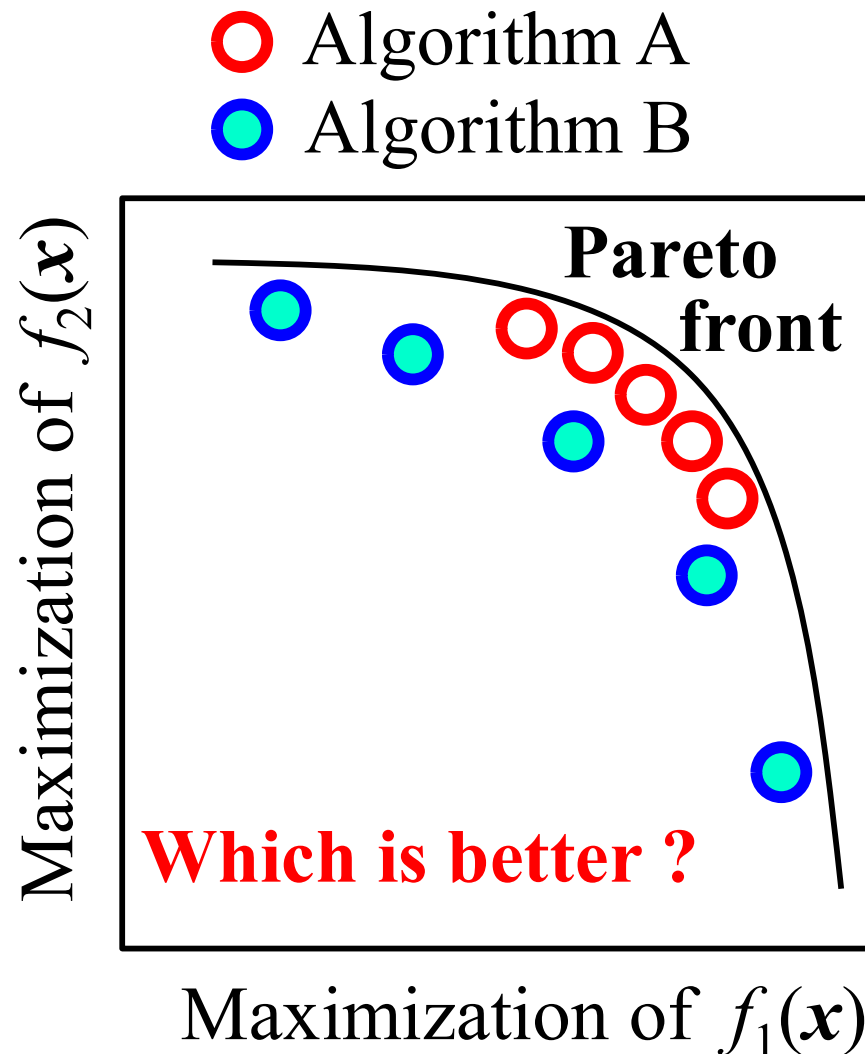
for algorithm evaluation (solution set evaluation)



In evolutionary multi-objective optimization (EMO)

Algorithm Evaluation (Solution Set Level):

Performance comparison is not very clear.

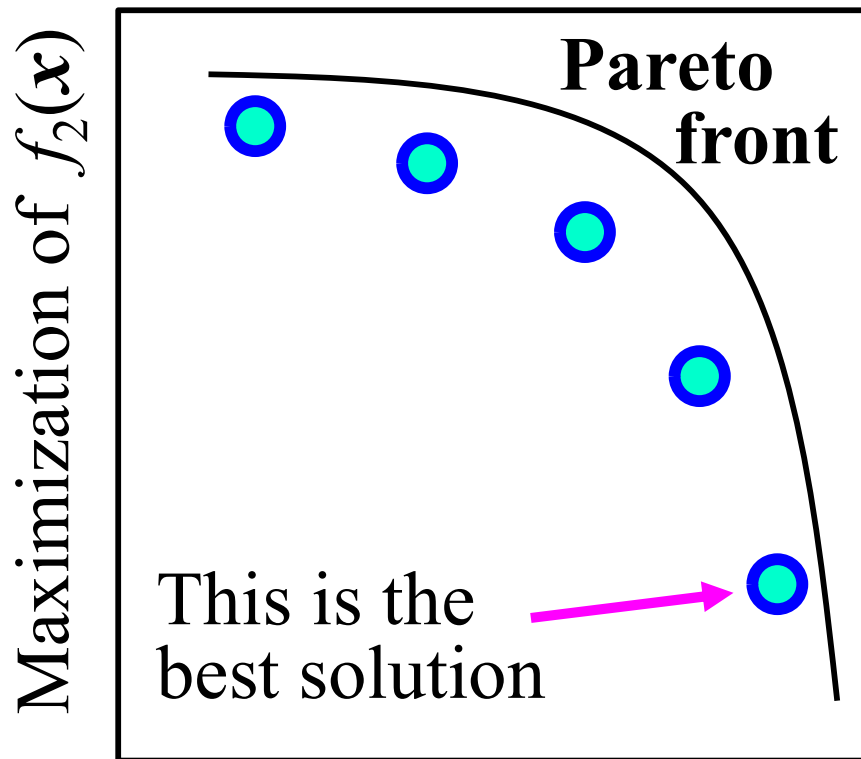


In evolutionary multi-objective optimization (EMO)

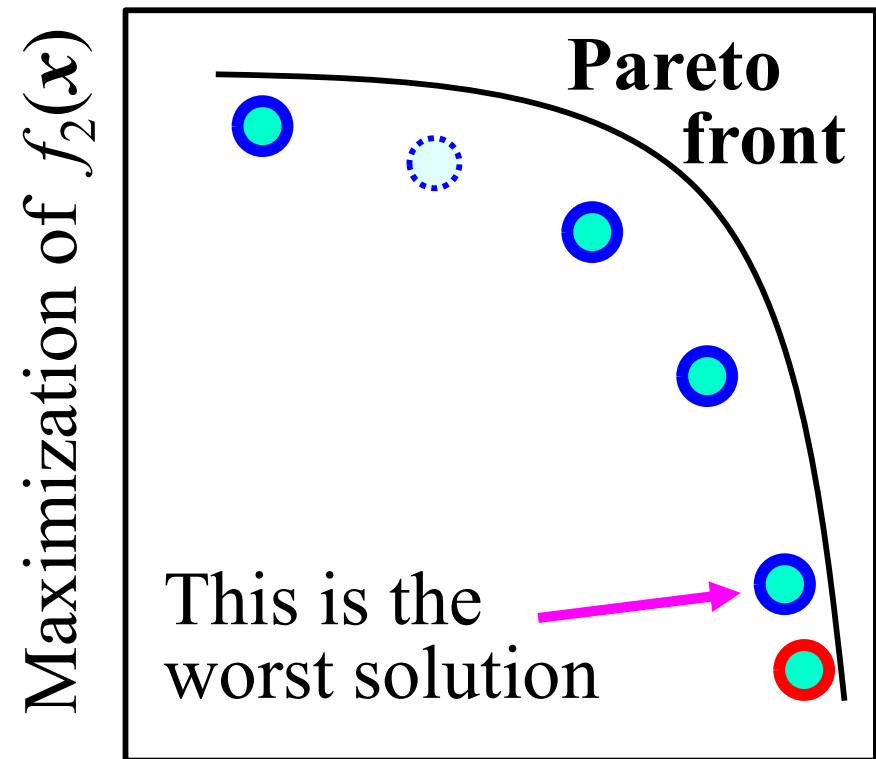
Evaluation of each solution:

based on the relation with the other solutions.

$\Rightarrow$ The same solution can have different evaluation results.



Maximization of $f_1(\mathbf{x})$



Maximization of $f_1(\mathbf{x})$

Similar Case: Genetics-based machine learning

Michigan Approach (Similar to EMO):

Solution (Individual) = A single rule

Fitness = Performance of a single rule

Population = A set of rules = A single classifier

Algorithm evaluation = Performance of the final population
(Performance of a set of individuals).

Classifier

If x is ... then class 1
If x is ... then class 1
If x is ... then class 2
...
If x is ... then class 3

Pittsburgh Approach (Similar to Standard Optimization):

Solution (Individual) = A set of rules = A single classifier

Fitness = Performance of a classifier

Population = A set of classifiers

Algorithm evaluation = Performance of the best classifier
(Performance of the best individual)

Population

Classifier

If x is ... then class 1
If x is ... then class 1
If x is ... then class 2
...
If x is ... then class 3

Classifier

If x is ... then class 1
If x is ... then class 1
If x is ... then class 2
...
If x is ... then class 3

...

Classifier

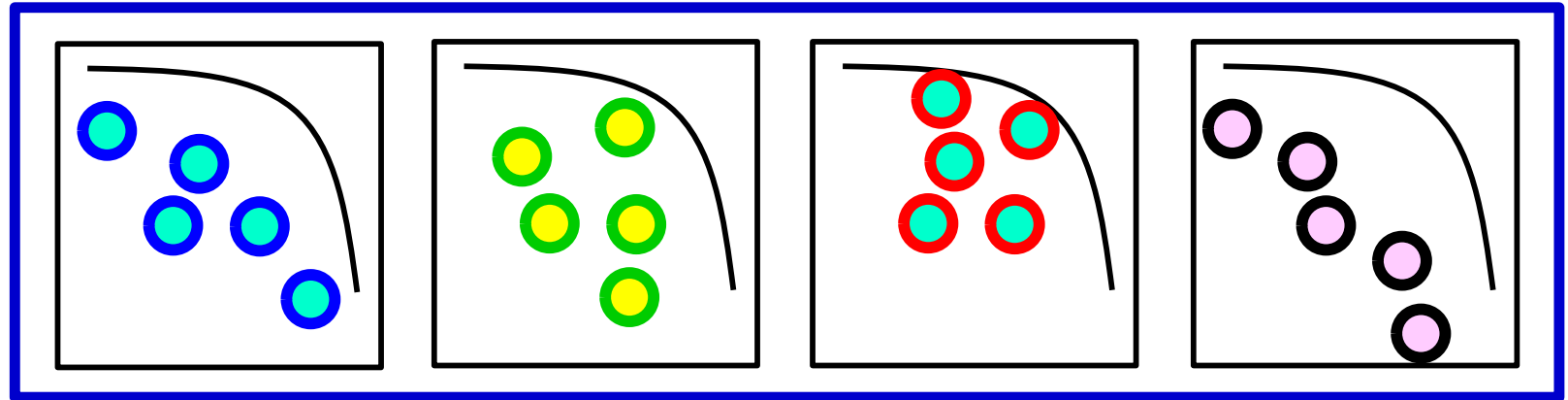
If x is ... then class 1
If x is ... then class 1
If x is ... then class 2
...
If x is ... then class 3

Set-based EMO Framework (Similar to Pittsburgh Approach)

Individual = A set of solutions (i.e., solution set)

Fitness evaluation = Performance of a solution set

Population



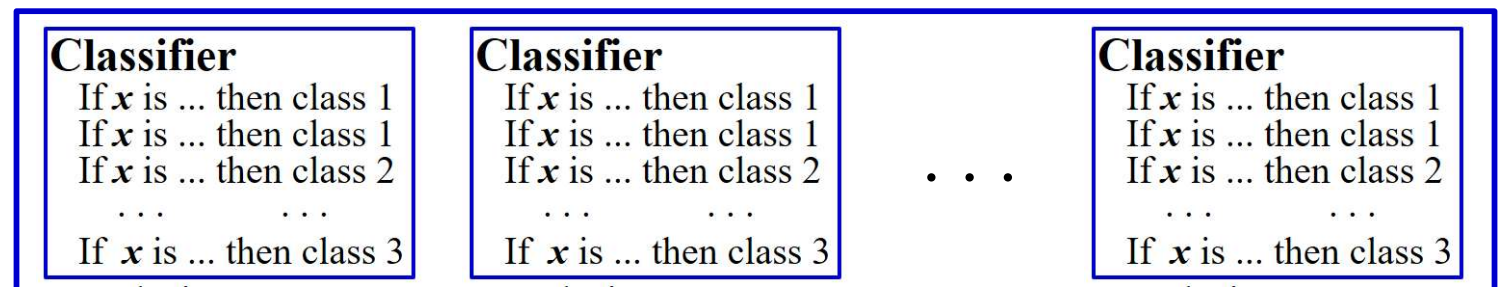
Pittsburgh Approach (Similar to Standard Optimization):

Solution (Individual) = A set of rules = A single classifier

Fitness = Performance of a classifier

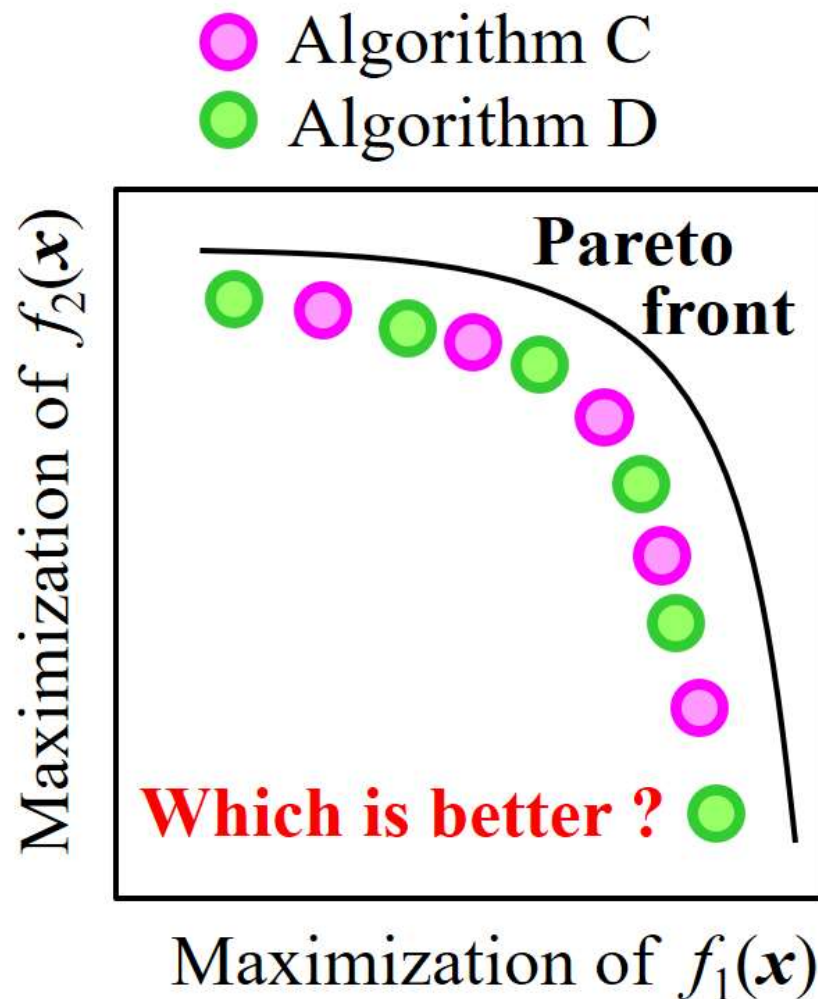
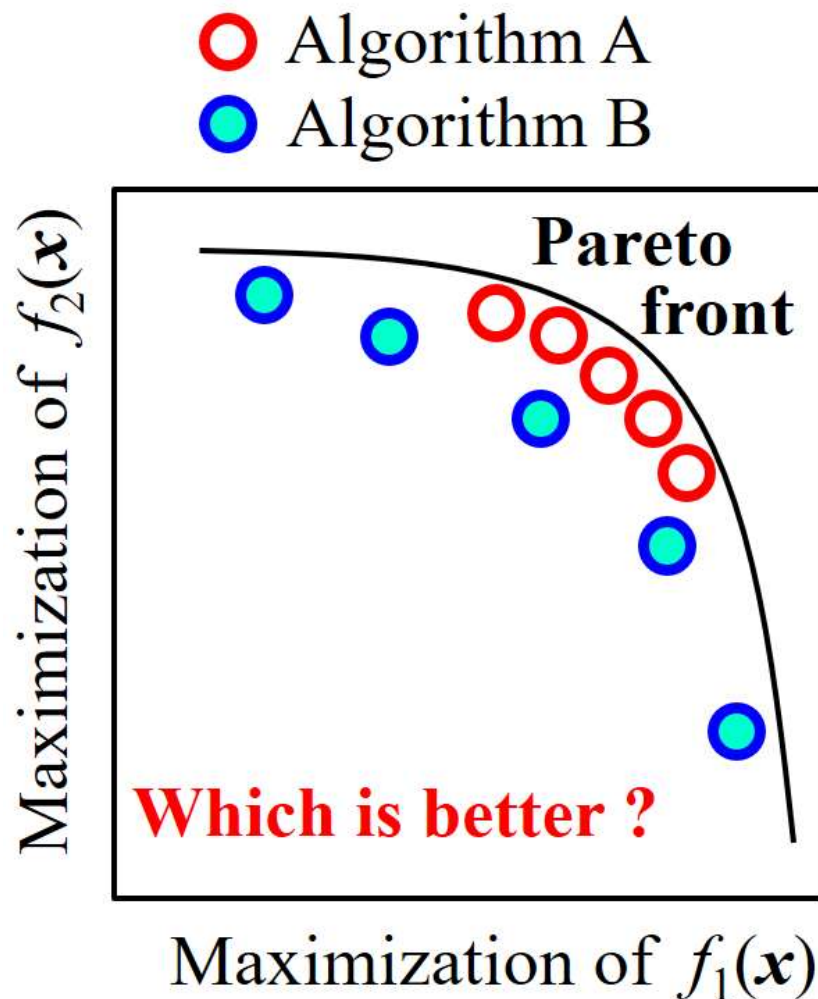
Population = A set of classifiers

Population



Indicators: Performance evaluation of a solution set

- Convergence to the Pareto front.
- Spread of a solution set (Broadness): Diversity of solutions.
- Uniformity of solutions: Diversity of solutions.
- Overall performance.



Indicators: Performance evaluation of a solution set

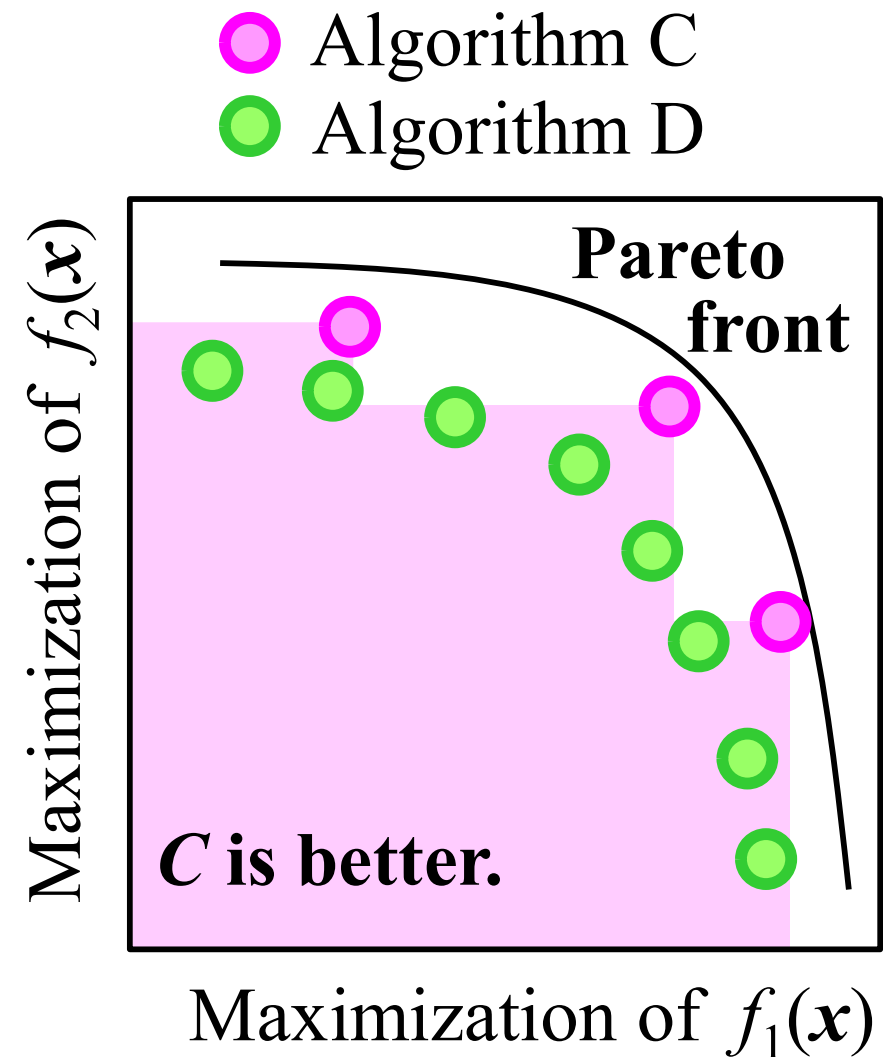
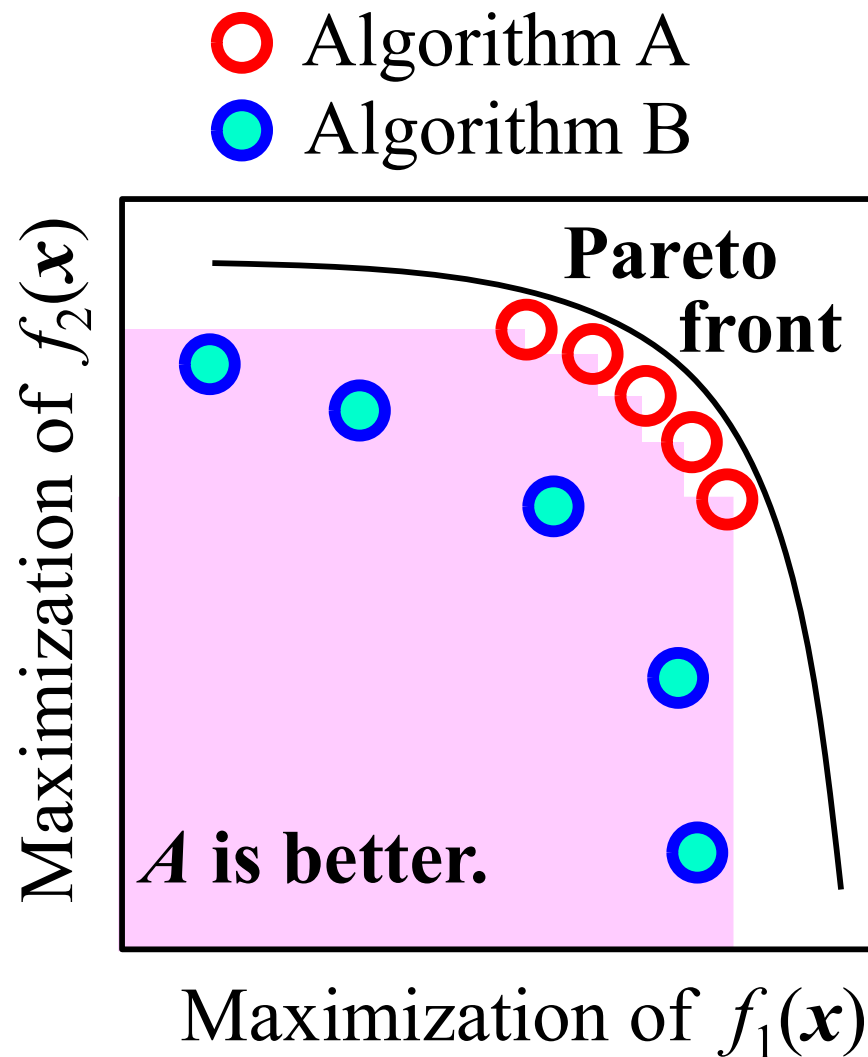
- Convergence to the Pareto front.
- Spread of a solution set (Broadness): Diversity of solutions.
- Uniformity of solutions: Diversity of solutions.
- Overall performance.



One Important Feature of Indicators

Pareto Compliance

If a solution set A is better than another solution set B ,
 A is always evaluated as a better solution set than B by an indicator.

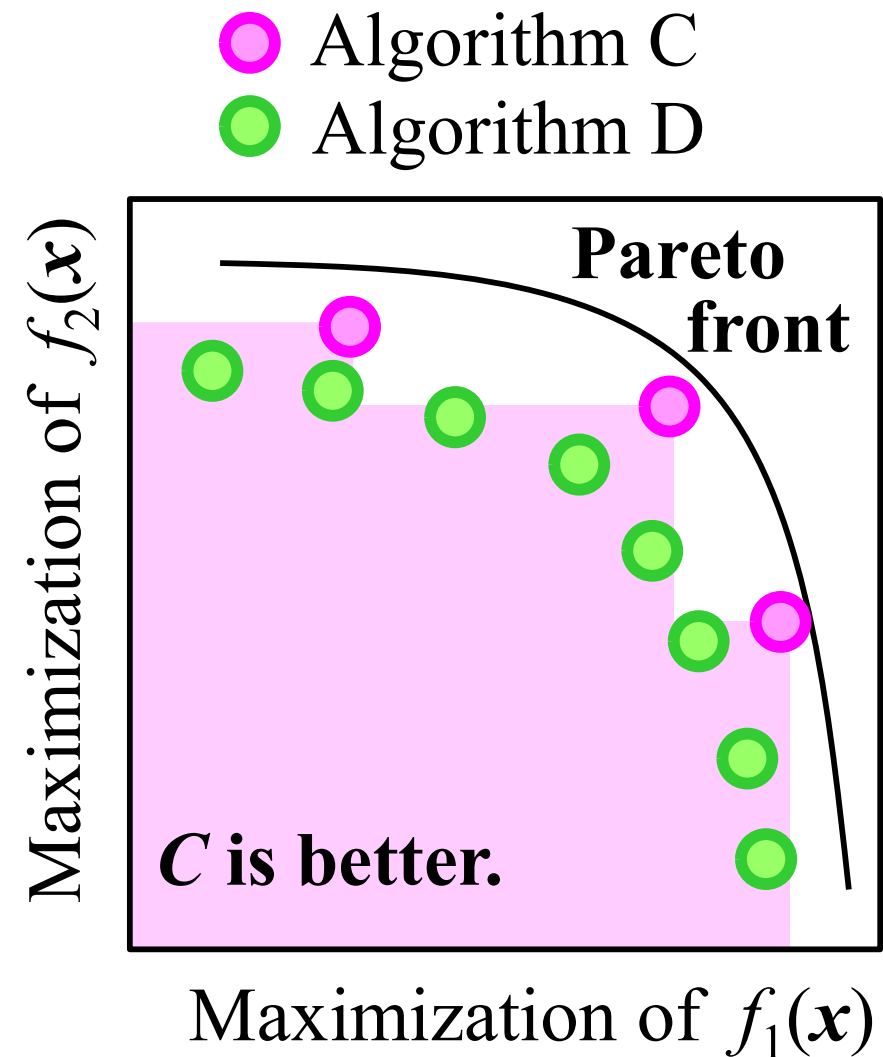
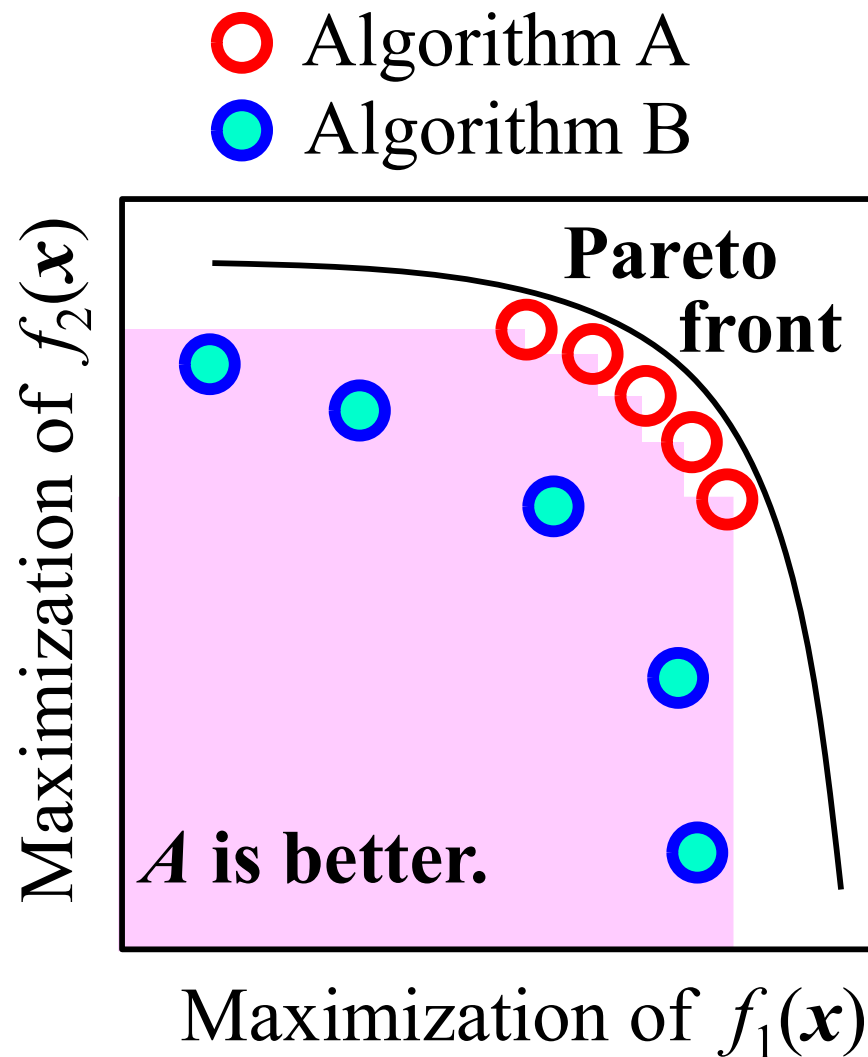


One Important Feature of Indicators

Pareto Compliance

If **a solution set A is better than another solution set B** ,
 A is always evaluated as a better solution set than B by an indicator.

How to define?



Dominance Relation between Solutions $\mathbf{x}$ and $\mathbf{y}$

Maximize $f(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$

Pareto Dominance: $\mathbf{x}$ is dominated by $\mathbf{y}$ ($\mathbf{y}$ is better than $\mathbf{x}$)

$$f(\mathbf{x}) \prec f(\mathbf{y}) \Leftrightarrow \forall i, f_i(\mathbf{x}) \leq f_i(\mathbf{y}) \text{ and } \exists j, f_j(\mathbf{x}) < f_j(\mathbf{y}).$$

Weak Pareto Dominance: $\mathbf{x}$ is weakly dominated by $\mathbf{y}$
($\mathbf{y}$ is not worse than $\mathbf{x}$)

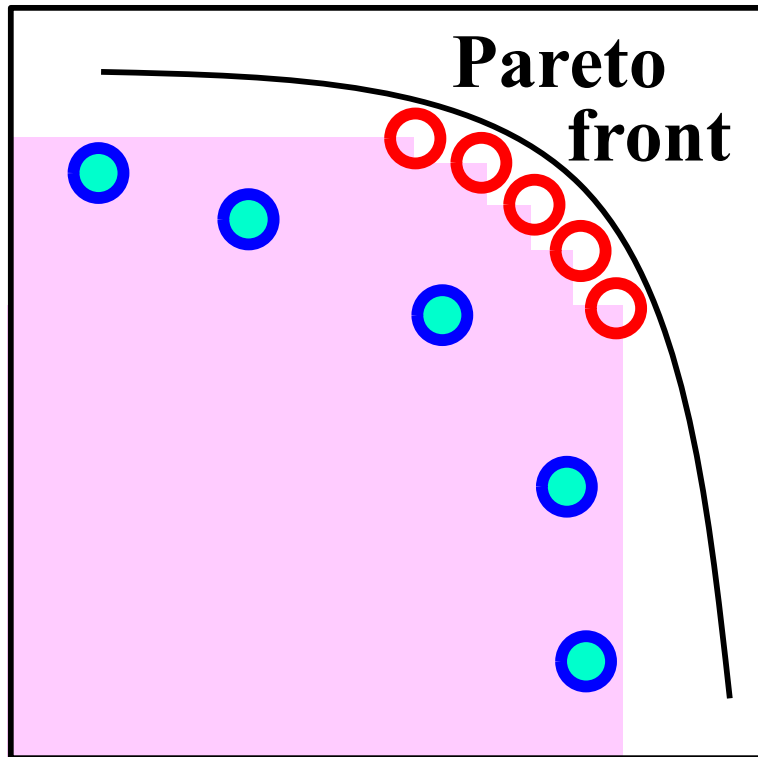
$$f(\mathbf{x}) \preceq f(\mathbf{y}) \Leftrightarrow \forall i, f_i(\mathbf{x}) \leq f_i(\mathbf{y}). \text{ This relation includes } f(\mathbf{x}) = f(\mathbf{y})$$

Another explanation of Pareto dominance (the same meaning):

$$f(\mathbf{x}) \prec f(\mathbf{y}) \Leftrightarrow f(\mathbf{x}) \preceq f(\mathbf{y}) \text{ and } f(\mathbf{x}) \neq f(\mathbf{y}).$$

Relation between Non-Dominated Solution Sets A and B

○ : Solution Set A ● : B



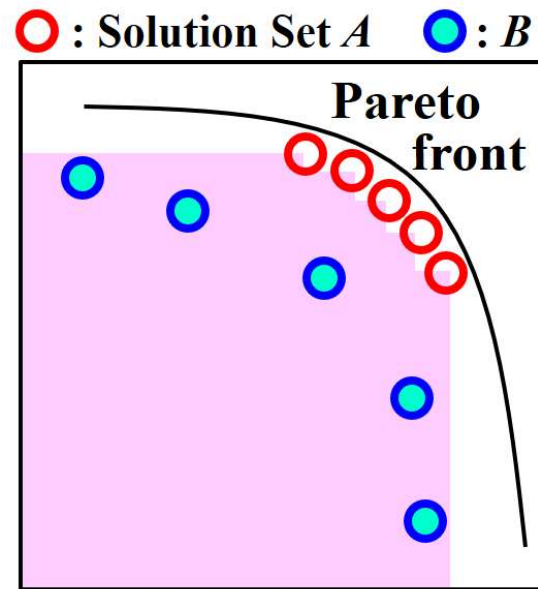
Important:

A and B are non-dominated solution sets. That is, all solutions are non-dominated in each solution set (as shown in this figure).

Weak Pareto Dominance: $f(\mathbf{x}) \preceq f(\mathbf{y}) \Leftrightarrow \forall i, f_i(\mathbf{x}) \leq f_i(\mathbf{y})$.

Pareto Dominance: $f(\mathbf{x}) \prec f(\mathbf{y}) \Leftrightarrow f(\mathbf{x}) \preceq f(\mathbf{y})$ and $f(\mathbf{x}) \neq f(\mathbf{y})$.

Relation between Non-Dominated Solution Sets A and B



Weak Pareto Dominance (B is weakly dominated by A).

$$B \preceq A \Leftrightarrow \forall \mathbf{b} \in B, \exists \mathbf{a} \in A, \mathbf{b} \preceq \mathbf{a}. \quad \text{This relation includes } A = B.$$

Better Relation (A is better than B).

$$B \triangleleft A \Leftrightarrow B \preceq A \text{ and } A \neq B. \quad \text{This relation excludes } A = B.$$

Weak Pareto Dominance: $f(\mathbf{x}) \preceq f(\mathbf{y}) \Leftrightarrow \forall i, f_i(\mathbf{x}) \leq f_i(\mathbf{y})$.

Pareto Dominance: $f(\mathbf{x}) \prec f(\mathbf{y}) \Leftrightarrow f(\mathbf{x}) \preceq f(\mathbf{y}) \text{ and } f(\mathbf{x}) \neq f(\mathbf{y})$.

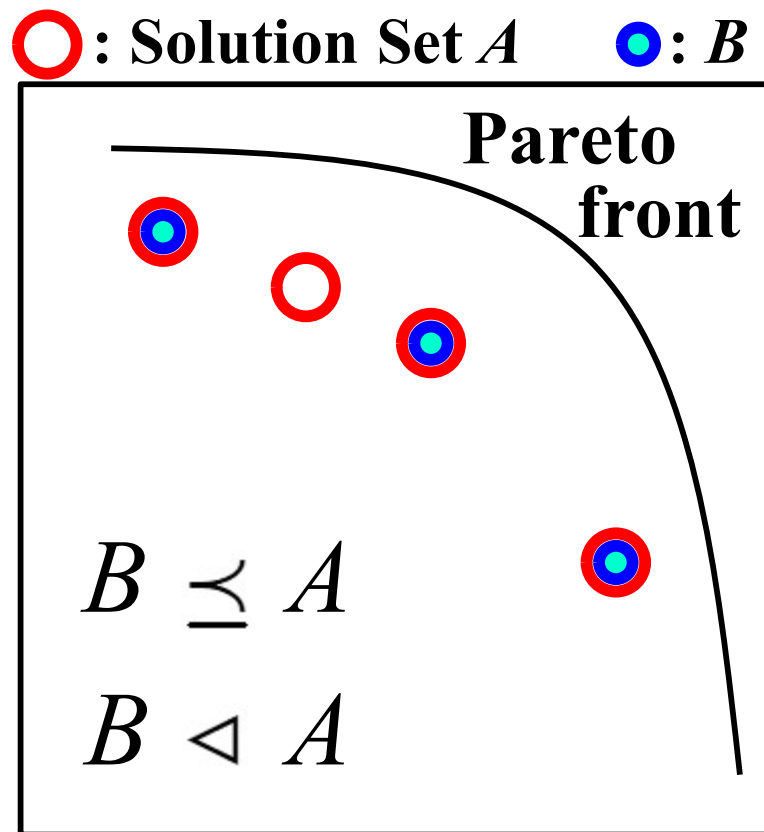
Important Note: A and B are non-dominated solution sets

Weak Pareto Dominance (B is weakly dominated by A).

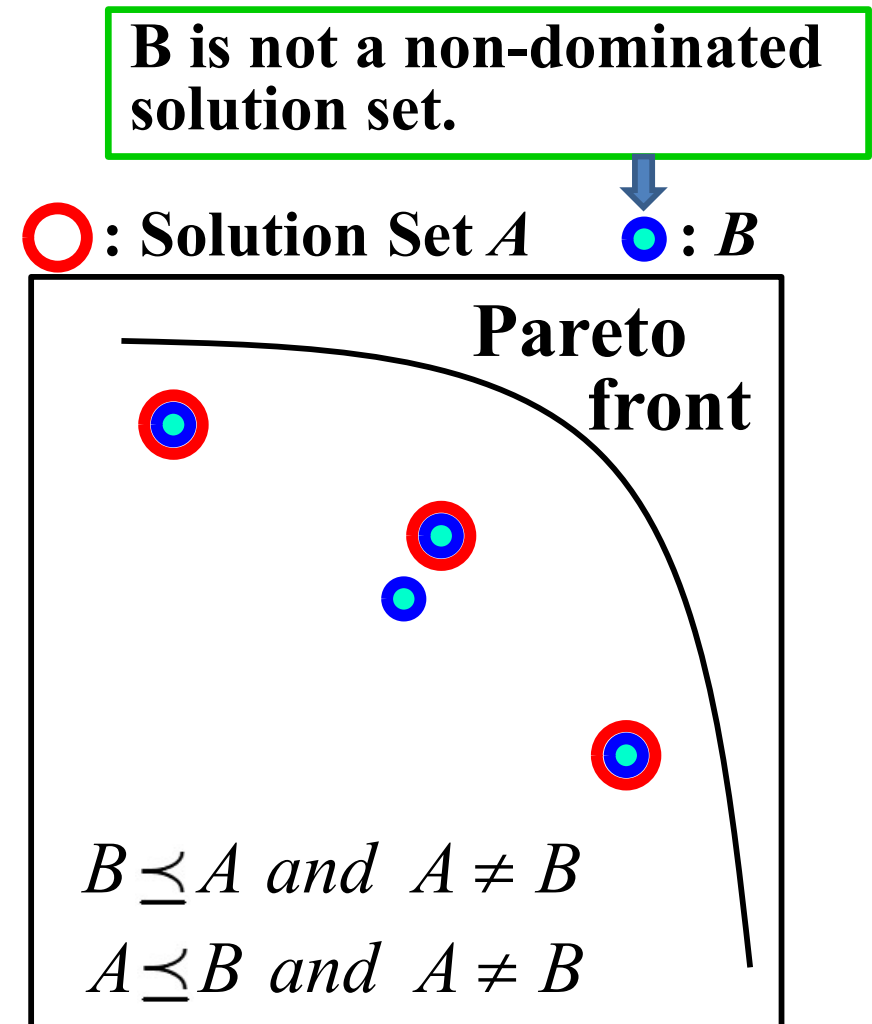
$$B \preceq A \Leftrightarrow \forall b \in B, \exists a \in A, b \preceq a. \quad \text{This relation includes } A = B.$$

Better Relation (A is better than B).

$$B \triangleleft A \Leftrightarrow B \preceq A \text{ and } A \neq B.$$



A is better than B .

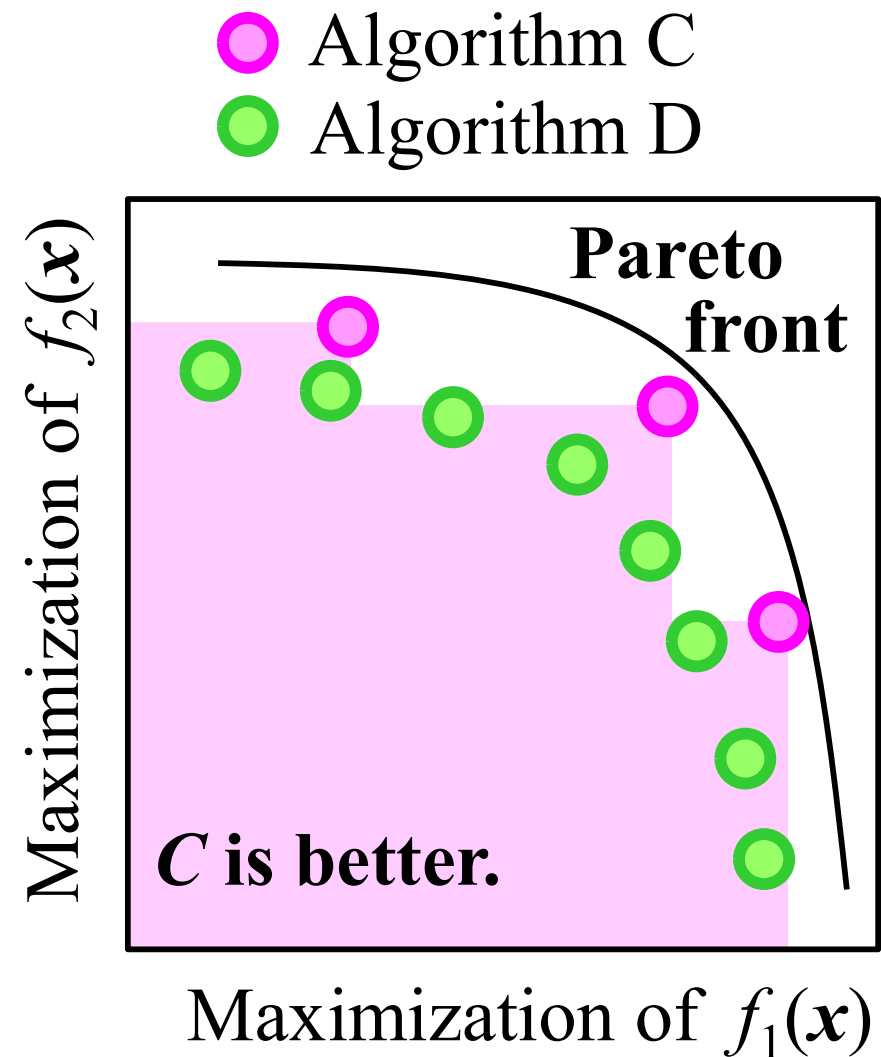
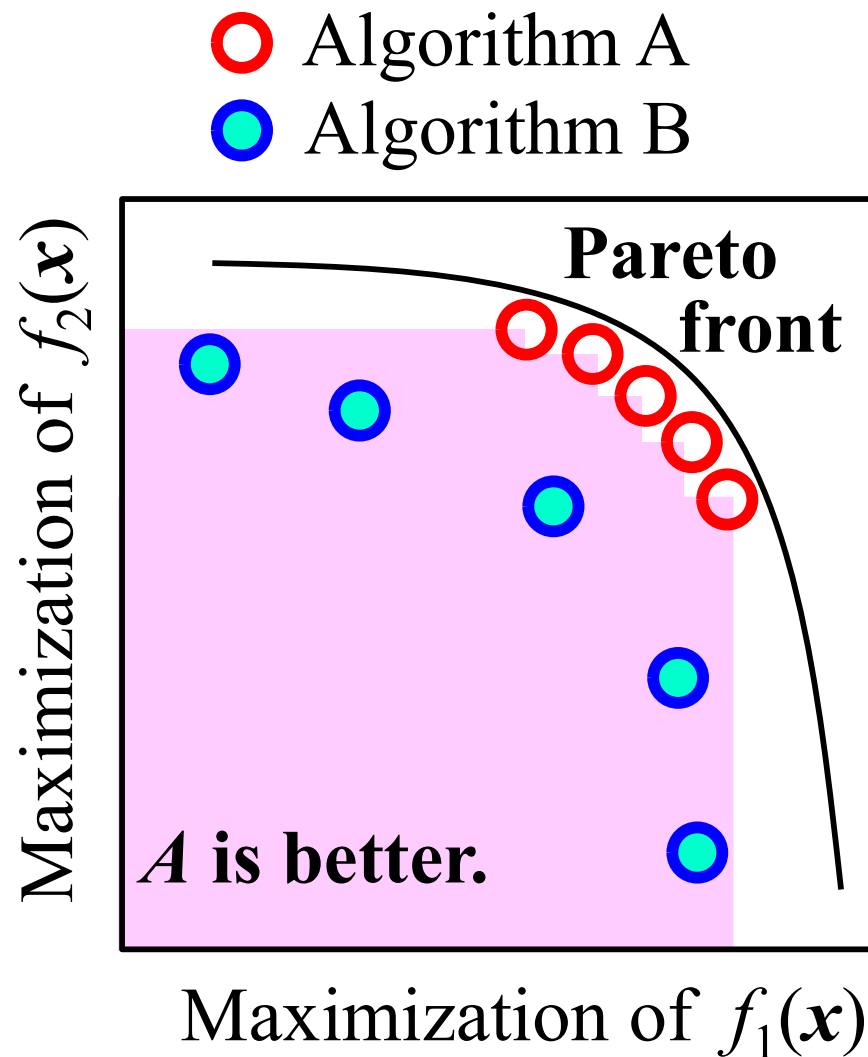


A is not better than B .

One Important Feature of Indicators

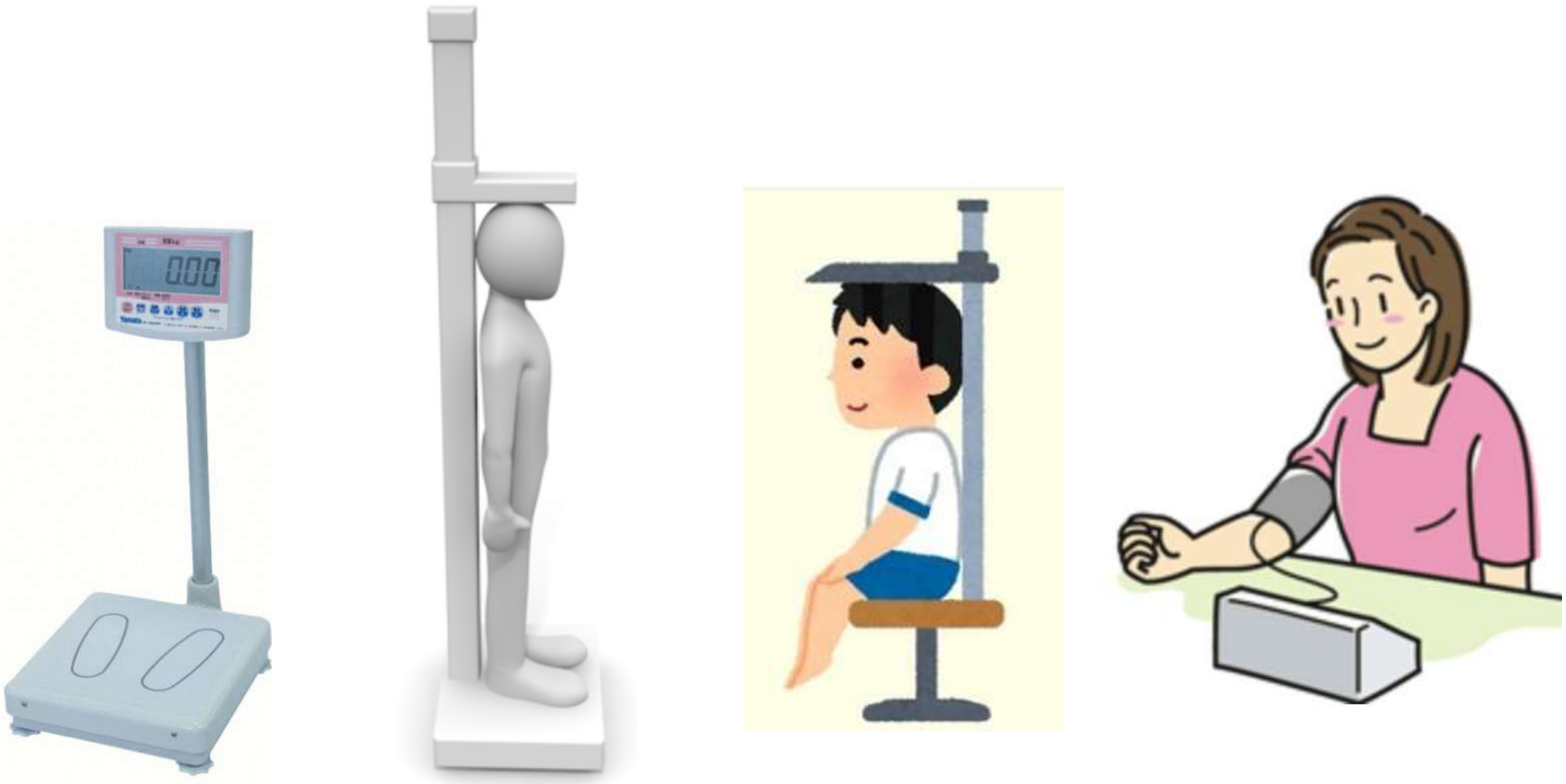
Pareto Compliance (Pareto Compliant Indicator)

If a solution set A is better than another solution set B ,
 A is always evaluated as a better solution set than B by an indicator.



Indicators: Performance evaluation of a solution set

- **Convergence to the Pareto front.**
- **Spread of a solution set (Broadness): Diversity of solutions.**
- **Uniformity of solutions: Diversity of solutions.**
- **Overall performance.**

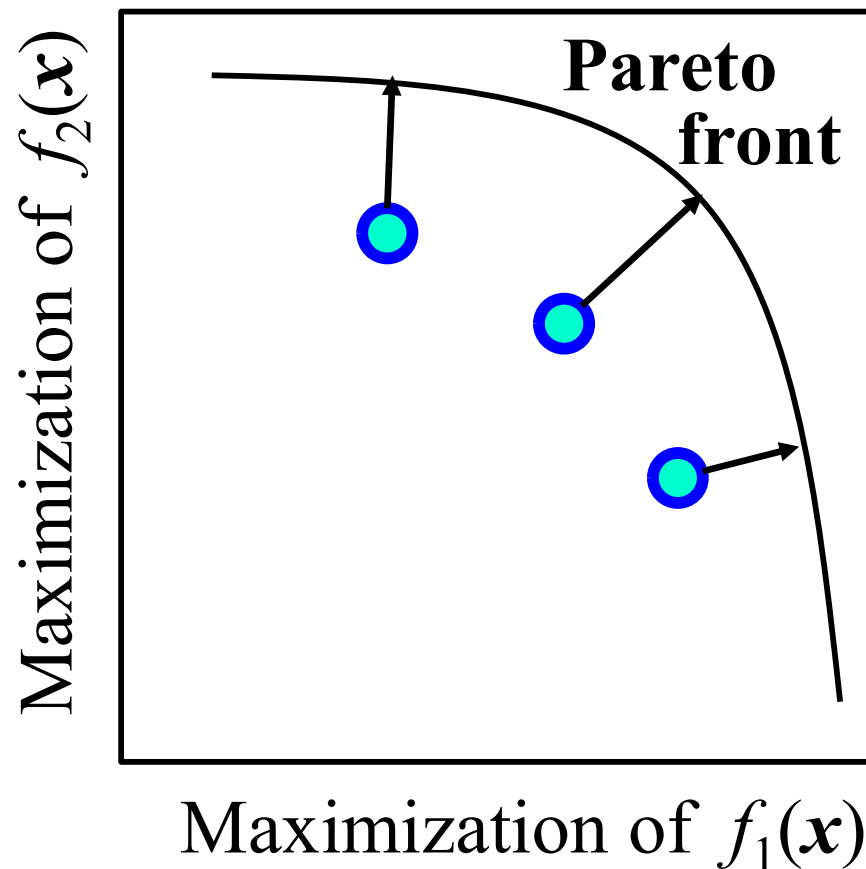


Performance Indicator for Convergence

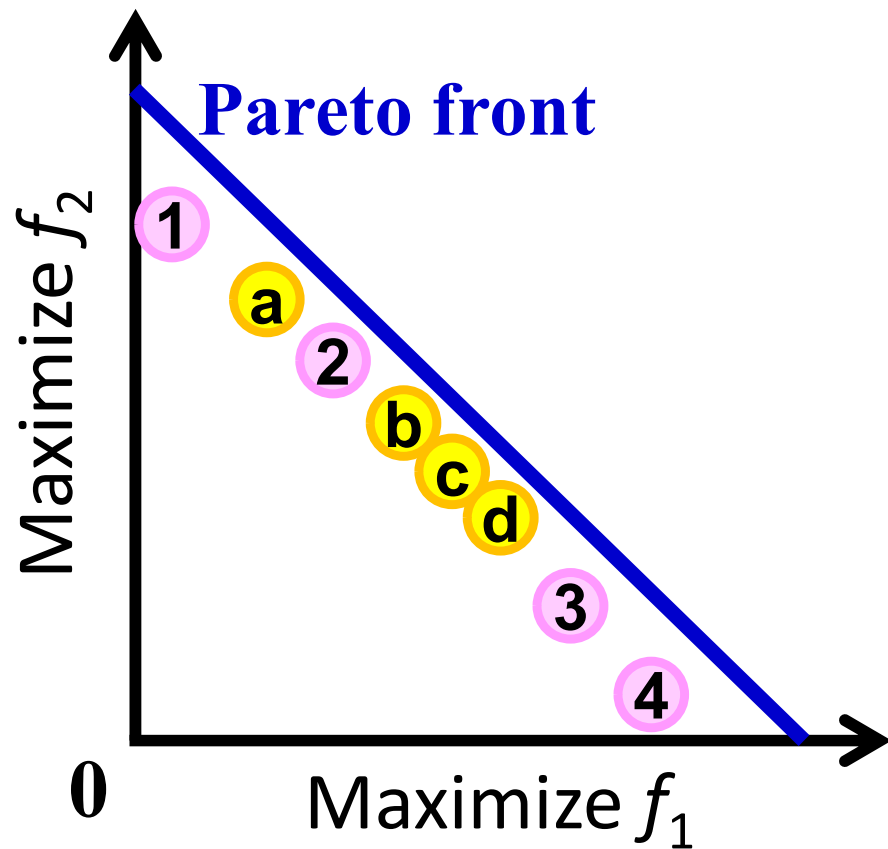
GD (Generational Distance): **The smaller, the better**

Basic Idea:

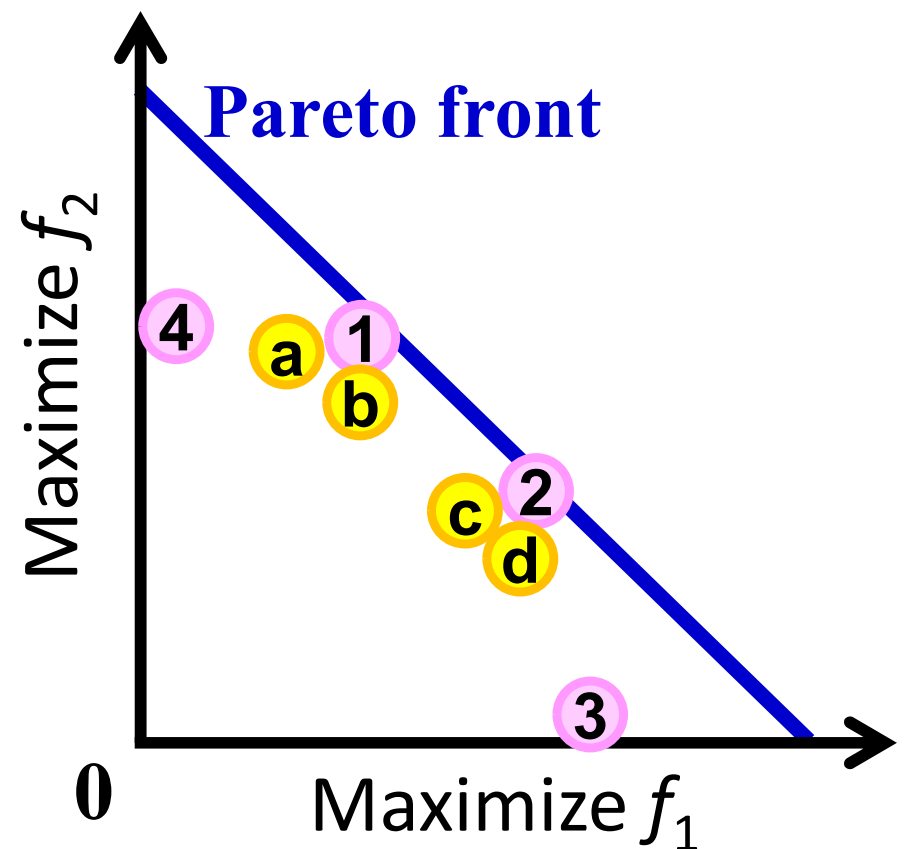
The average distance from each solution to the Pareto front.



Which is better based on GD: Pink or Yellow ?



Example 1



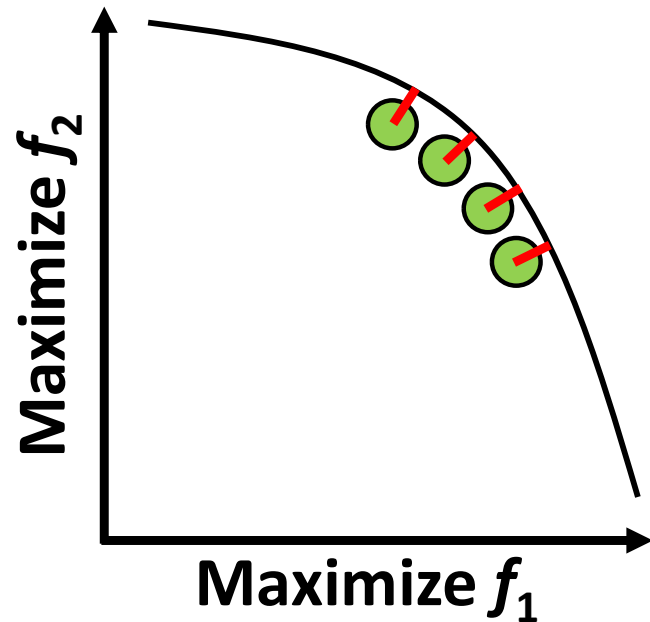
Example 2

Performance Indicator for Convergence

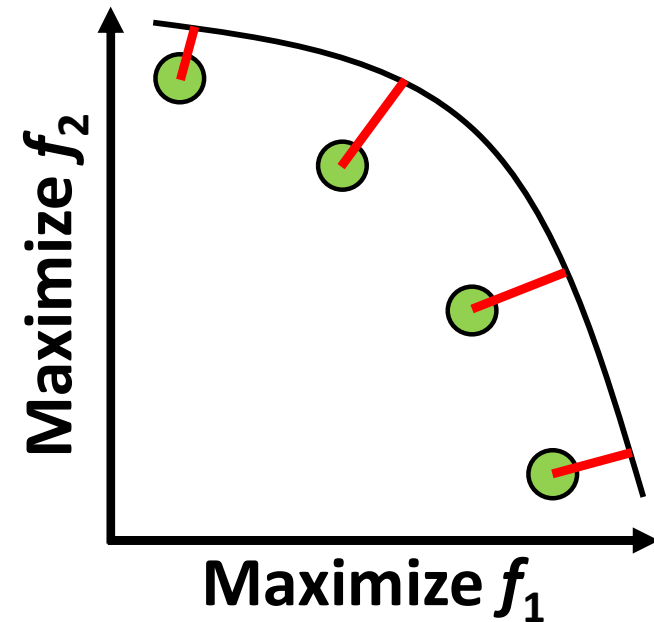
GD (Generational Distance)

Basic Idea:

The average distance from each solution to the Pareto front.



Small GD ==>
Good Convergence



Large GD ==>
Poor Convergence

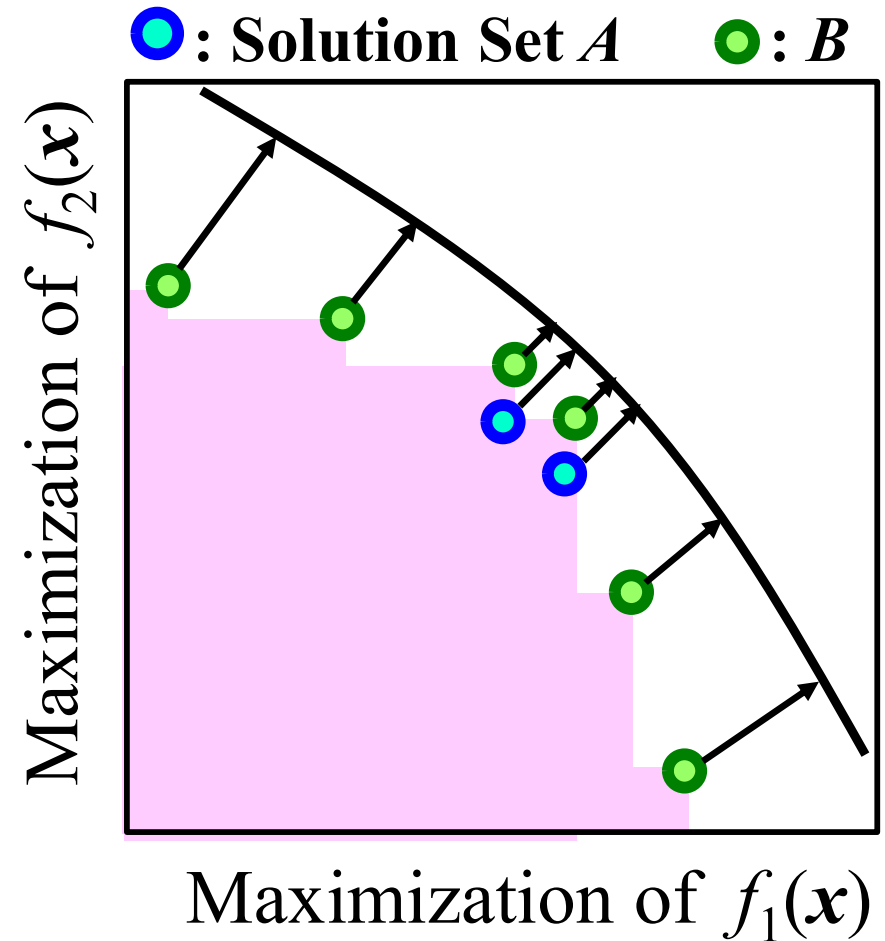
Performance Indicator for Convergence

GD (Generational Distance)

The main difficulty of GD:

GD is not Pareto compliant.

Each solution in A is dominated by at least one solution in B and $A \neq B$ (i.e., B is better than A). However, A has a better (smaller) GD value.



Performance Indicator for Convergence

GD (Generational Distance)

Related difficulty of GD:

GD favors a smaller solution set.

Smaller solution sets often have better GD values (i.e., GD values are improved by removing solutions that are not close to the Pareto front).

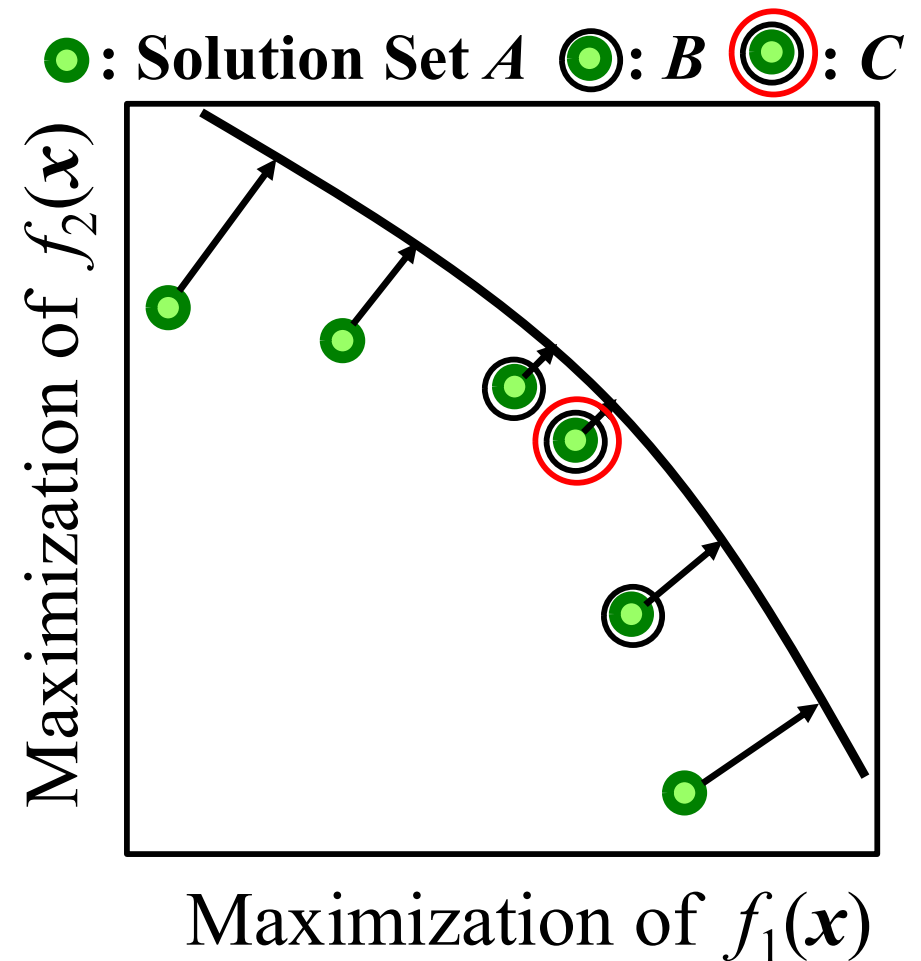
In the figure:

$$A \supset B \supset C$$

$$GD(A) > GD(B) > GD(C)$$

Pareto dominance-based evaluation:

A is better than B and C ,
and B is better than C .

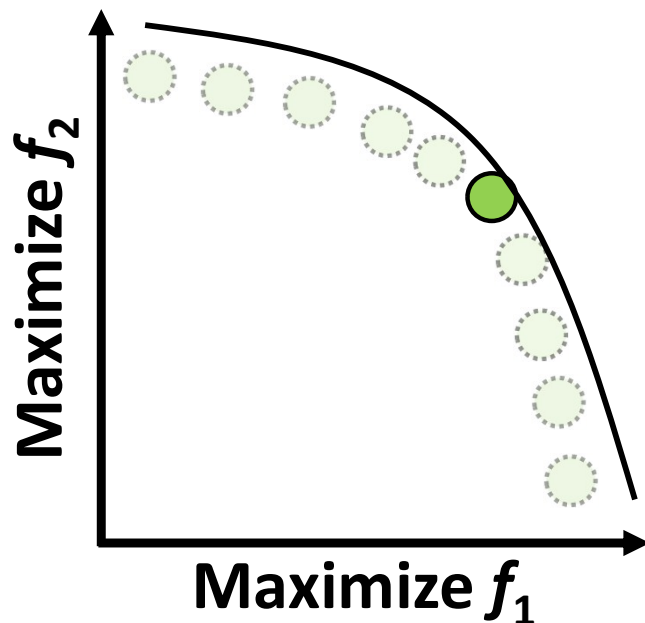


Performance Indicator for Convergence

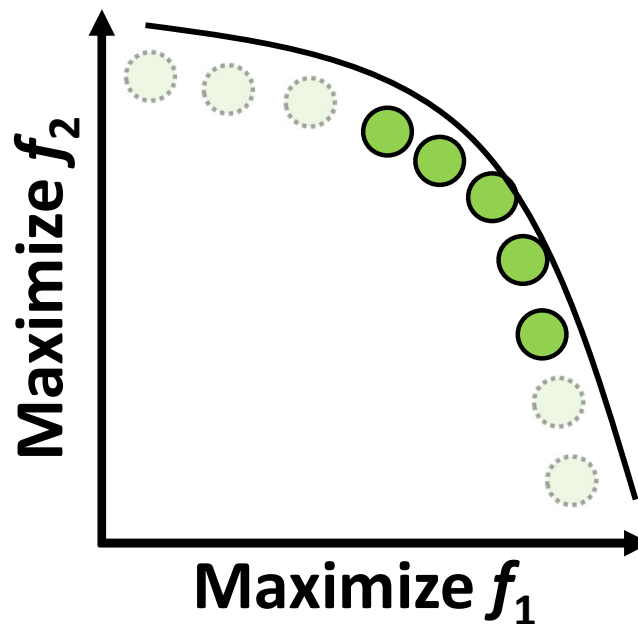
GD (Generational Distance)

Related difficulty of GD:

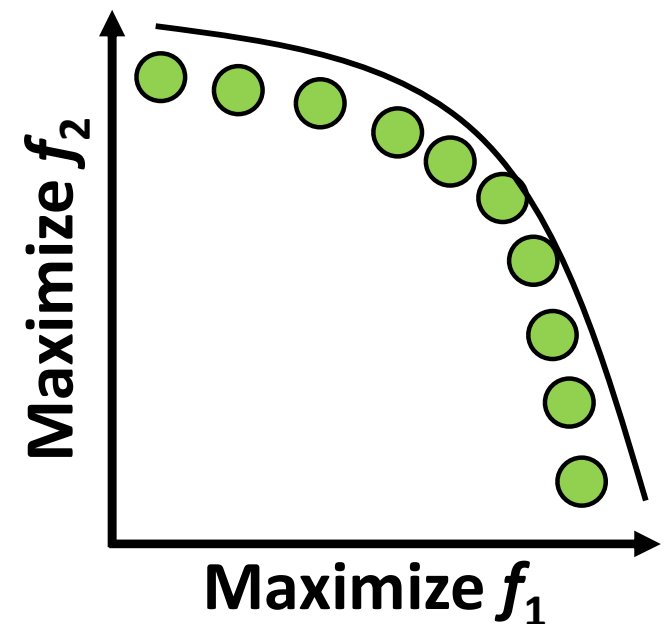
GD favors a smaller solution set.



Best GD



2nd Best GD



Worst GD

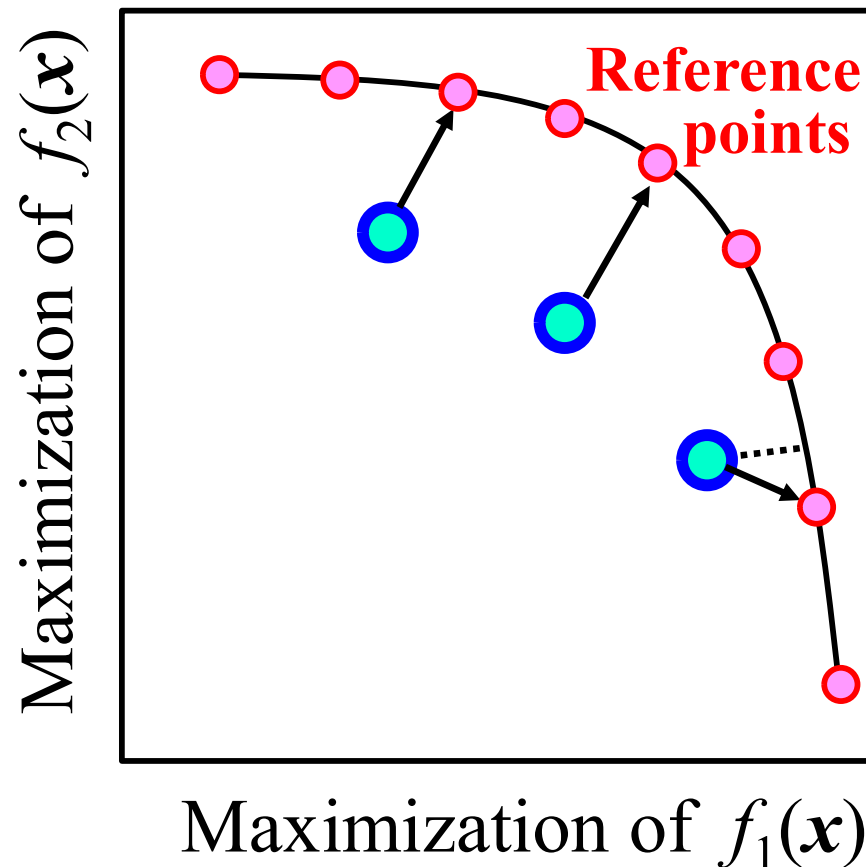
Performance Indicator for Convergence

GD (Generational Distance)

When the distance calculation to the Pareto front is difficult:

Step 1: Generate a number of reference points on the Pareto front.

Step 2: Calculate the average distance from each solution to the nearest reference point.

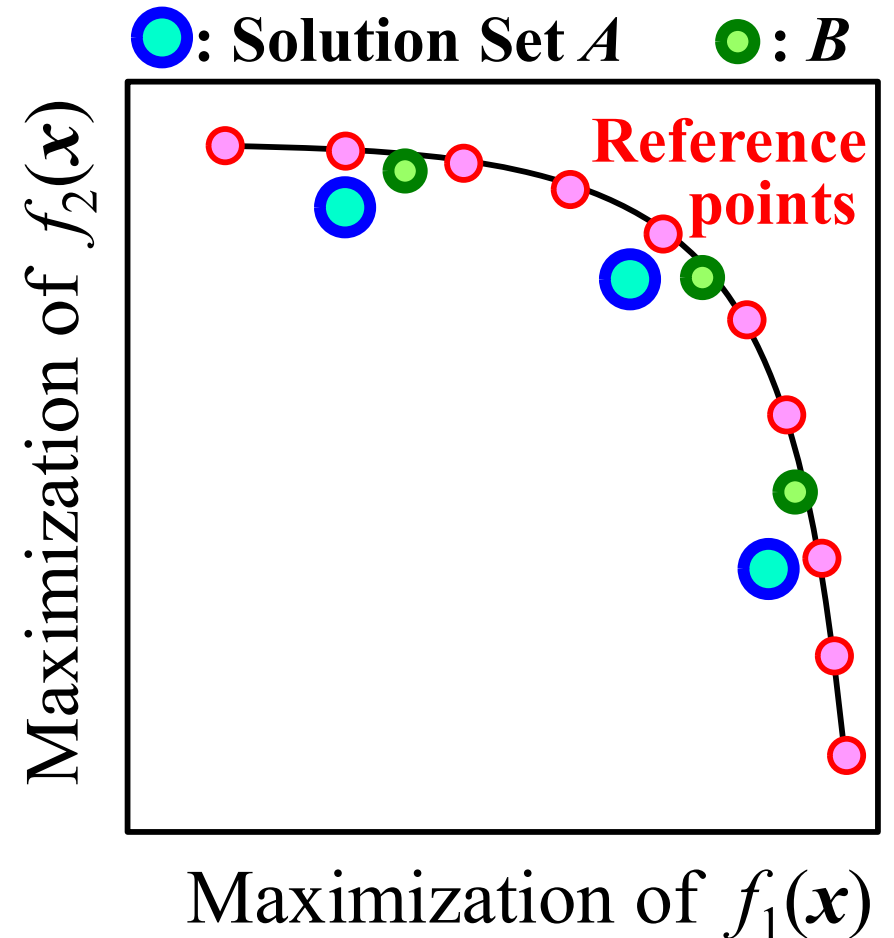


Performance Indicator for Convergence

GD (Generational Distance)

One potential difficulty when we use the reference points:
Dependency on the reference points

While solution set B is clearly closer to the Pareto front than solution set A (i.e., B has a better convergence), solution set A has a better GD value.



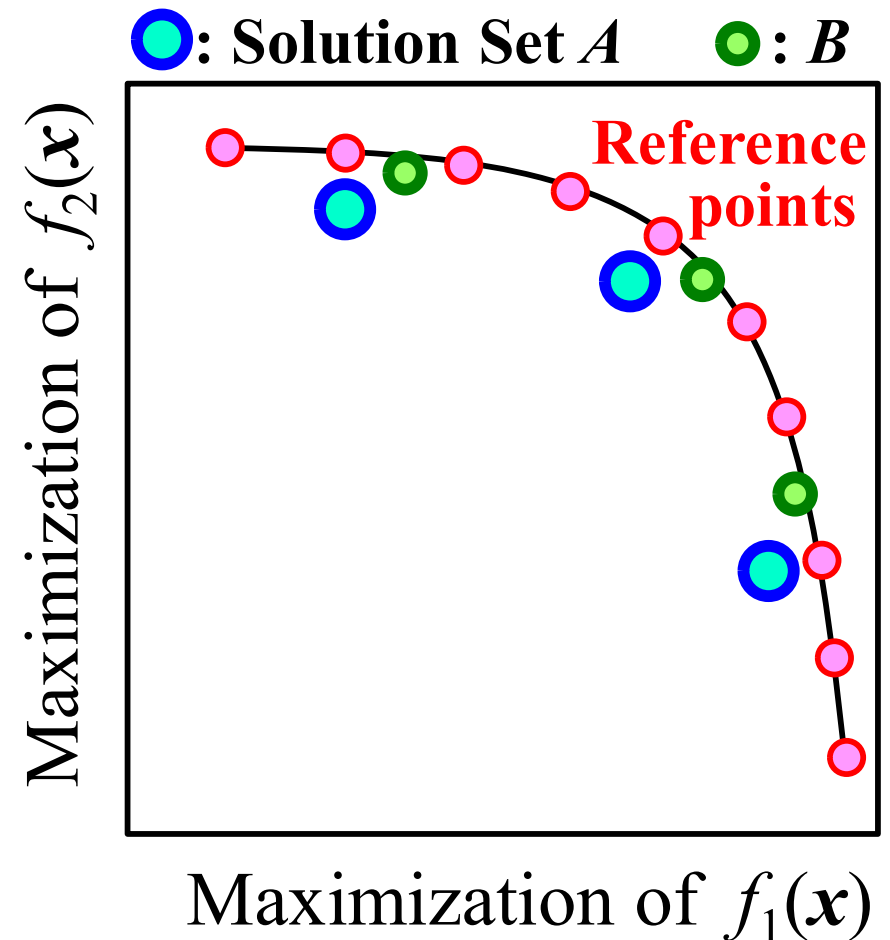
Performance Indicator for Convergence

GD (Generational Distance)

One potential difficulty when we use the reference points:
Dependency on the reference points

While solution set B is clearly closer to the Pareto front than solution set A (i.e., B has a better convergence), solution set A has a better GD value.

It is advisable to calculate the distance to the Pareto front (without using reference points).

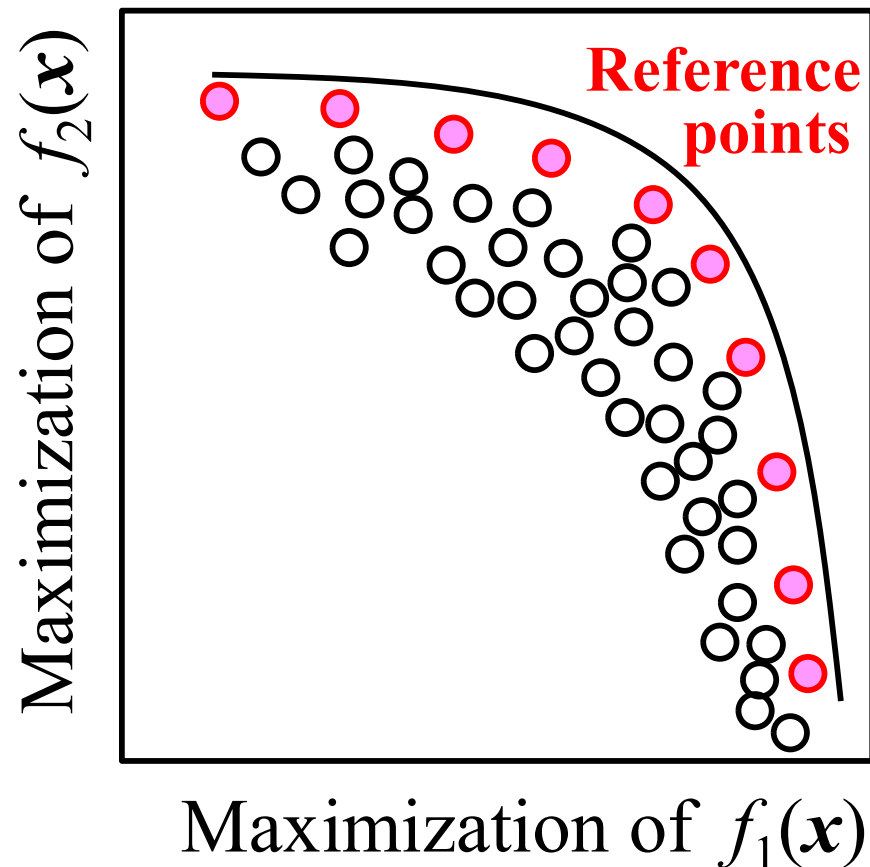


Performance Indicator for Convergence

GD (Generational Distance)

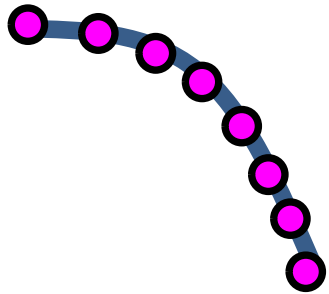
When the Pareto front is unknown,
how can we generate reference points for GD calculation?

1. Merge the obtained solutions by all algorithms into a single set.
2. Choose only non-dominated solutions from the merged solution set.

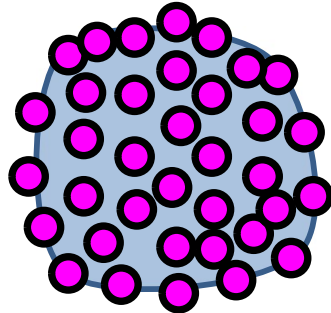


Difficulty: It is very difficult to create enough reference points for many-objective problems

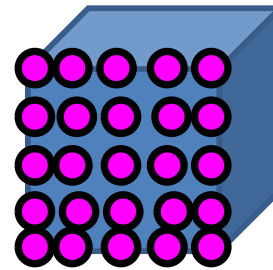
2-Objective problem



3-Objective problem

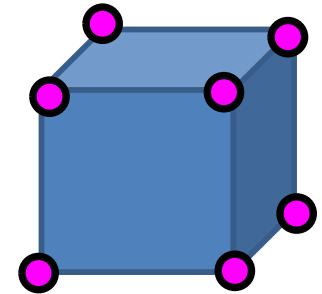


4-Objective problem



...

11-Objective problem



1D Pareto front
(100 solutions)

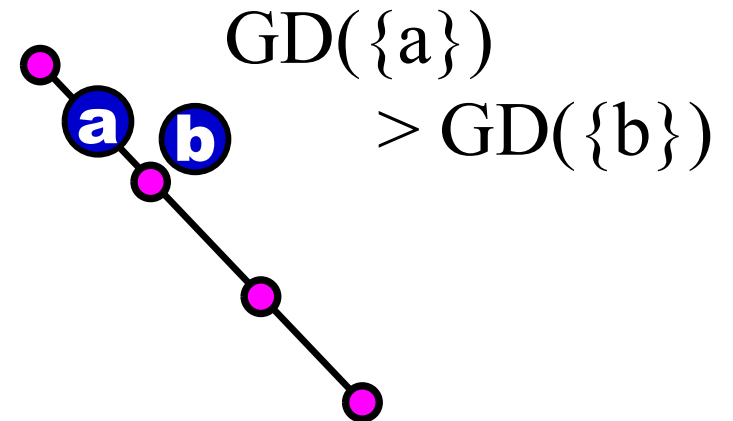
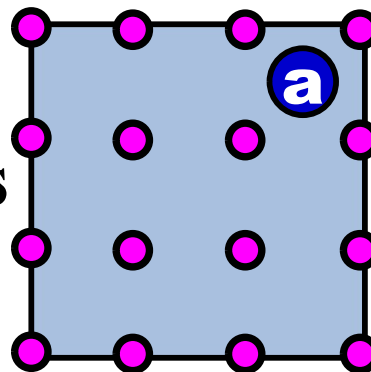
2D Pareto front
(100 x 100)

3D Pareto front
(100 x 100 x 100)

10D Pareto front
(1024 Corners)

11-Objective problem

$4^{10} = 1,048,576$ solutions



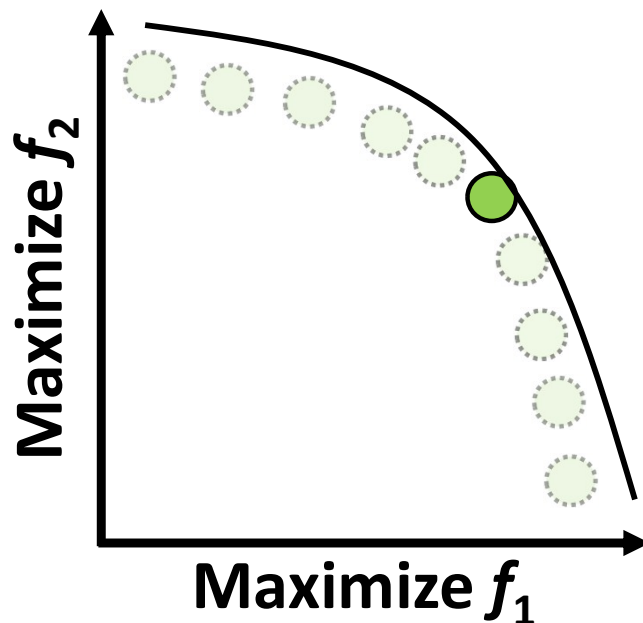
Performance Indicator for Convergence

GD (Generational Distance)

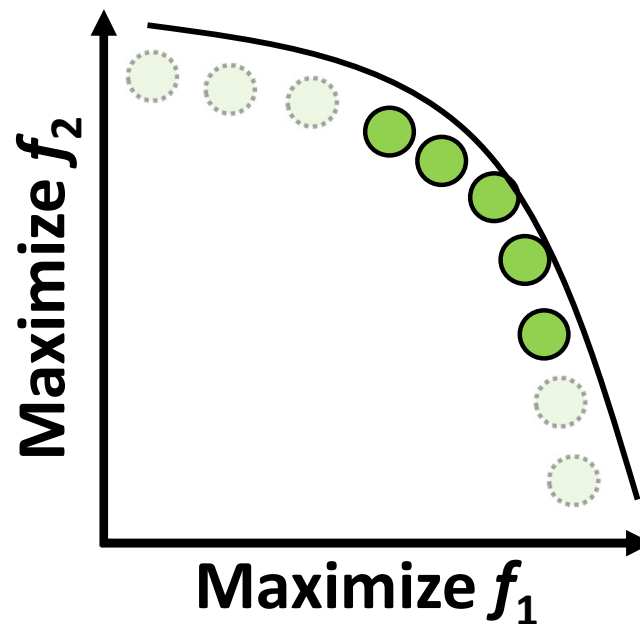


Important Note:

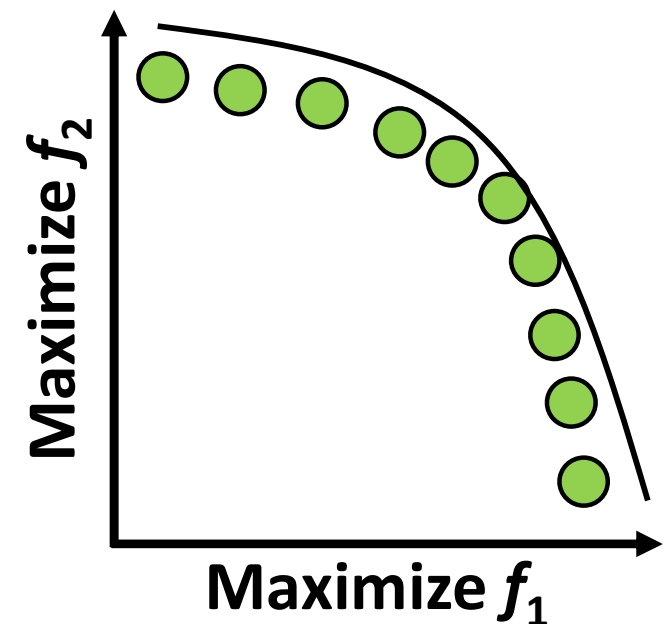
GD can evaluate only the convergence (not the overall performance).



Best GD



2nd Best GD



Worst GD

Other Ideas to Evaluate the Convergence

For Discrete (Combinatorial) Optimization Problem

When all the Pareto optimal solutions are known,

(1) Percentage of the obtained Pareto optimal solutions.

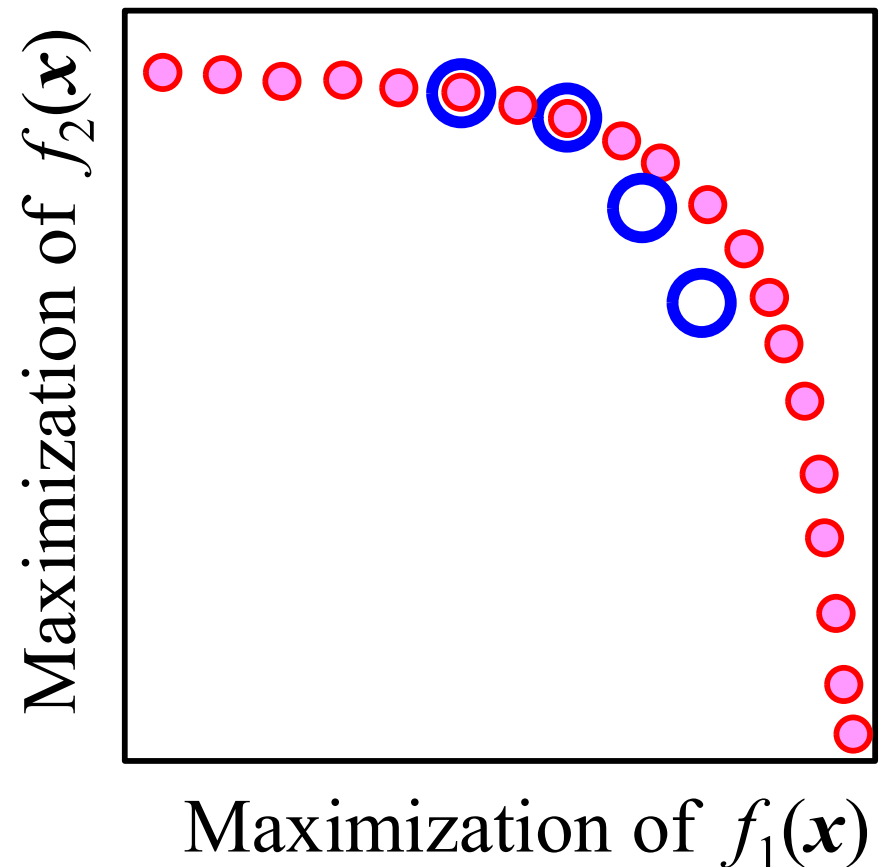
(2) Percentage of the Pareto optimal solutions in the final population.

(1) 10% (2 out of 20) of the Pareto optimal solutions are obtained by a single run of an EMO algorithm.
This is an overall performance indicator.

(2) 50% (2 out of 4) of the final population of a single run of an EMO algorithm are Pareto optimal solutions.
This is a pure convergence indicator.

The larger, the better

●: Pareto ○: Final population

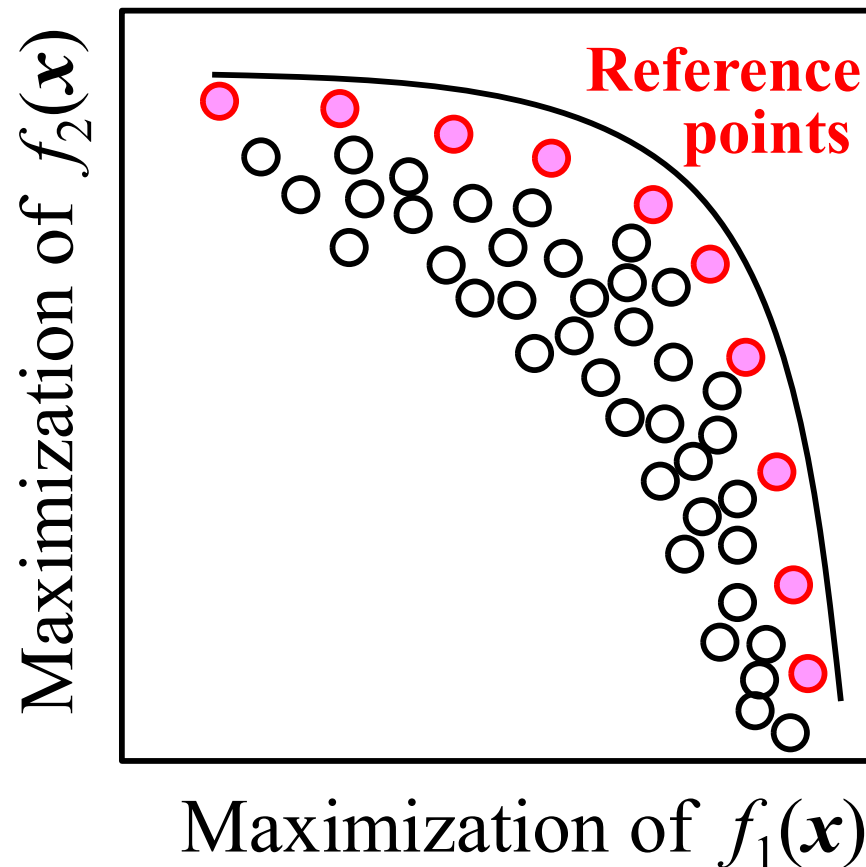


Other Ideas to Evaluate the Convergence

For Discrete (Combinatorial) Optimization Problem

When Pareto optimal solutions are unknown:

Using all the obtained solutions by all algorithm and choosing only non-dominated solutions, we can construct an approximate Pareto optimal solution set (in the same manner as the construction of the reference point set for GD calculation). Then we can calculate the percentage.



Performance Indicator for Diversity (**Spread**)

Maximum Spread: The larger, the better

Basic Idea:

The length of the diagonal of a hyperbox including all solutions.

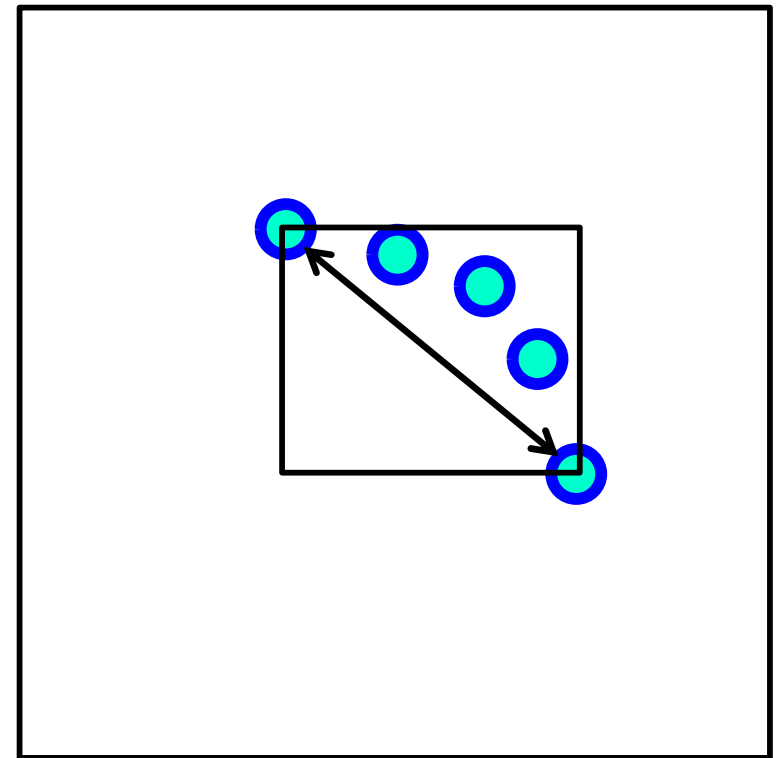
Solution Set: $A = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N\}$ where $\mathbf{a}_i = (a_{i1}, a_{i2}, \dots, a_{im})$

Maximum Spread:

$$D = \sqrt{\sum_{j=1}^m \left(\max_{\mathbf{a}_i \in A} \{a_{ij}\} - \min_{\mathbf{a}_i \in A} \{a_{ij}\} \right)^2}$$

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

Maximization of $f_2(\mathbf{x})$



Maximization of $f_1(\mathbf{x})$



Performance Indicator for Diversity (Spread)

Maximum Spread

The maximum spread indicator is not Pareto compliant.

It can evaluate only the spread (not the overall performance).

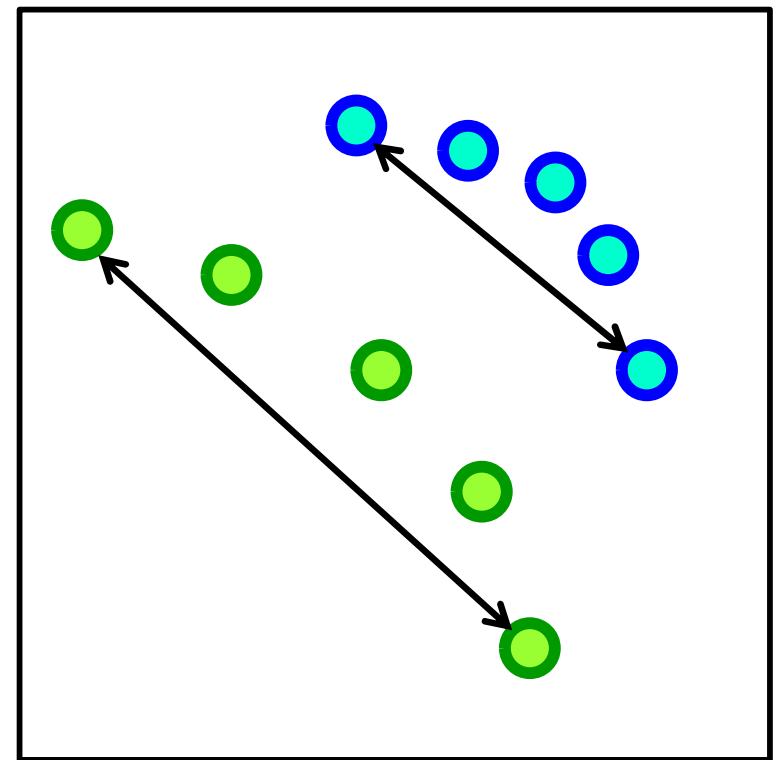
Solution Set: $A = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N\}$ where $\mathbf{a}_i = (a_{i1}, a_{i2}, \dots, a_{im})$

Maximum Spread:

$$D = \sqrt{\sum_{j=1}^m \left(\max_{\mathbf{a}_i \in A} \{a_{ij}\} - \min_{\mathbf{a}_i \in A} \{a_{ij}\} \right)^2}$$

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

Maximization of $f_2(\mathbf{x})$



Maximization of $f_1(\mathbf{x})$



Indicators: Performance evaluation of a solution set

- Convergence to the Pareto front: **GD**
- Spread of a solution set (Broadness): **Maximum Spread**
- **Uniformity of solutions**
- Overall performance.



Performance Indicator for Diversity (Uniformity)

Spacing: The smaller, the better

Basic Idea:

Standard deviation of the distance from each solution to the nearest neighbor.



Solution Set: $A = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_N\}$ where $\mathbf{a}_i = (a_{i1}, a_{i2}, \dots, a_{im})$

Spacing
$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N (d_i - \bar{d})^2}$$

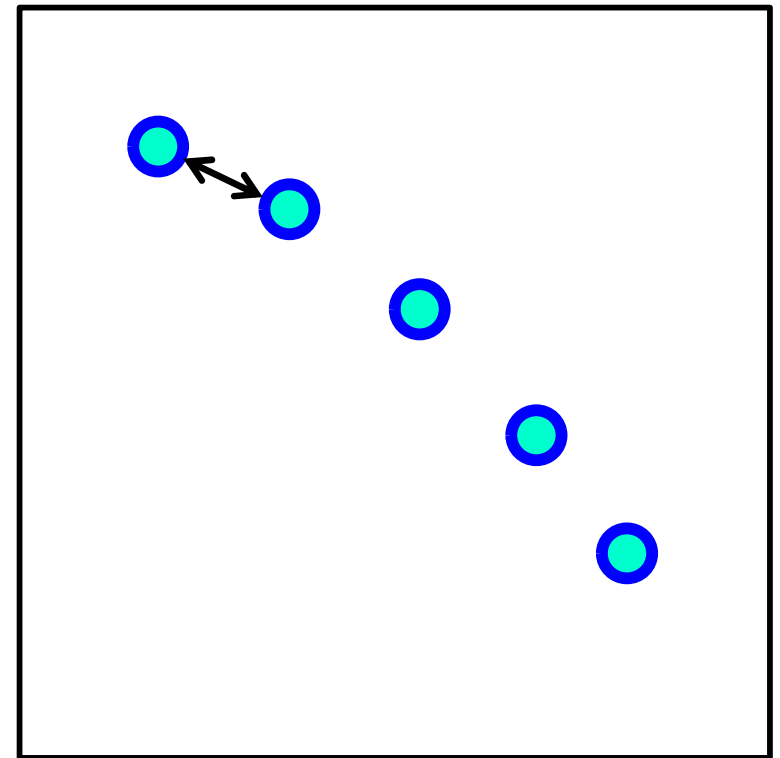
where

$$d_i = \min_{\substack{\mathbf{a}_k \in A \\ k \neq i}} \text{distance}(\mathbf{a}_i, \mathbf{a}_k)$$

$$\bar{d} = \frac{1}{N} \sum_{i=1}^N d_i$$

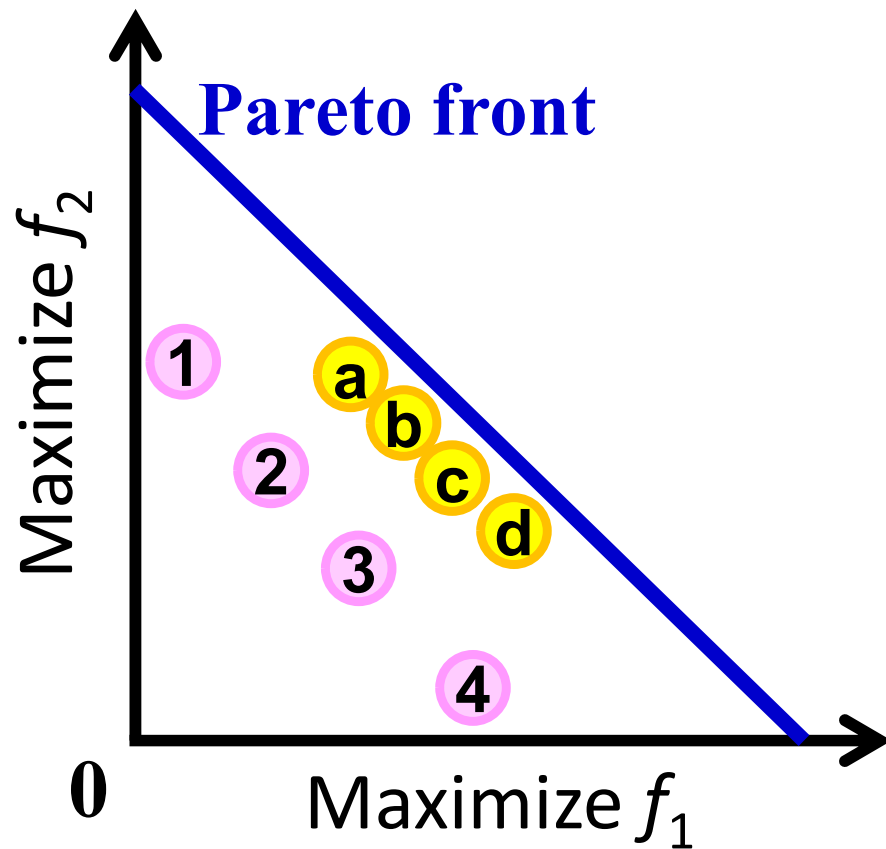
$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x}))$$

Maximization of $f_2(\mathbf{x})$

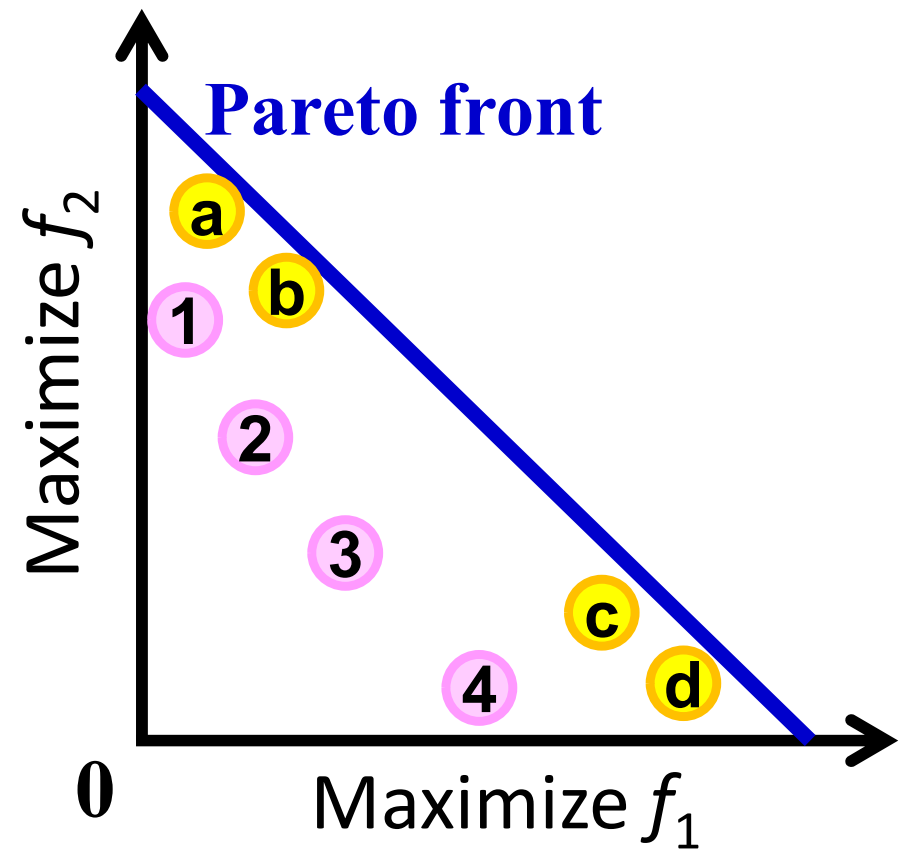


Maximization of $f_1(\mathbf{x})$

Which is better based on Spacing: Pink or Yellow ?



Example 1



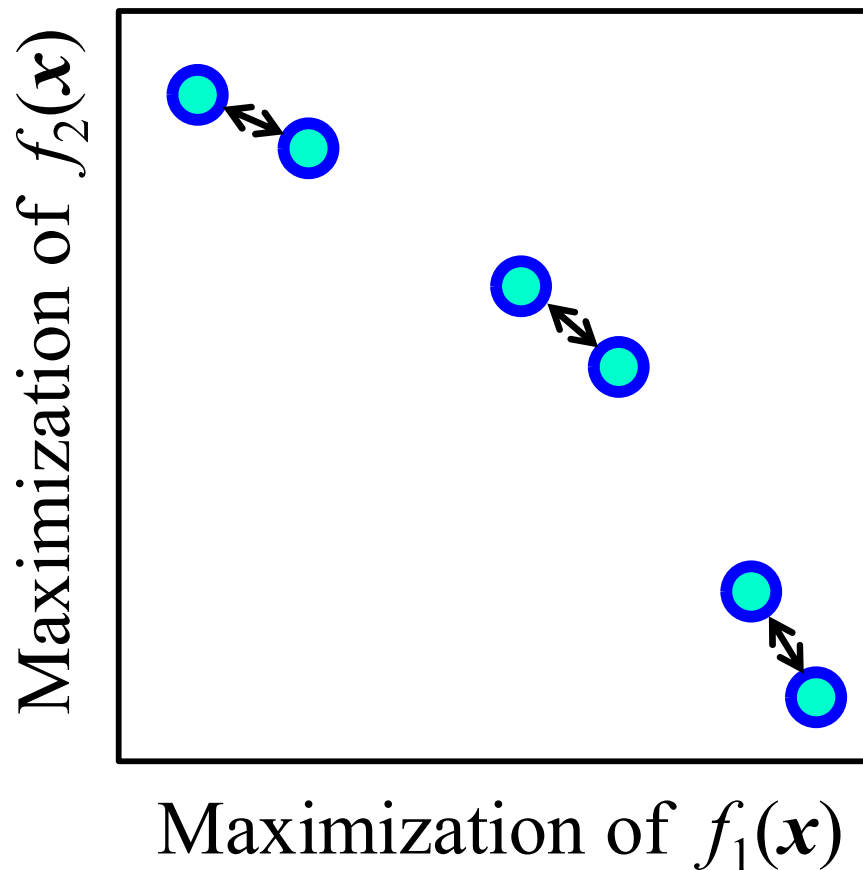
Example 2

Performance Indicator for Diversity (Uniformity)

Spacing

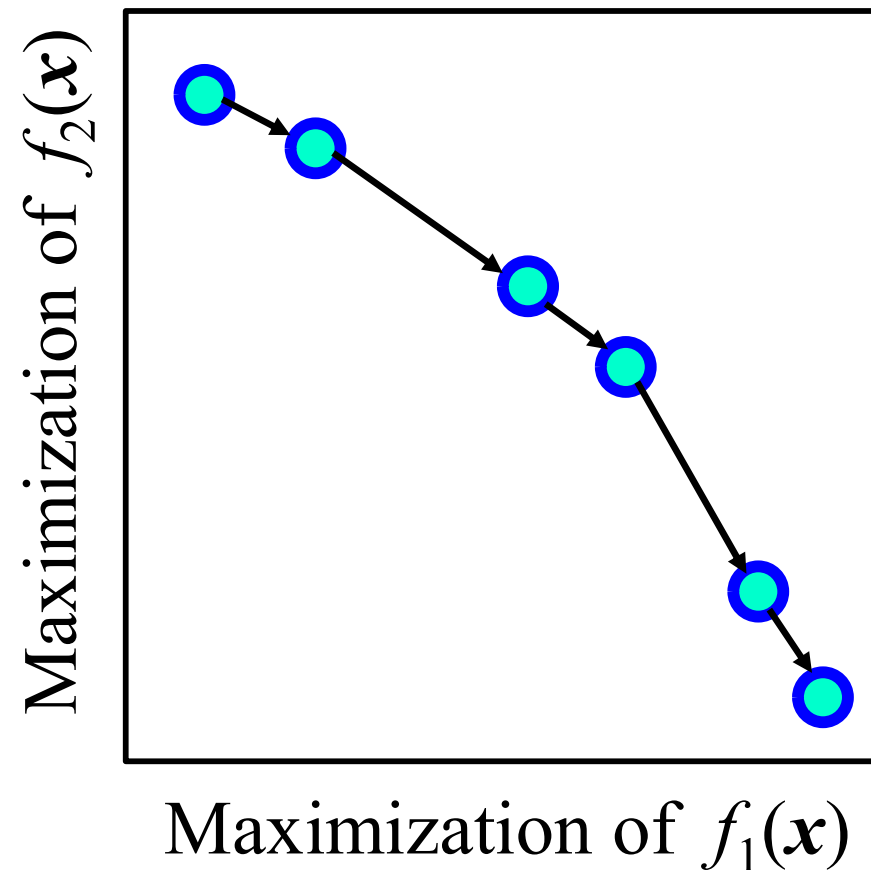
This definition does not work as intended.

Nearest Neighbor Distance
(based on the definition)



Intention (?)

The distance to the next solution

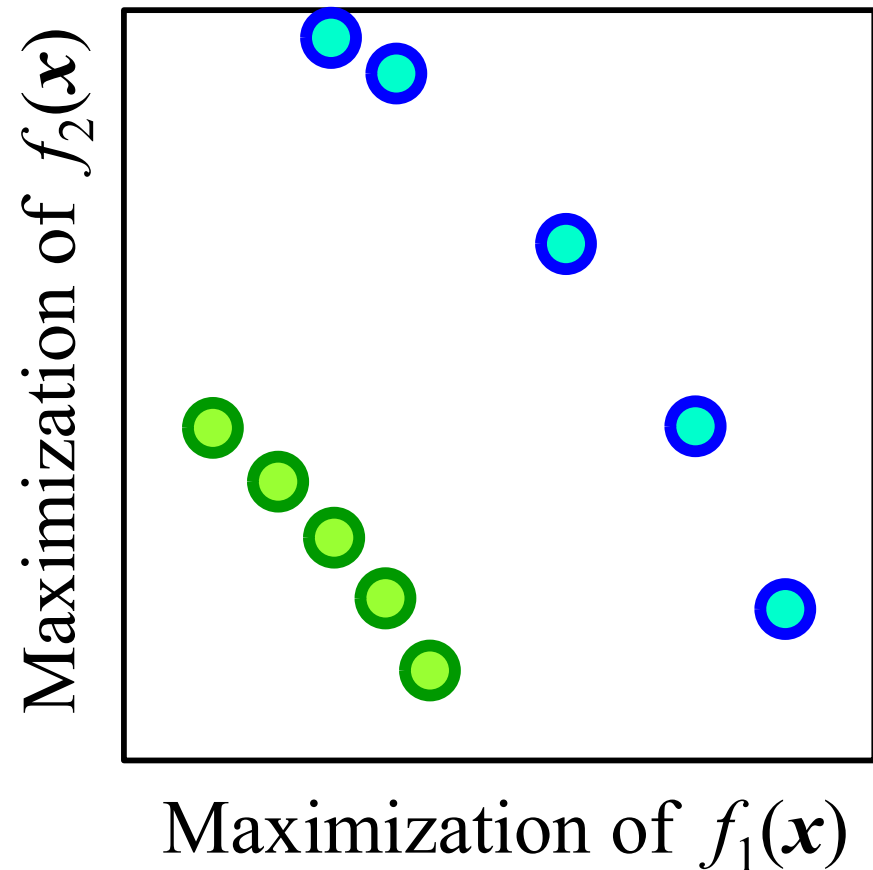
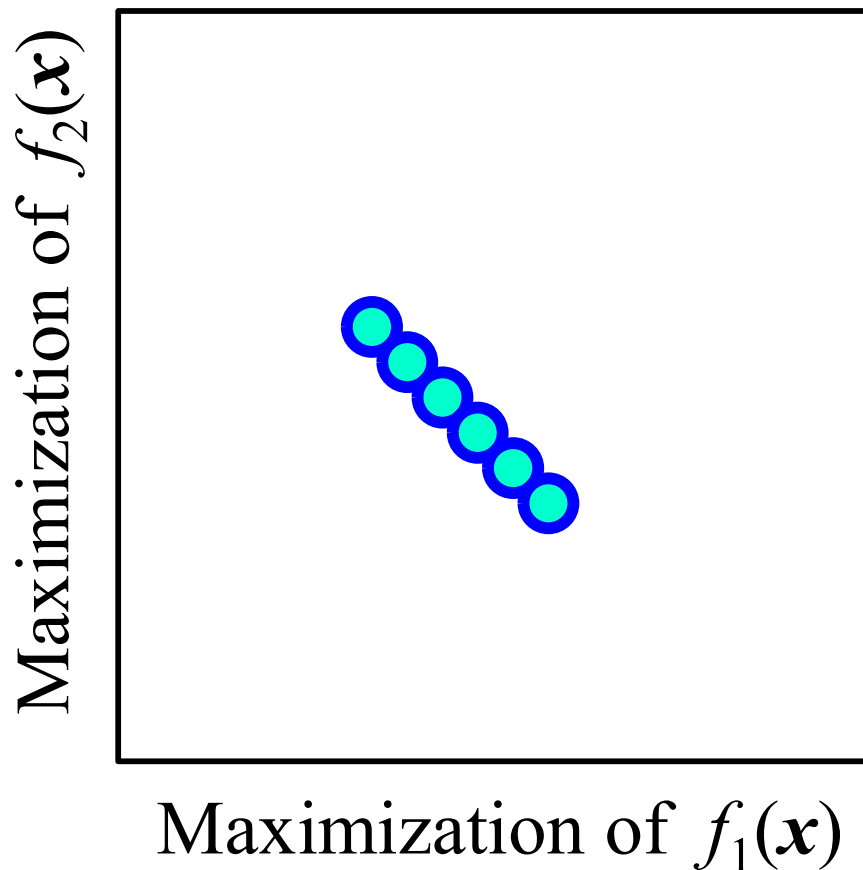


Performance Indicator for Diversity (Uniformity)

Spacing

Other difficulties:

- (1) A concentrated solution set has a good value.
- (2) The spacing indicator is not Pareto compliant.



Performance Indicator for Diversity (Uniformity)

Spread: The smaller, the better

$$\Delta = \frac{d_1^e + d_2^e + \sum_{i=1}^{N-1} |d_i - \bar{d}|}{d_1^e + d_2^e + (N-1)\bar{d}}$$

Positive comments:

- d_i is the distance to the next solution (**Uniformity**).
- The distance from each extreme point of the Pareto front is also used (**Spread**: Overall diversity)

Negative comments:

- The distance to the next point is not easy to define for the case of three or more objectives.
- Not Pareto compliant.

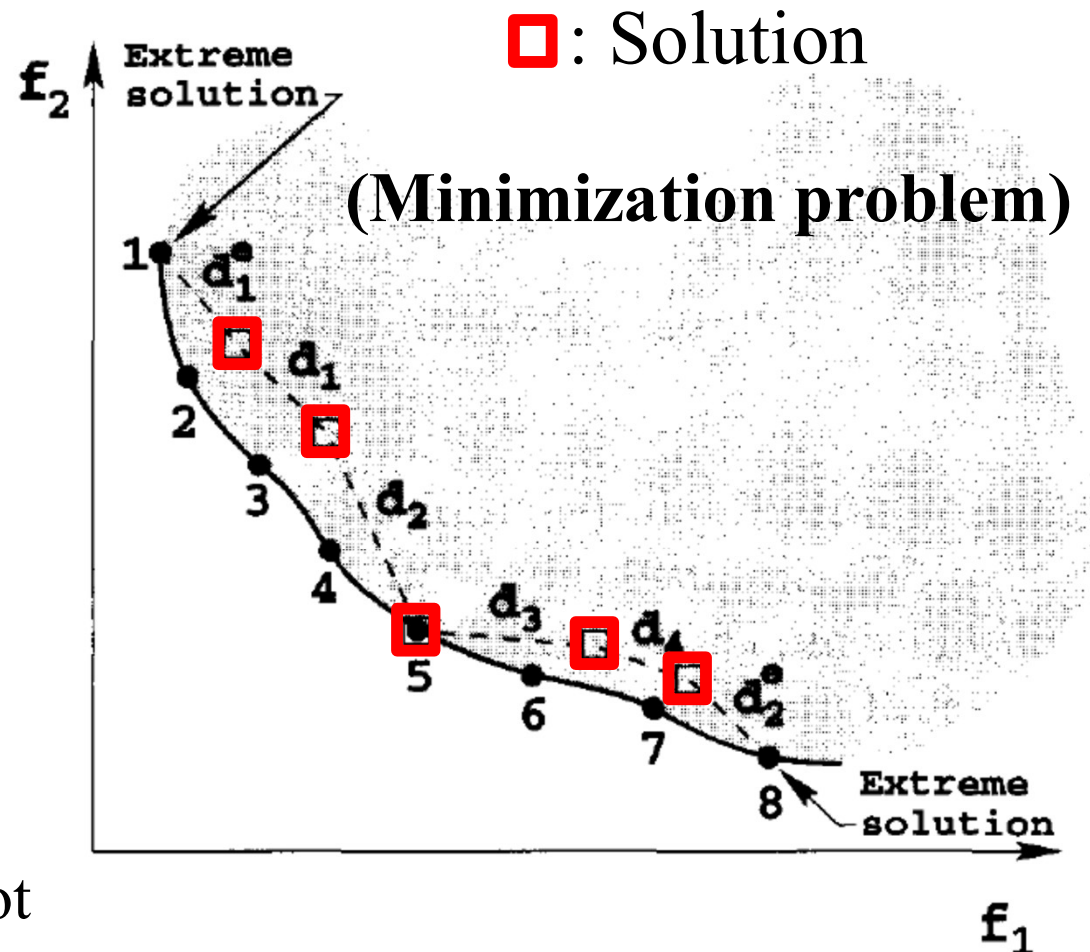


Figure 188 Distances from the extreme solutions.

K. Deb, *Multi-Objective Optimization Using Evolutionary Algorithms*, John Wiley & Sons, Chichester, 2001.

Performance Indicator for Diversity (Uniformity)

Spread

$$\Delta = \frac{d_1^e + d_2^e + \sum_{i=1}^{N-1} |d_i - \bar{d}|}{d_1^e + d_2^e + (N-1)\bar{d}}$$

Not Pareto compliant

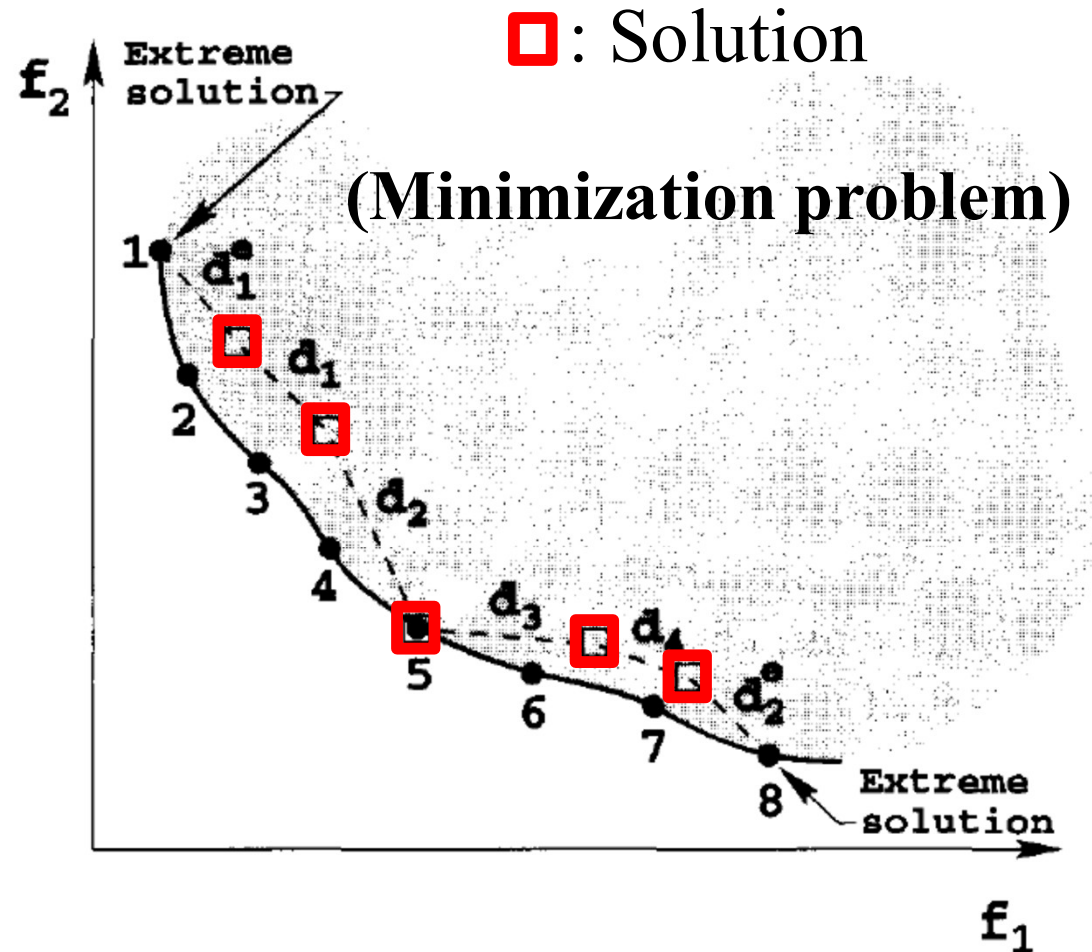
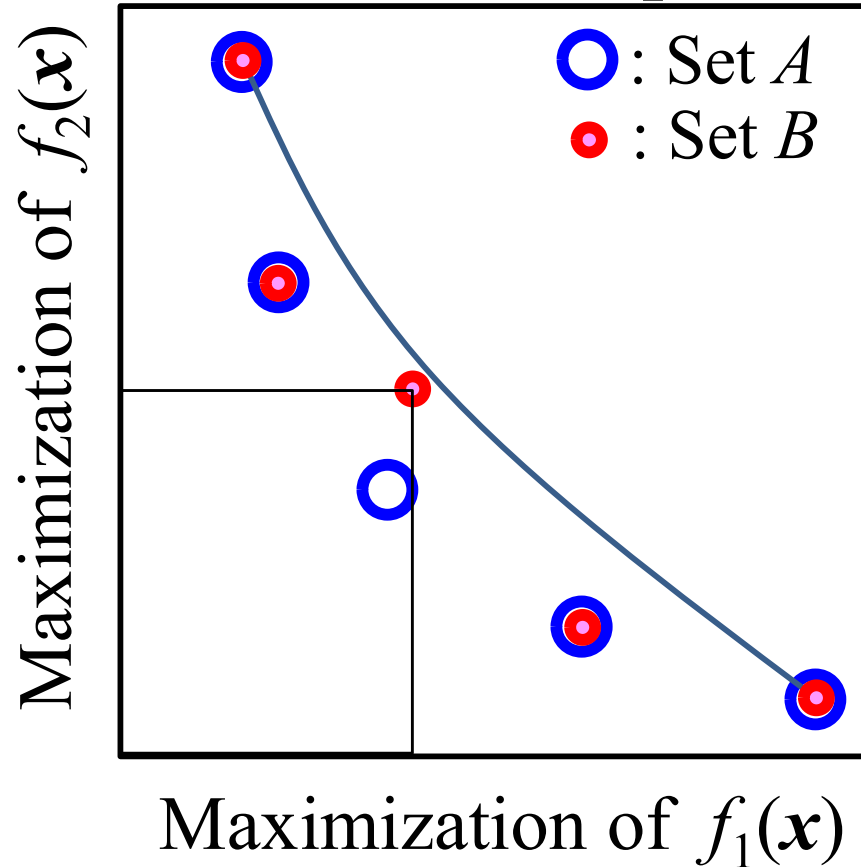
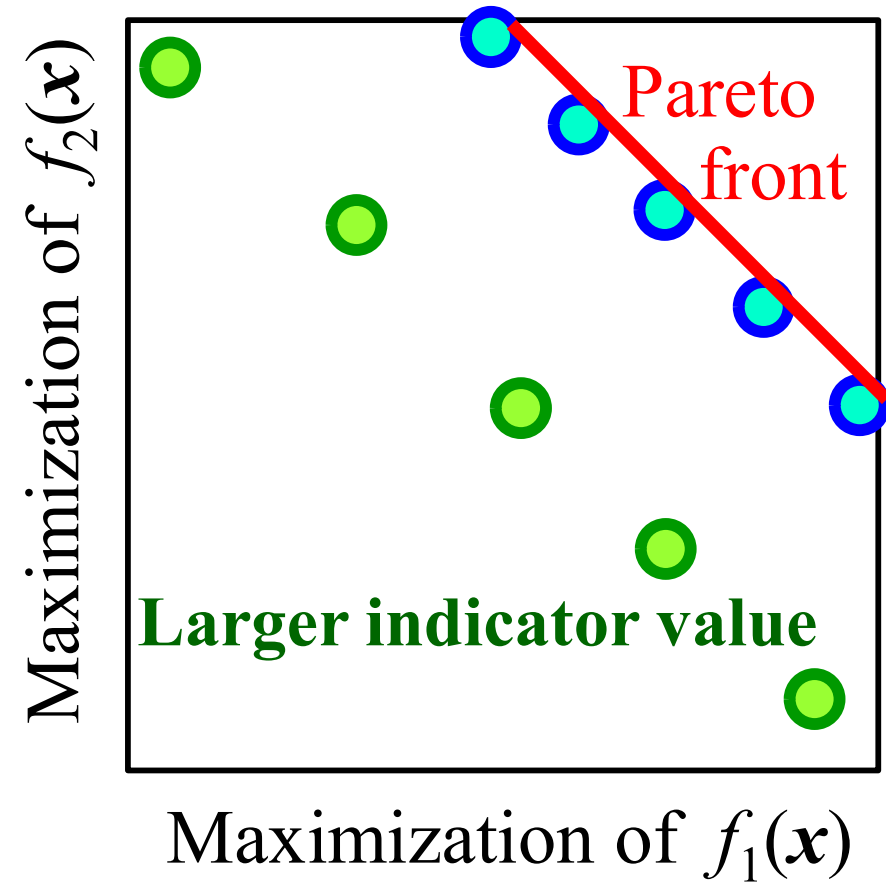
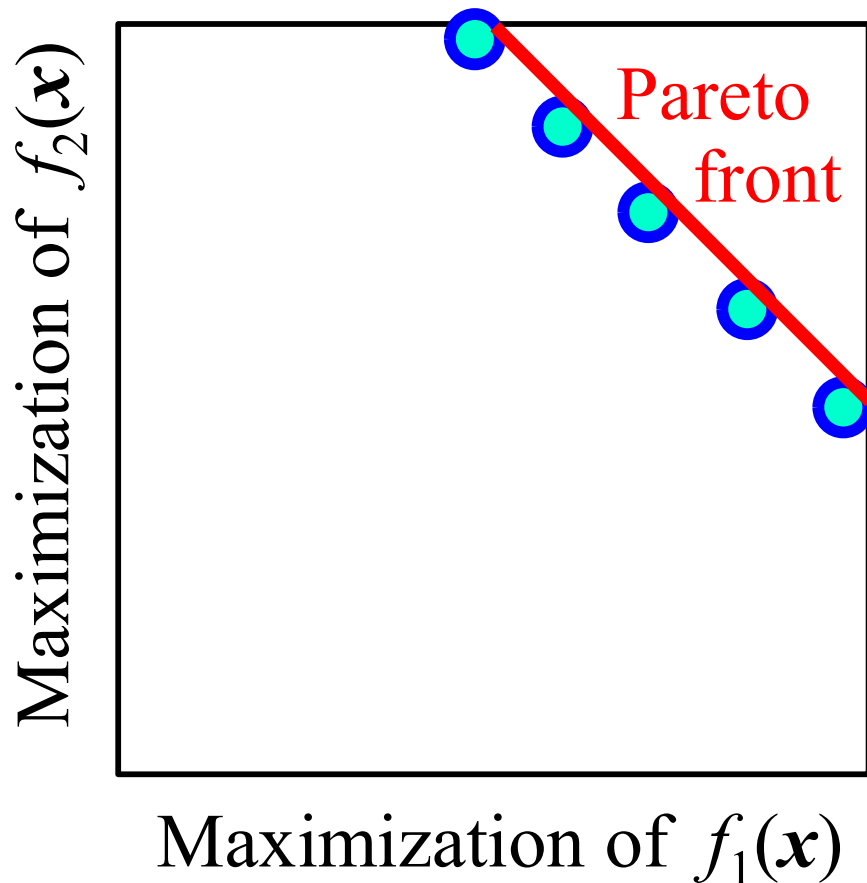


Figure 188 Distances from the extreme solutions.

Performance Indicator for Diversity (Uniformity): Uniformity

Minimum distance between solutions: The larger, the better

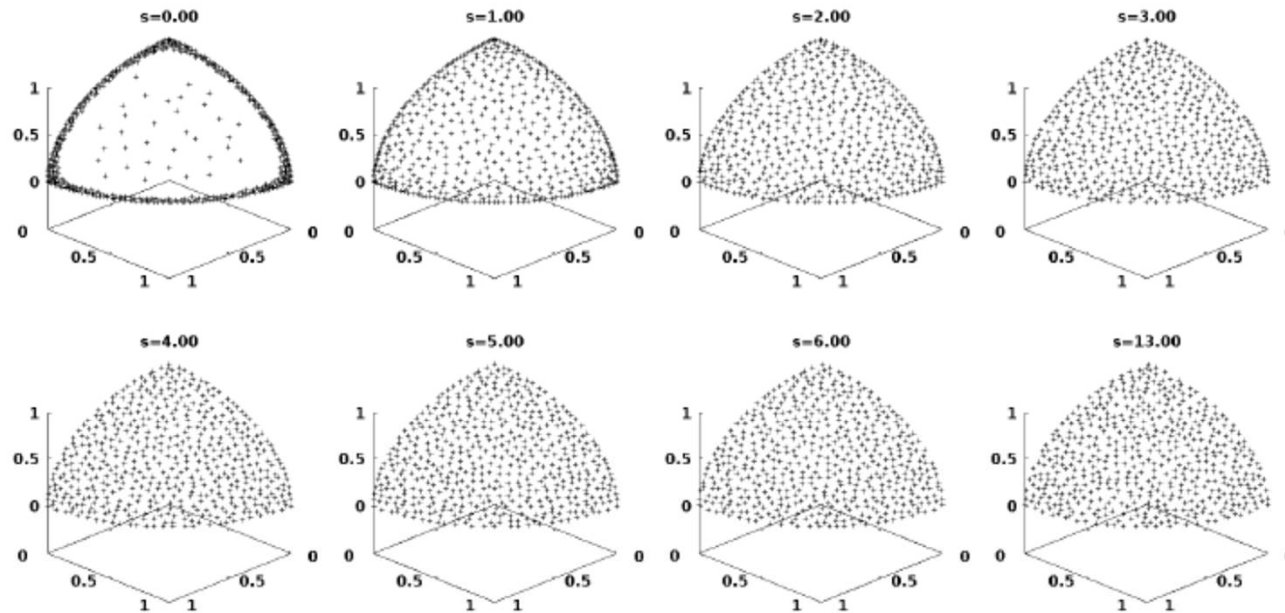
- (1) If all solutions are on the Pareto front, this is a good indicator.
The maximization of this indicator means the best uniformity.
- (2) If solutions are not converged, this is not a good indicator.



Performance Indicator for Diversity (Spread & Uniformity)

Riesz s -Energy

$$E_s(A) = \sum_{a \in A} \sum_{\substack{b \in A \\ b \neq a}} \frac{1}{\|a - b\|^s} \quad (s = m - 1 \text{ (suggested value)})$$



(c) DTLZ2 3D

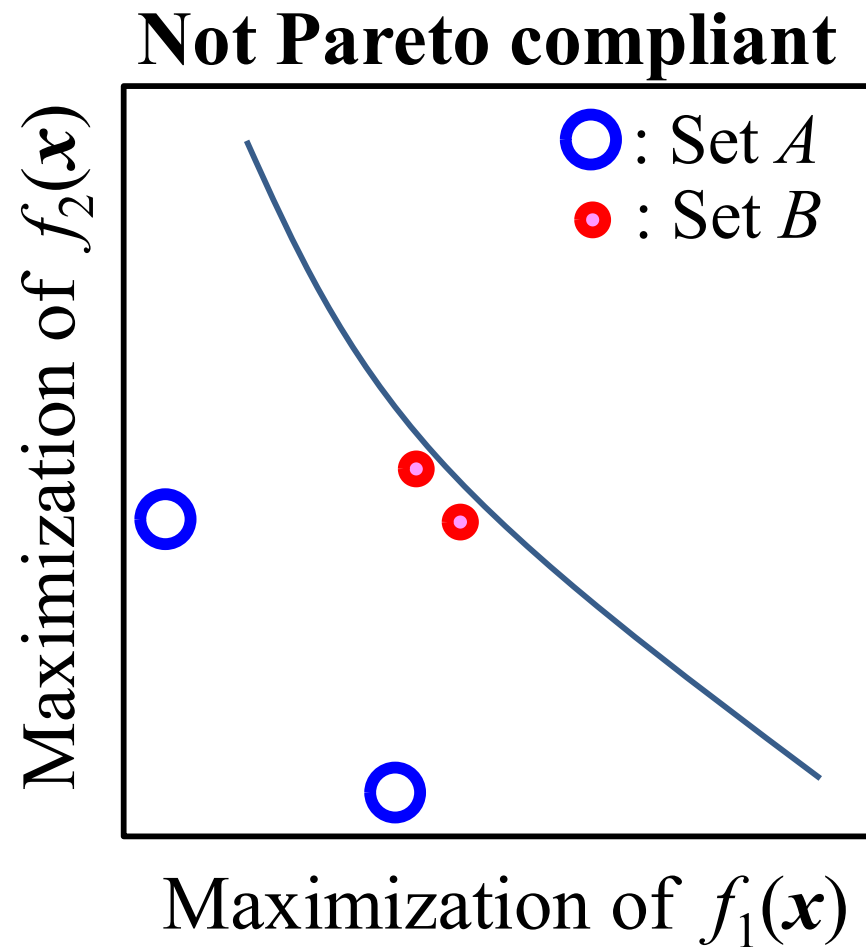
Fig. 3: Approximated Riesz s -energy optimal distributions, varying the value of the parameter s .

J. G. Falcon-Cardona, H. Ishibuchi, and C. A. C. Coello, "Riesz s -energy-based reference sets for multi-objective optimization," CEC 2020.

Performance Indicator for Diversity (Spread & Uniformity)

Riesz s -Energy

$$E_s(A) = \sum_{a \in A} \sum_{\substack{b \in A \\ b \neq a}} \frac{1}{\|a - b\|^s} \quad (s = m - 1 \text{ (suggested value)})$$



Performance Indicators

Convergence: GD (OK)

Spread: Maximum Spread (OK)

Uniformity: Spacing (Questionable)

Spread & Uniformity: Deb's spread for two-objective problems
A current active research topic for three and more objectives.

In Proceedings of EMO 2019

Kalyanmoy Deb, Sunith Bandaru, and Haitham Seada, **Generating Uniformly Distributed Points on a Unit Simplex for Evolutionary Many-Objective Optimization.**

Alberto Rodríguez Sánchez, Antonin Ponsich, Antonio López Jaimes, and Saúl Zapotecas Martínez: **A Parallel Tabu Search Heuristic to Approximate Uniform Designs for Reference Set Based MOEAs**

In Proceedings of IEEE CEC 2020

J. G. Falcon-Cardona, H. Ishibuchi, and C. A. C. Coello, **Riesz s -energy-based Reference Sets for Multi-objective Optimization.**

Overall Performance Indicators

IGD (inverted generational distance) indicator

IGD⁺ (inverted generational distance plus) indicator

ε -indicator

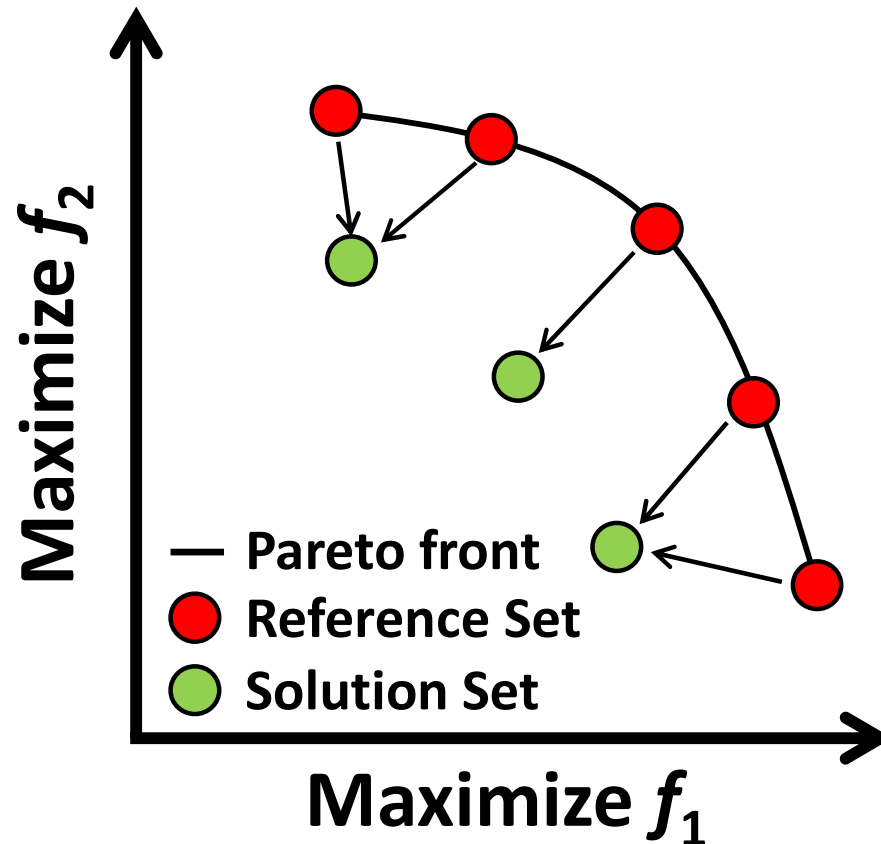
HV (hypervolume) indicator



Performance Indicator: IGD

IGD: Inverted Generational Distance

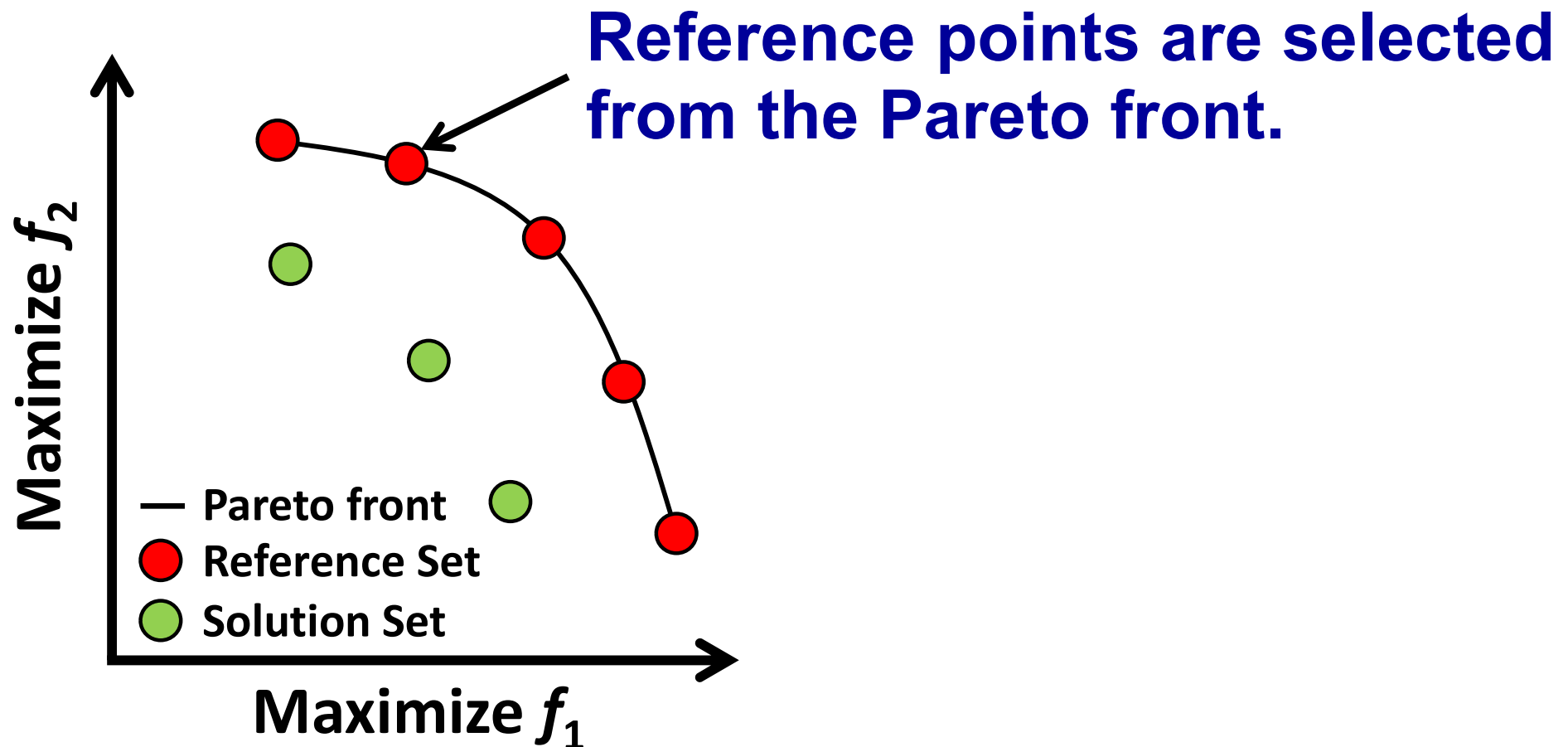
Average distance from each reference point on the Pareto front to the nearest solution.



Performance Indicator: IGD

IGD: Inverted Generational Distance

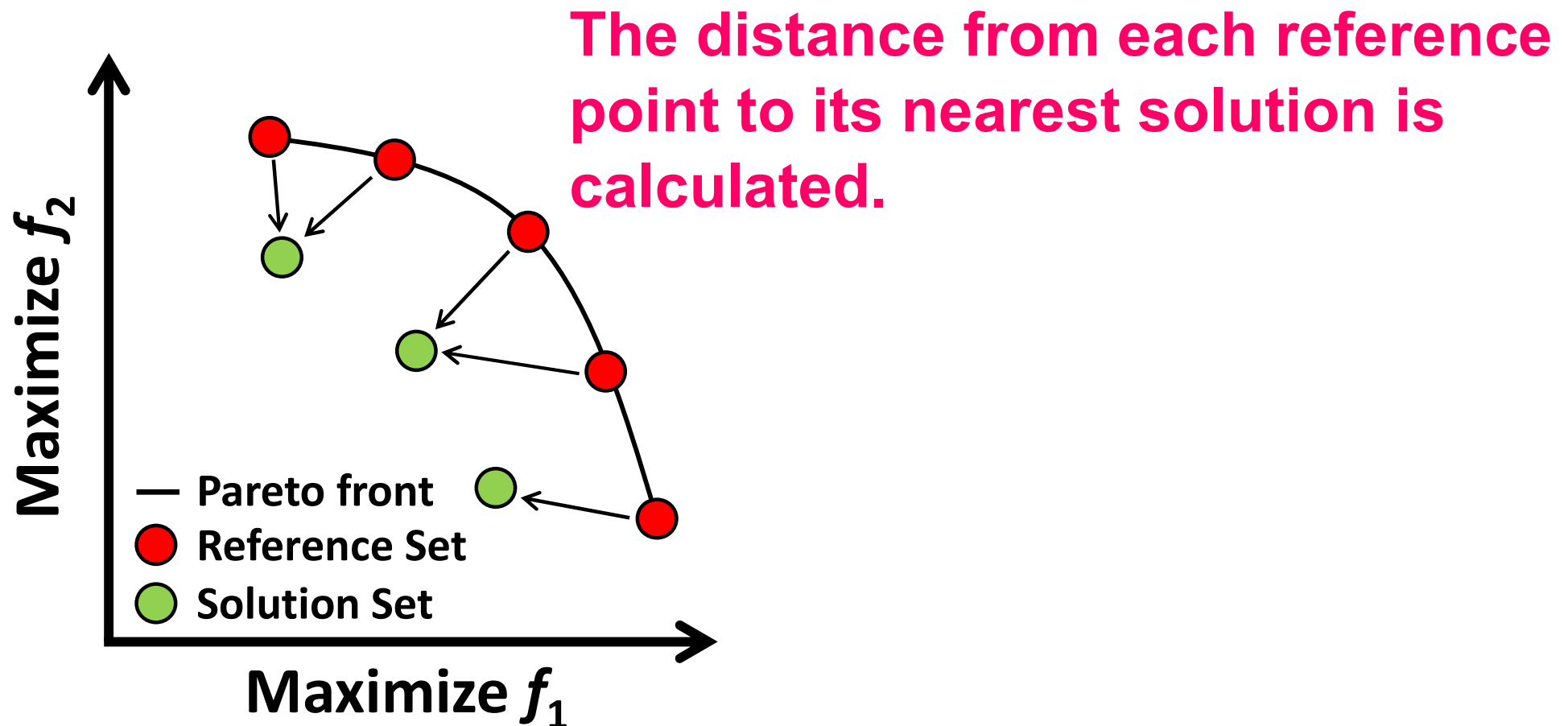
Average distance from each reference point on the Pareto front to the nearest solution.



Performance Indicator: IGD

IGD: Inverted Generational Distance

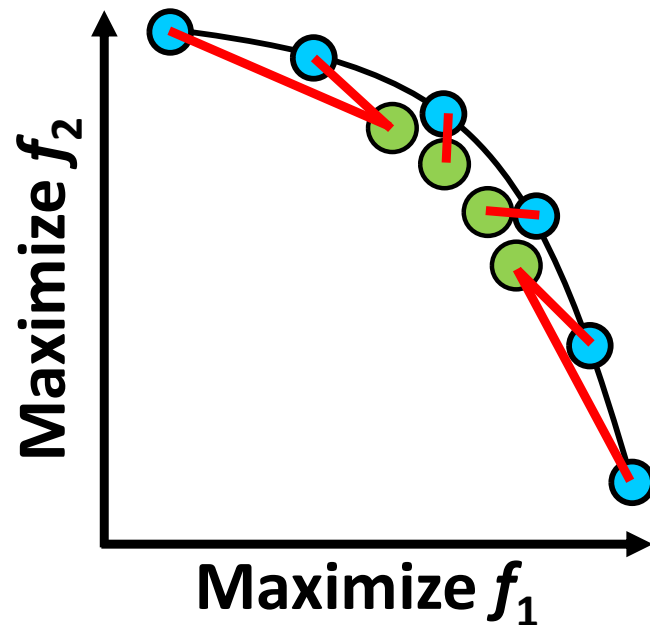
Average distance from each reference point on the Pareto front to the nearest solution.



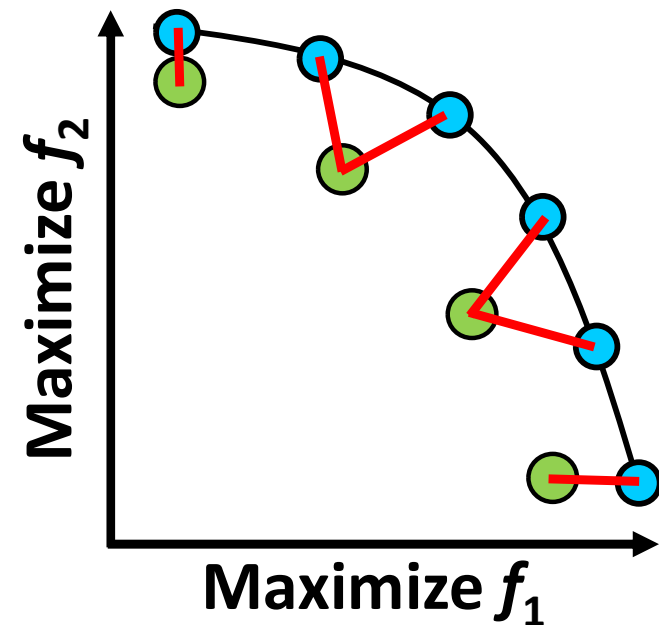
Performance Indicator: IGD

IGD: Inverted Generational Distance

Average distance from each reference point on the Pareto front to the nearest solution.



Due to poor diversity,
IGD is not very good.



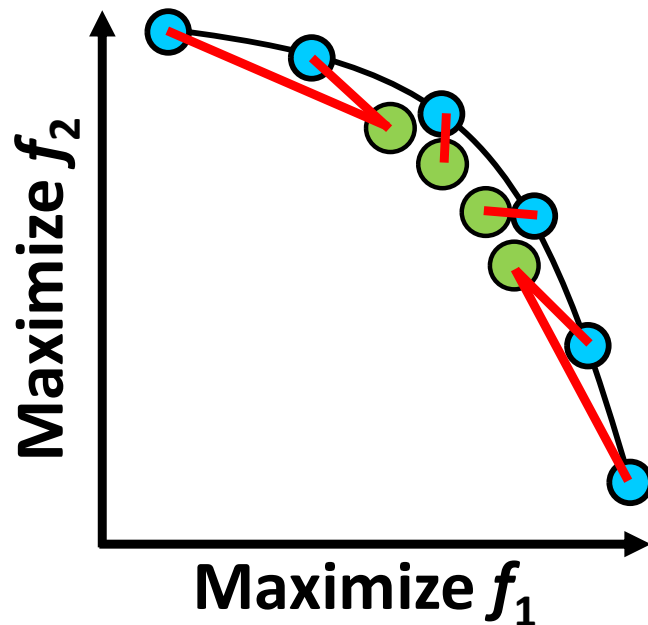
Due to poor convergence
IGD is not very good.

Performance Indicator: IGD

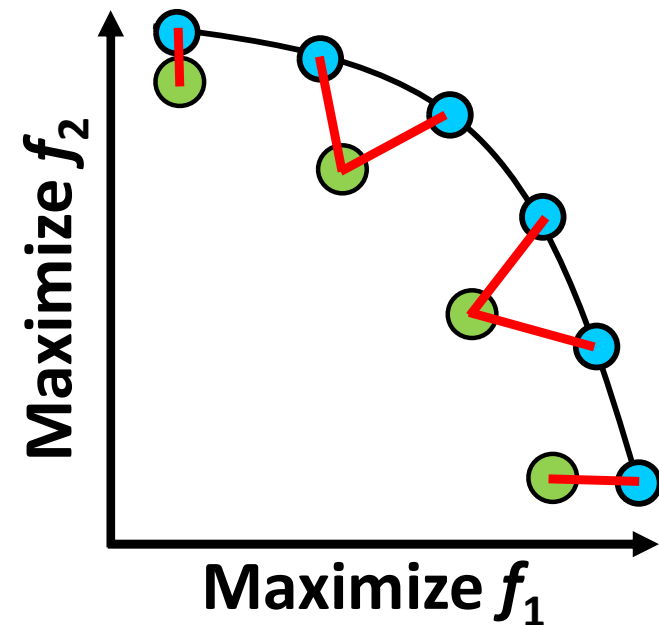
IGD: Inverted Generational Distance

Good IGD needs good convergence and good diversity.

==> The IGD can evaluate the overall performance.

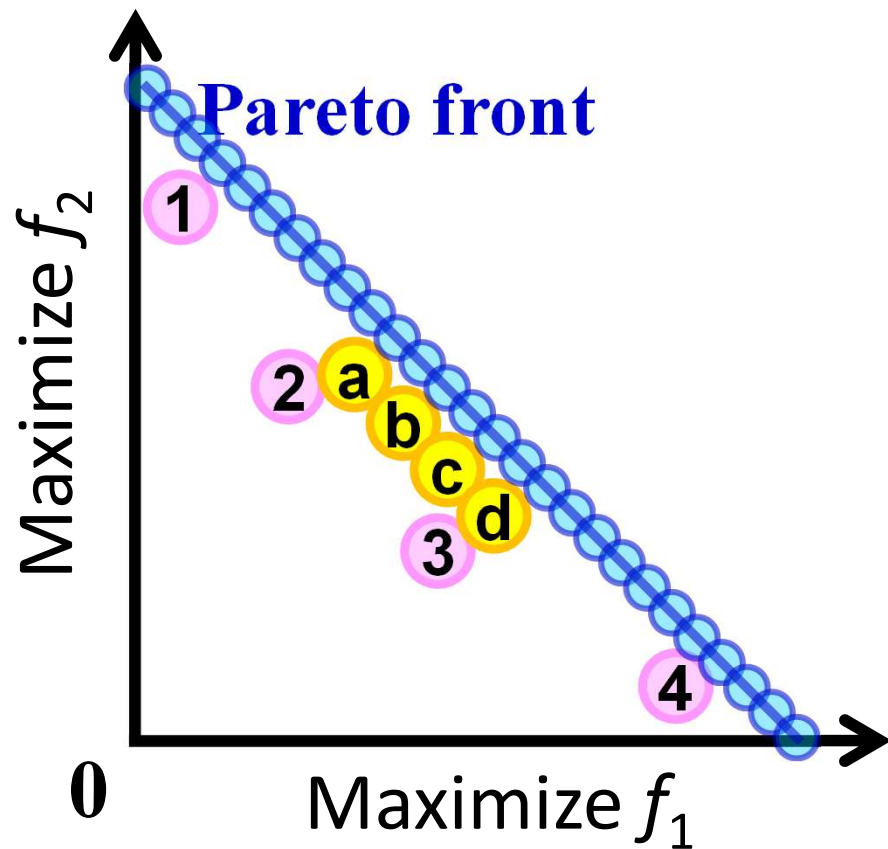


Due to poor diversity,
IGD is not very good.

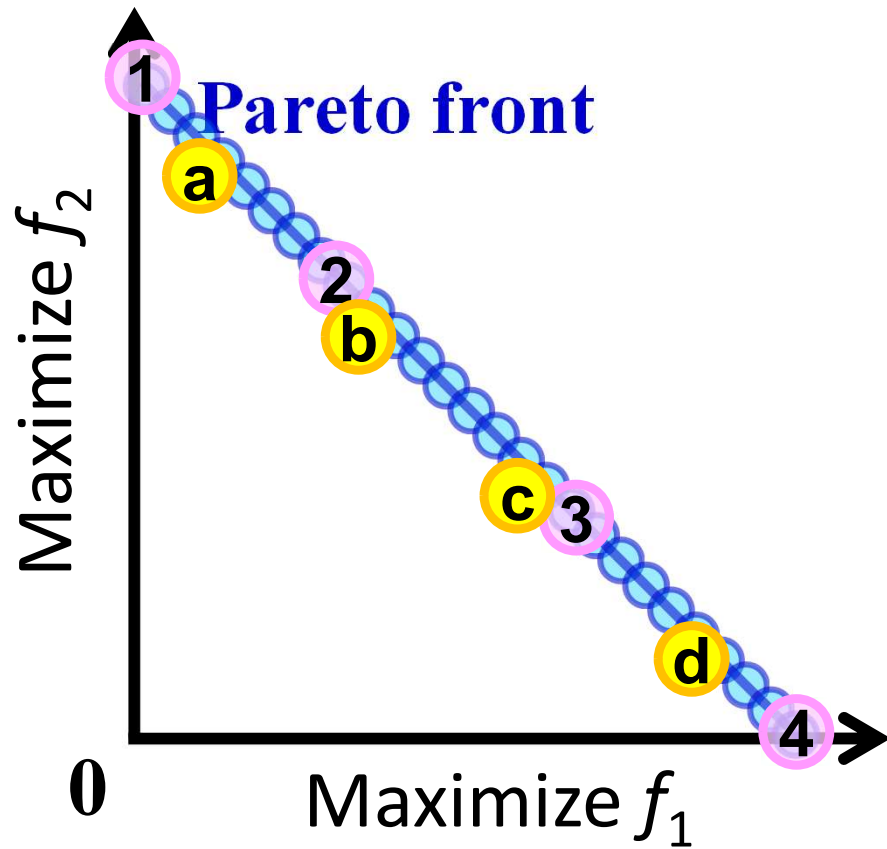


Due to poor convergence
IGD is not very good.

Which is better based on IGD: Pink or Yellow ?

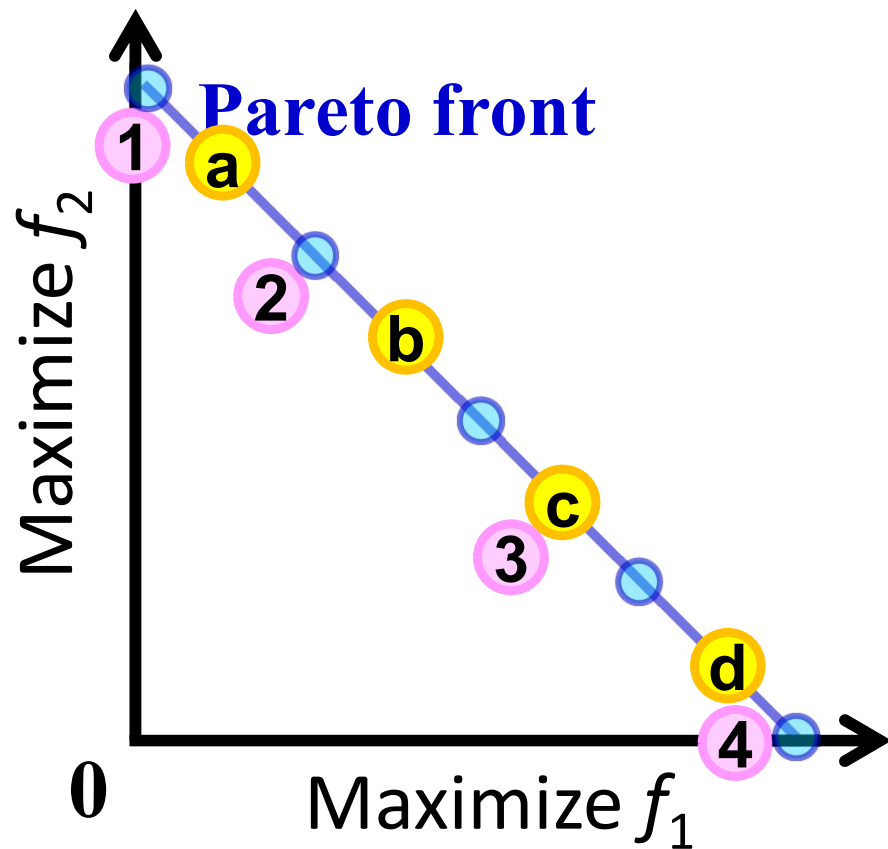


Example 1

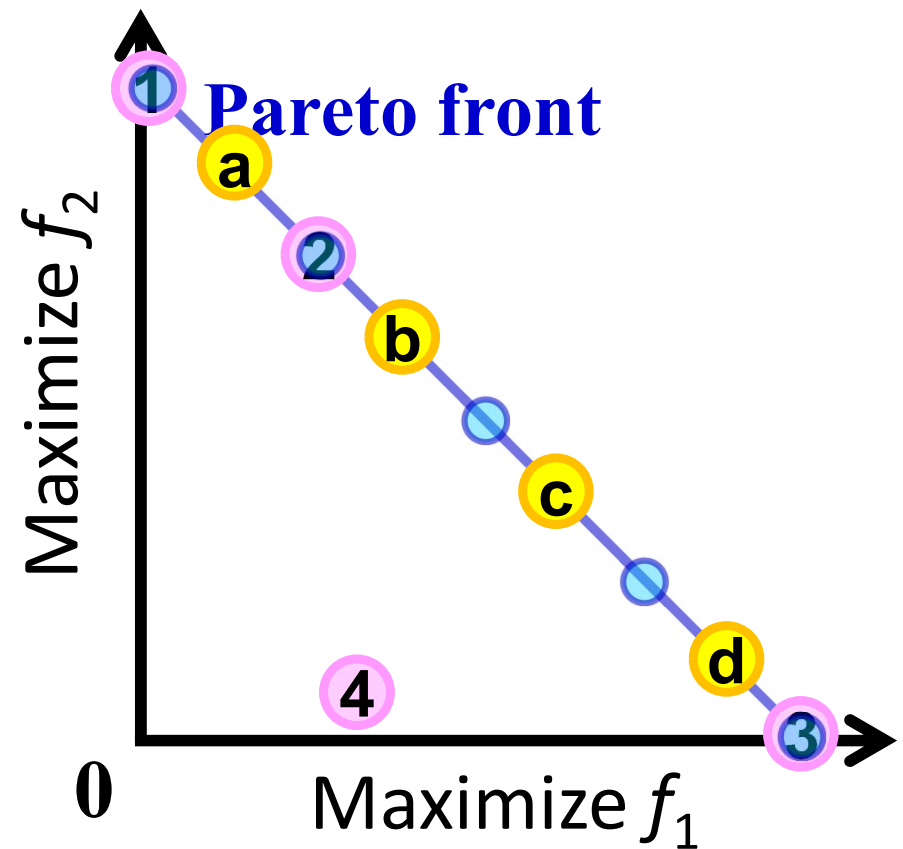


Example 2

Which is better based on IGD: Pink or Yellow ?



Example 1



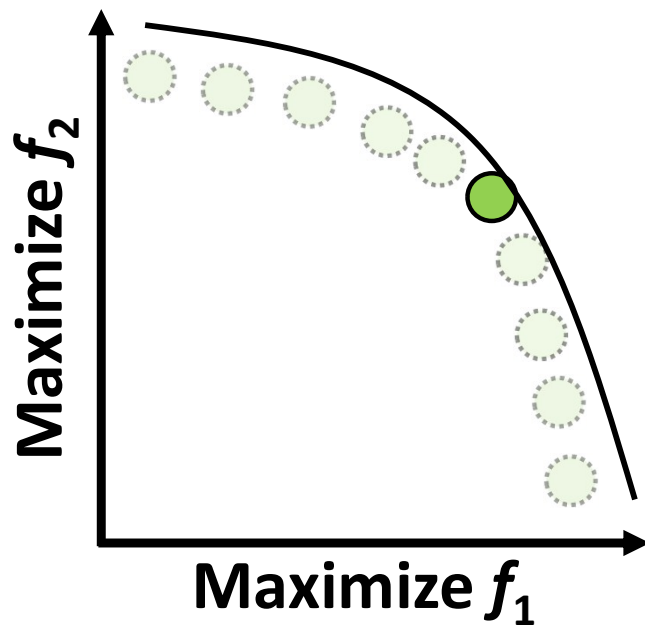
Example 2

Performance Indicator: IGD

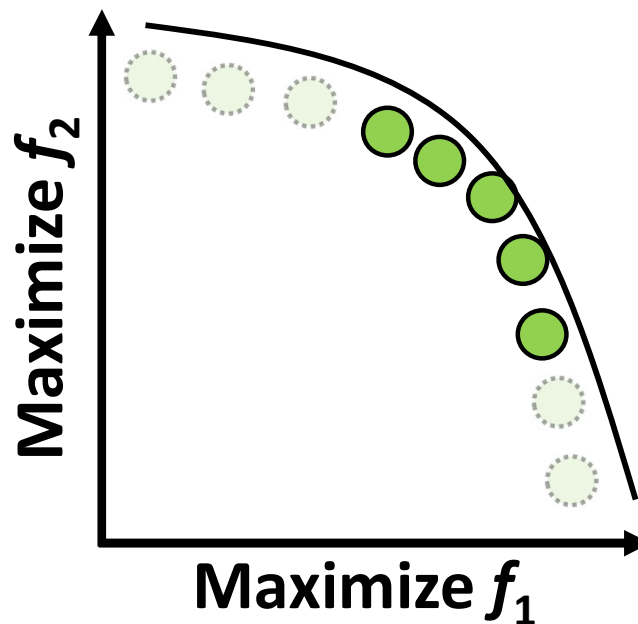
IGD: Inverted Generational Distance

Good IGD needs good convergence and good diversity.

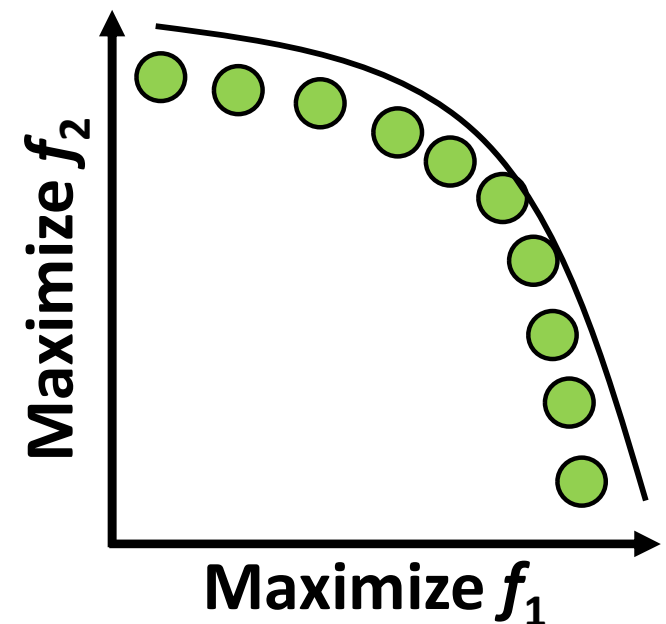
==> The IGD can evaluate the overall performance.



Worst IGD



2nd Best IGD

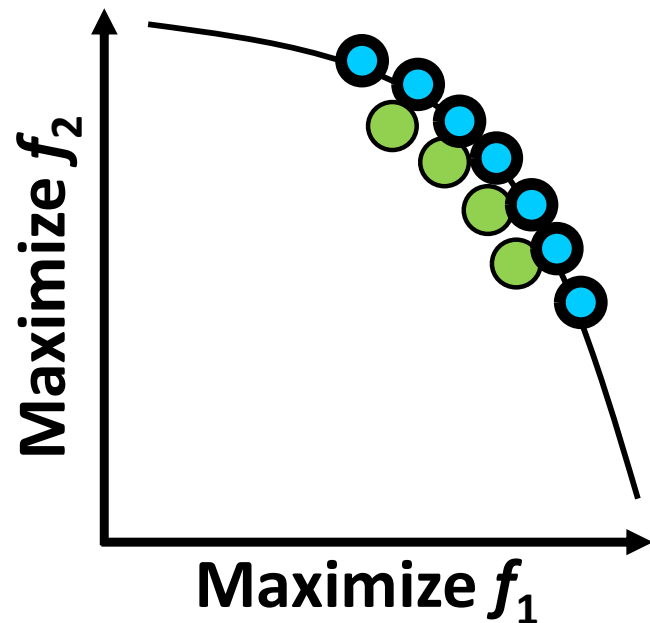


Best IGD

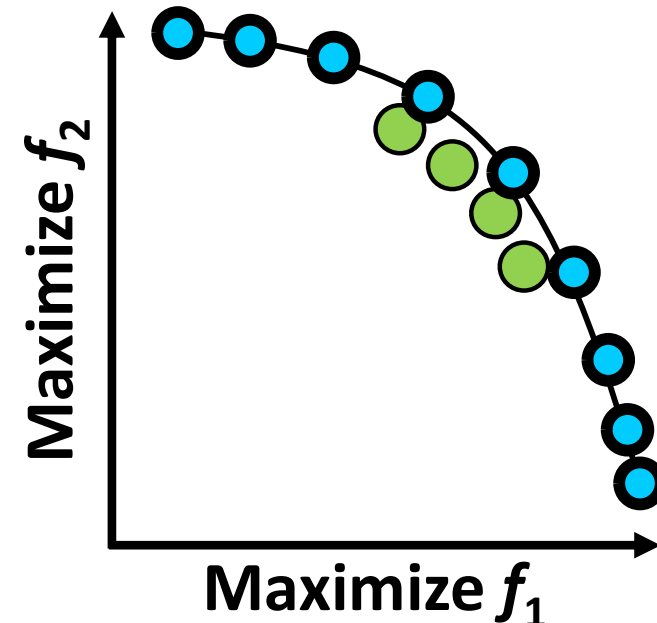
Performance Indicator: IGD

IGD: Inverted Generational Distance

Evaluation results strongly depend on reference points.



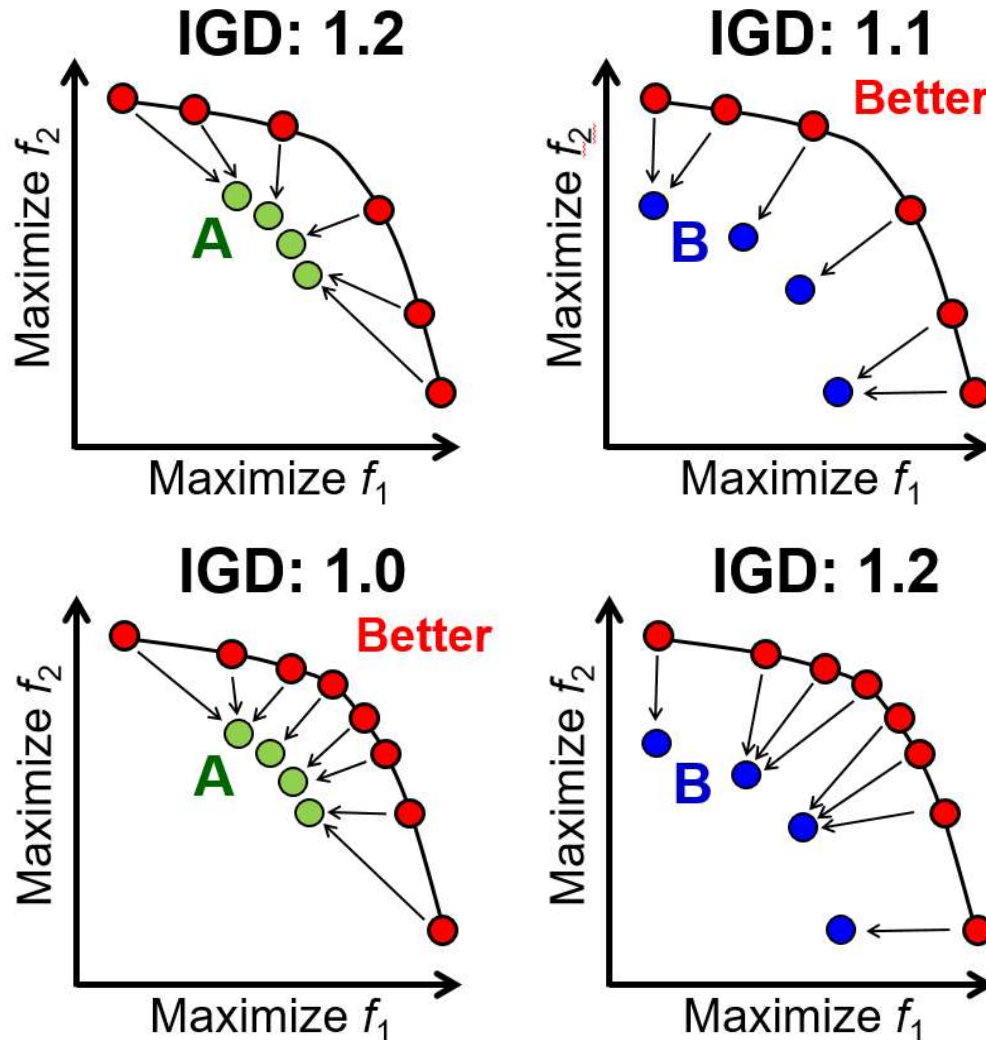
Good evaluation result



Poor evaluation result

IGD: Inverted Generational Distance

Evaluation results strongly depend on reference points.

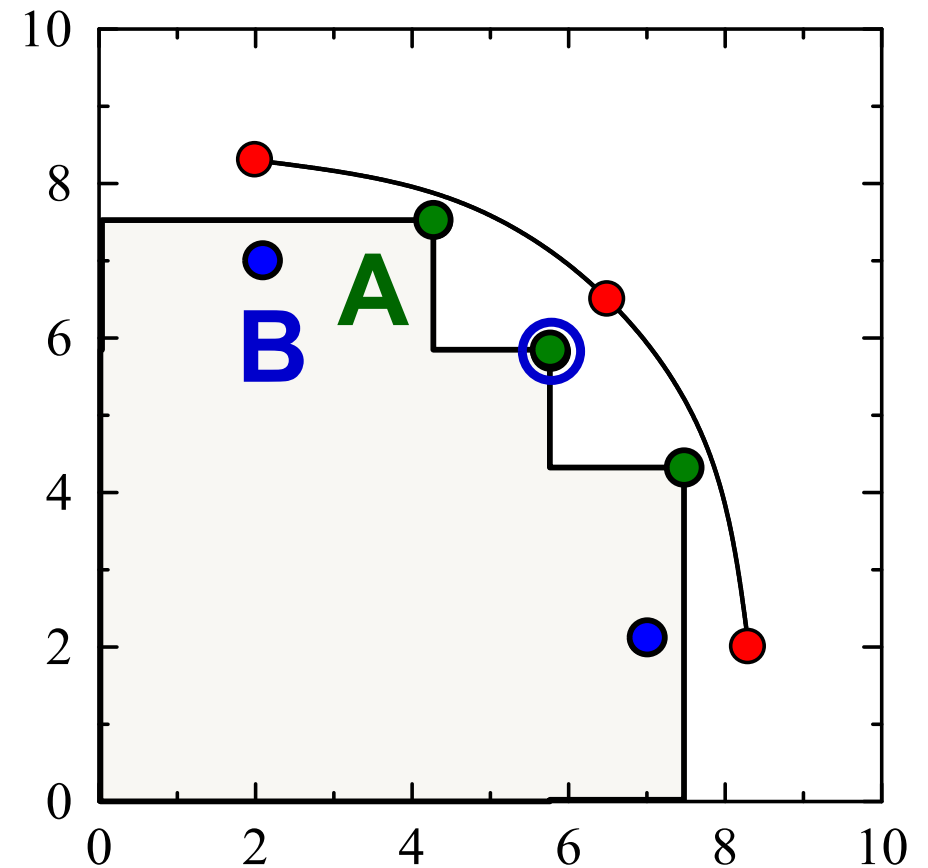
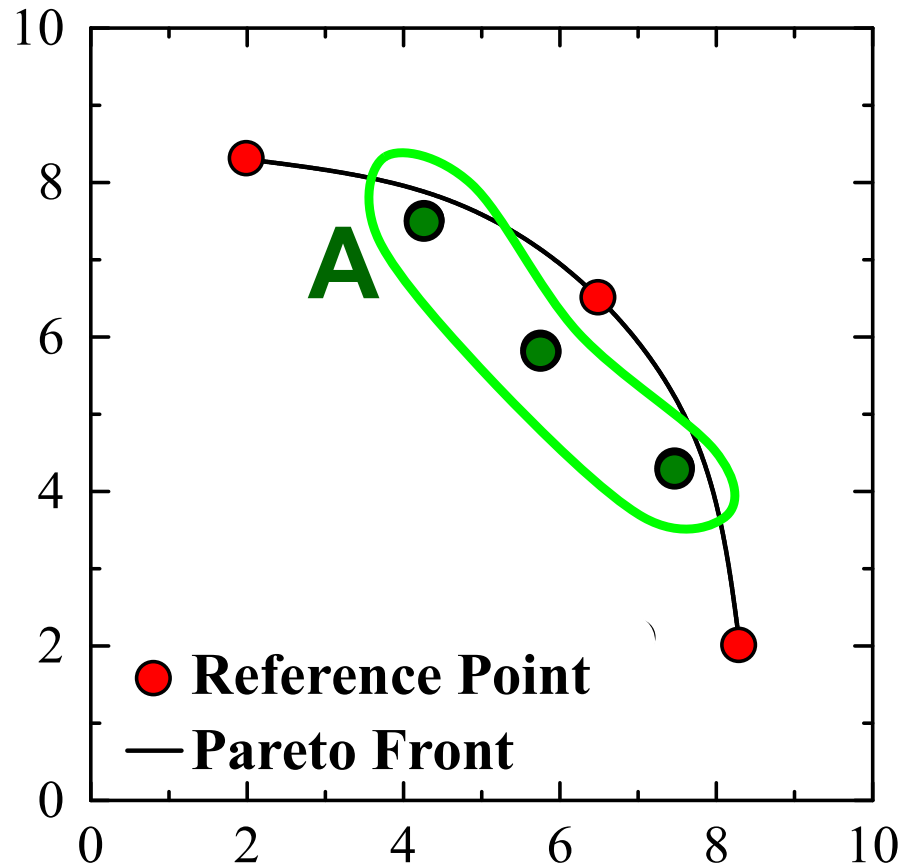


For many-objective problems, it is not easy to appropriately specify the reference point set for IGD calculation.

H. Ishibuchi et al., Reference point specification in inverted generational distance for triangular linear Pareto front, *IEEE TEVC* (2018).

IGD is not Pareto Compliant

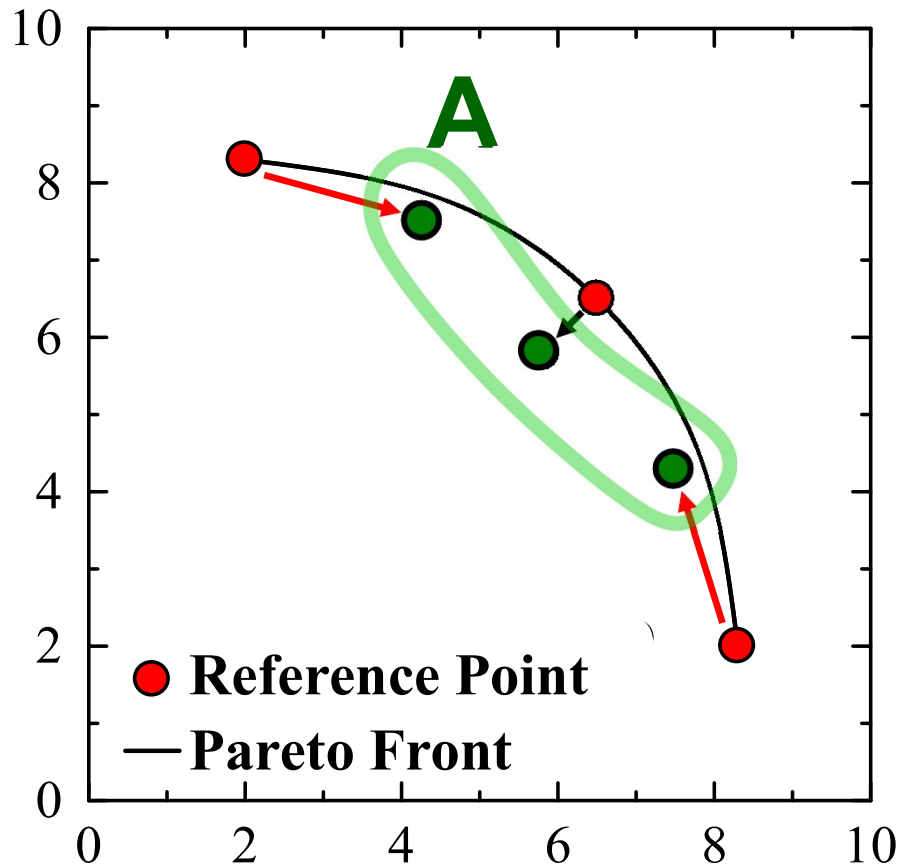
Solution set A dominates B



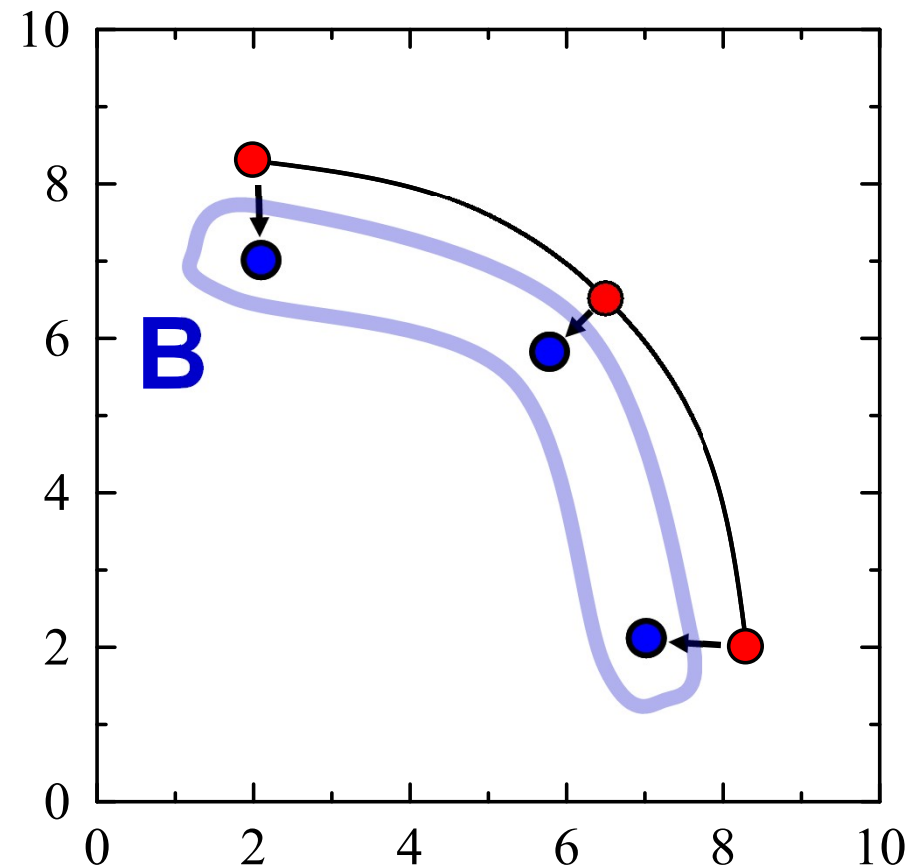
IGD is not Pareto Compliant

Solution set B is evaluated as being better than A

IGD is larger



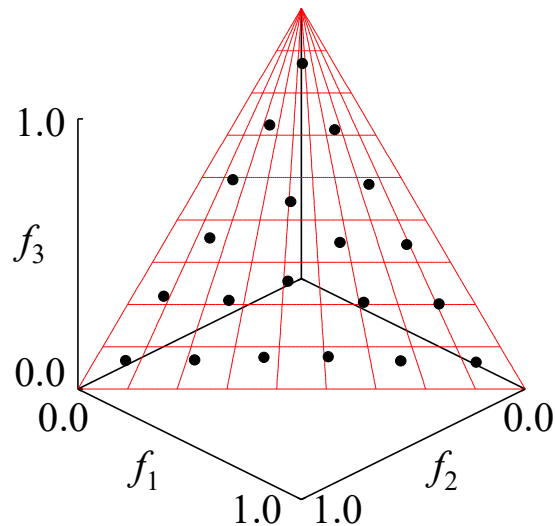
IGD is smaller



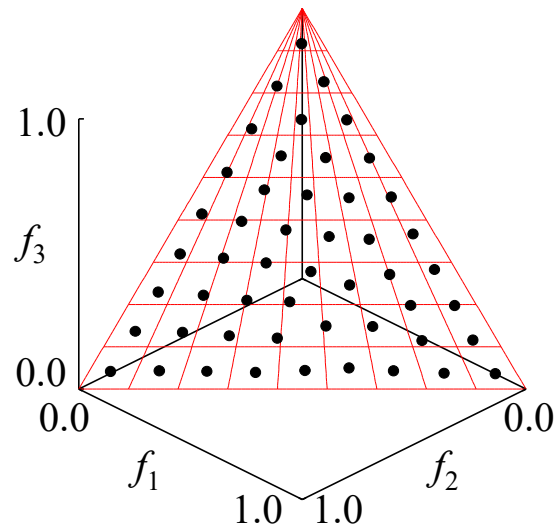
A dominates B (A is better than B).

However, B is evaluated as being better than A.

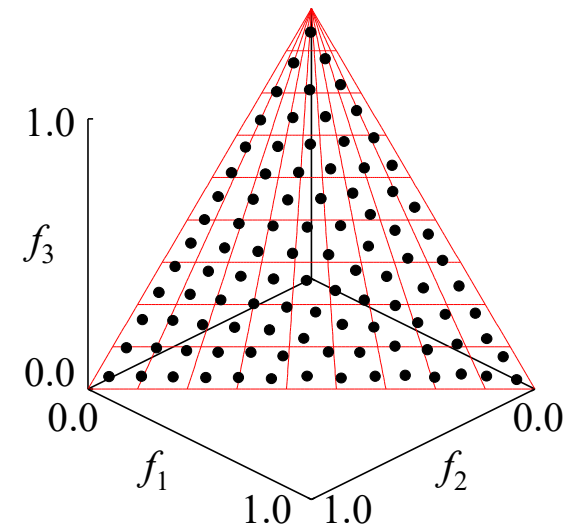
Optimal Distributions for IGD are not always intuitive



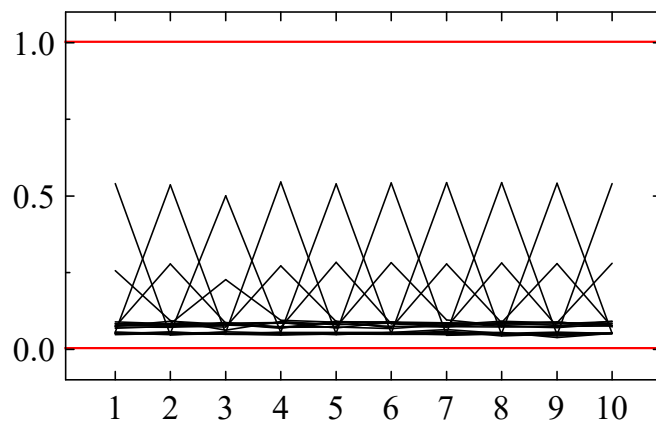
Population size 20



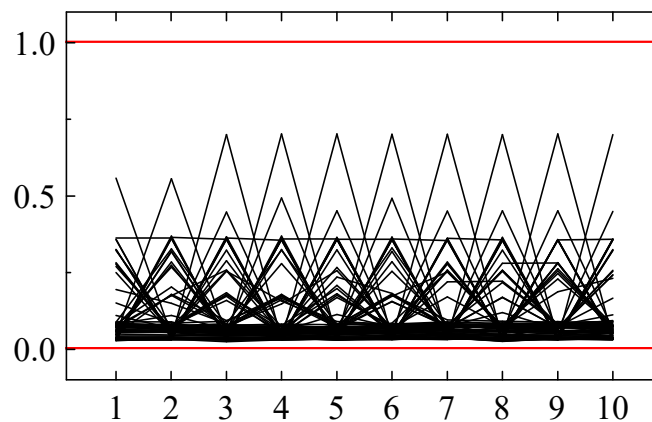
Population size 50



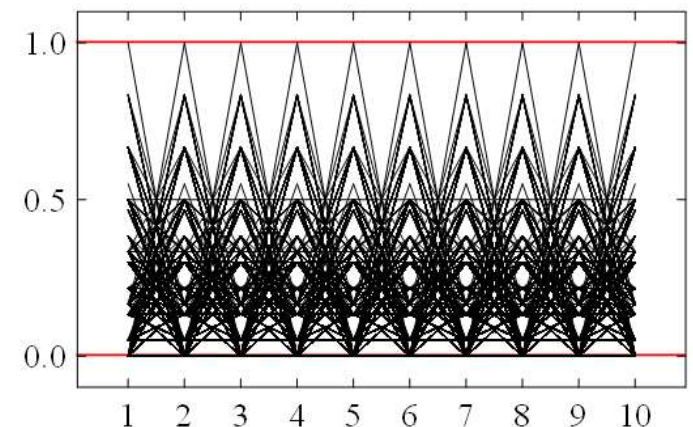
Population size 100



Population size 20



Population size 100



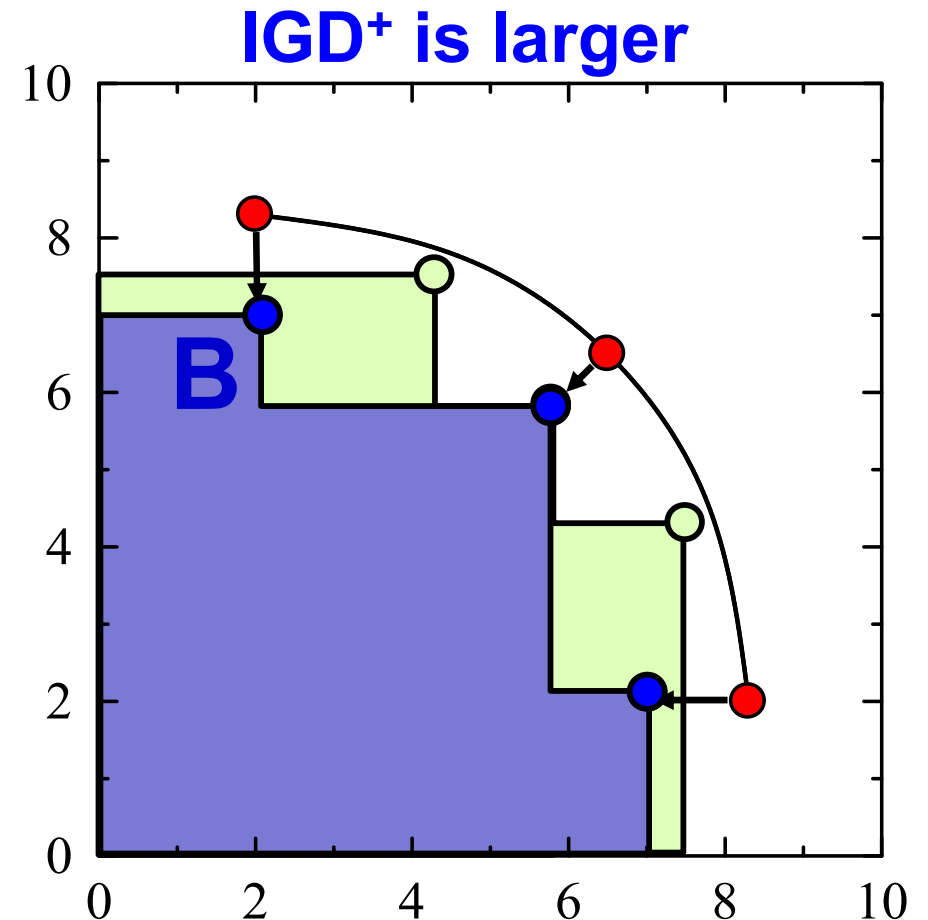
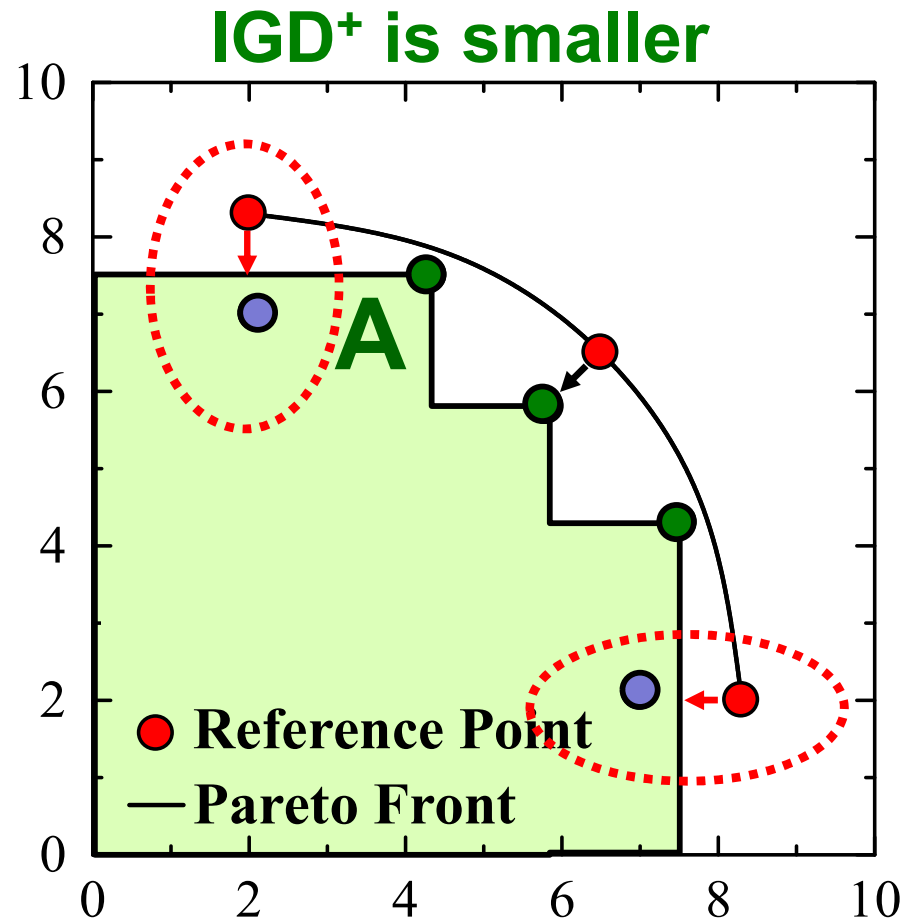
Reference point set

H. Ishibuchi et al., Reference point specification in inverted generational distance for triangular linear Pareto front, *IEEE TEVC* (2018).

IGD⁺ (Inverted Generational Distance Plus)

(Ishibuchi et al., EMO 2015, GECCO 2015)

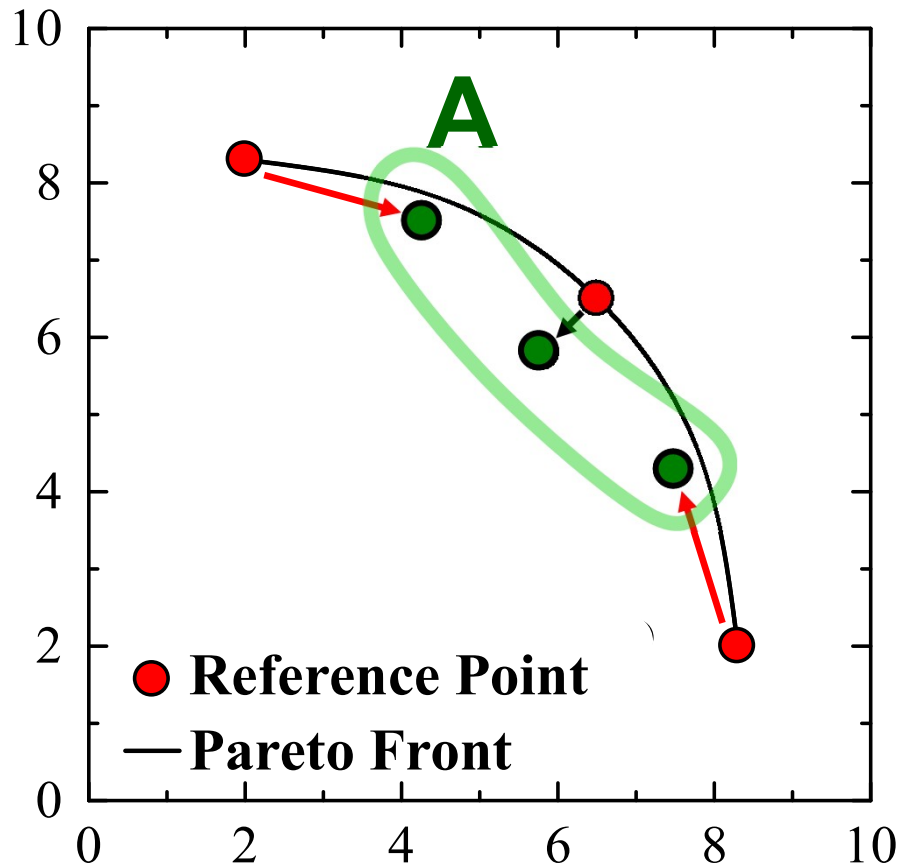
The calculation is from each reference point to the dominated region by the obtained solution set.



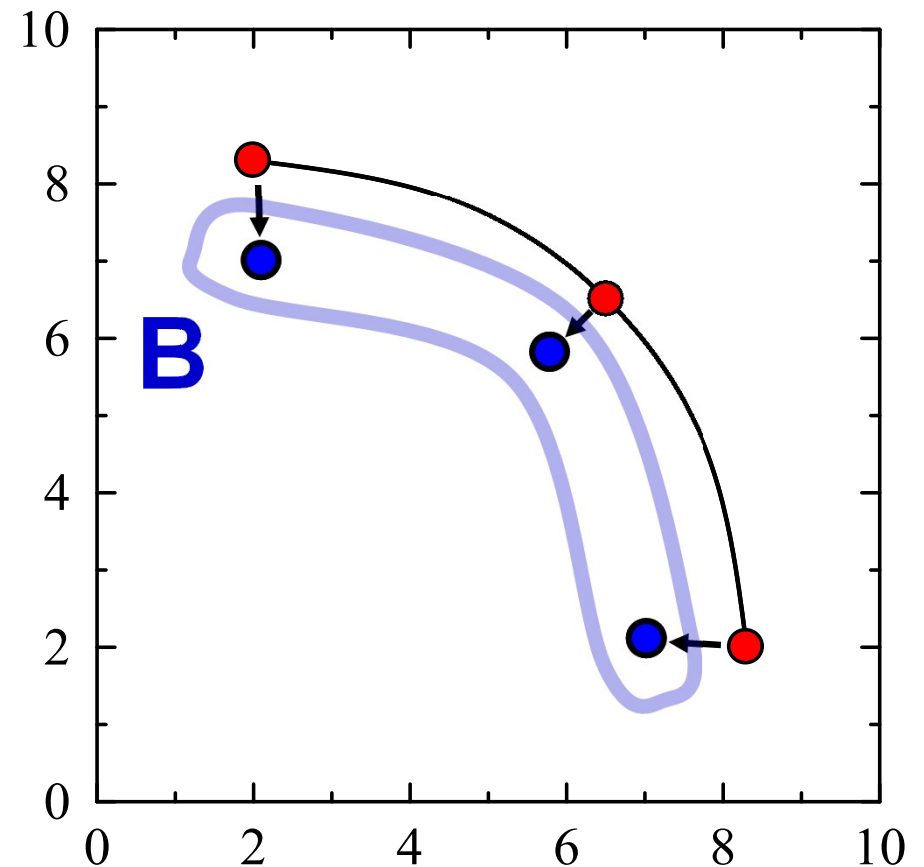
IGD is not Pareto Compliant

Solution set B is evaluated as being better than A

IGD is larger



IGD is smaller



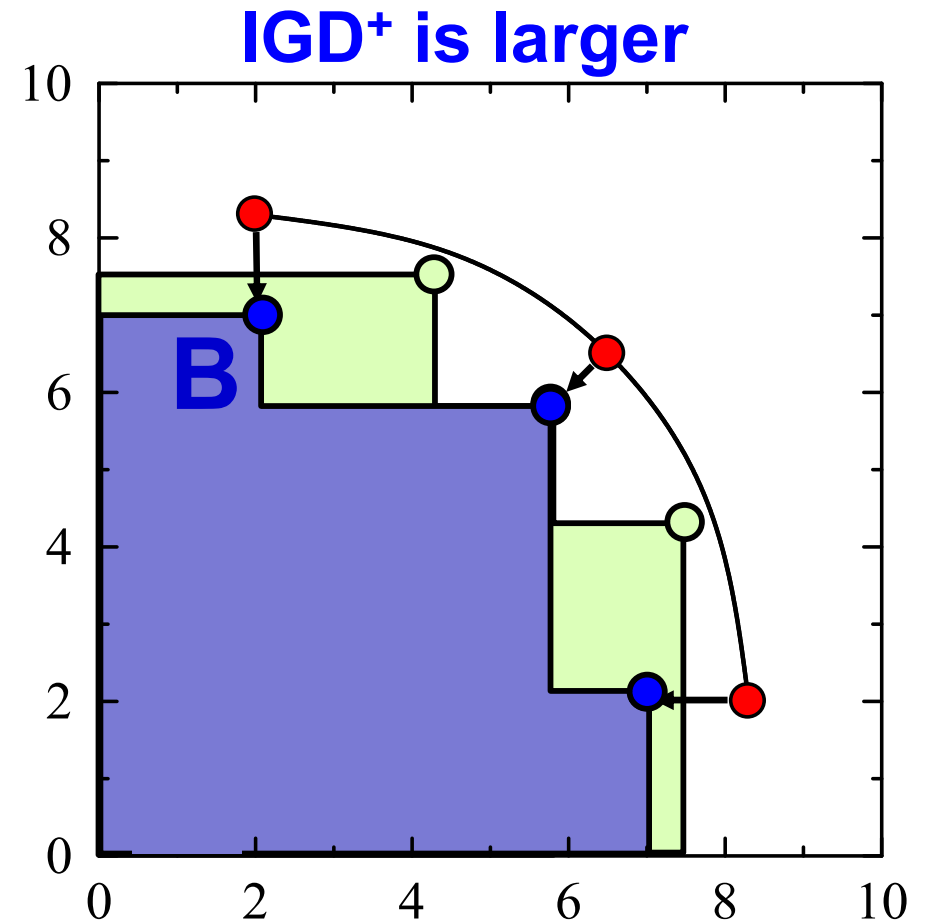
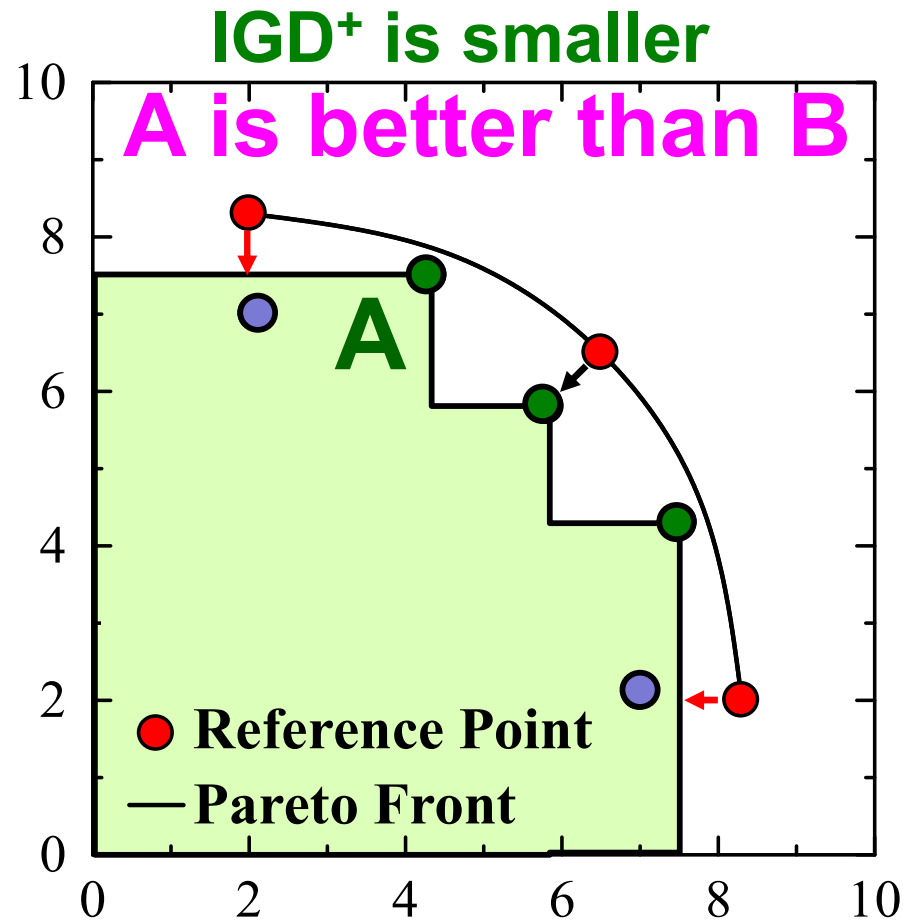
A dominates B (A is better than B).

However, B is evaluated as being better than A.

Weak Pareto Compliance

IGD^+ is not inconsistent with the Pareto dominance relation between solution sets.

IGD^+ is a weakly Pareto compliant indicator.



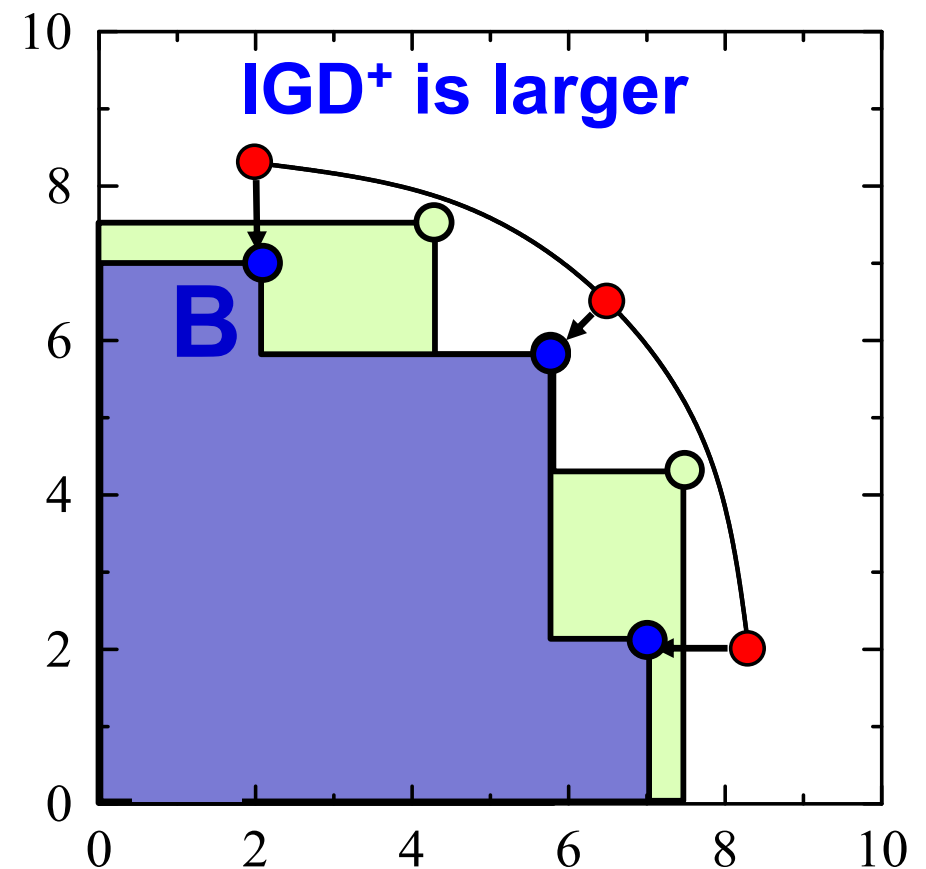
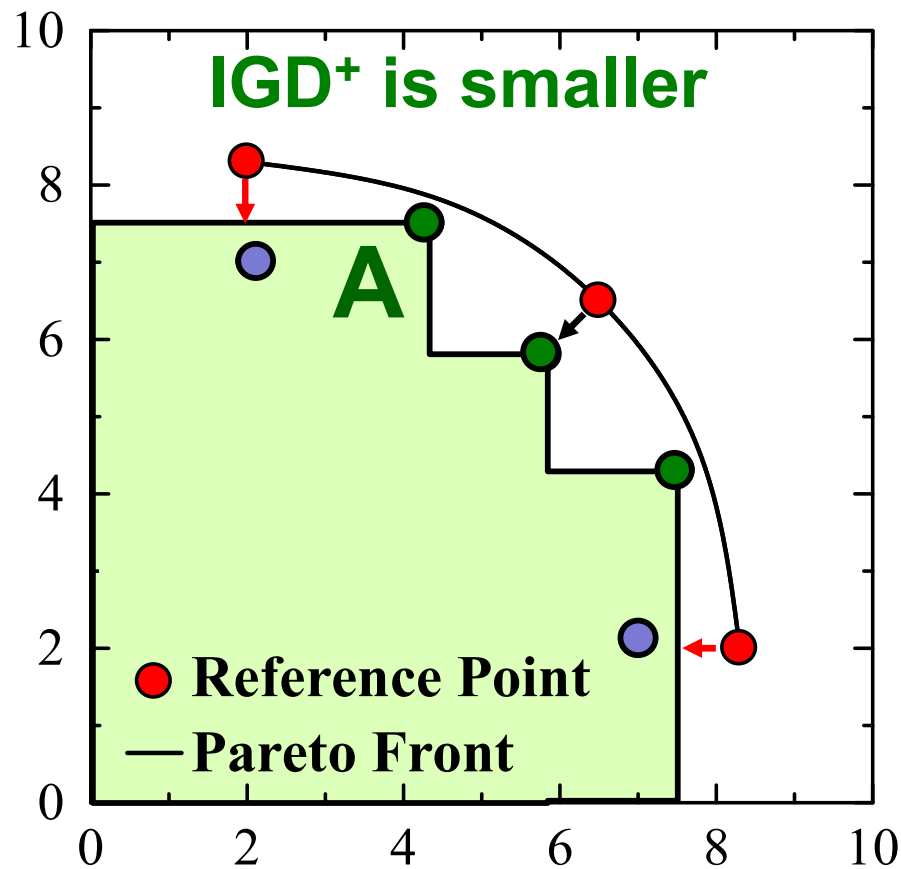
Weak Pareto Compliance

Pareto Compliance

$B \triangleleft A \Rightarrow IGD^+(B) > IGD^+(A)$ (This does not hold)

Weak Pareto Compliance.

$B \triangleleft A \Rightarrow IGD^+(B) \geq IGD^+(A)$ (This holds)



Weak Pareto Compliance

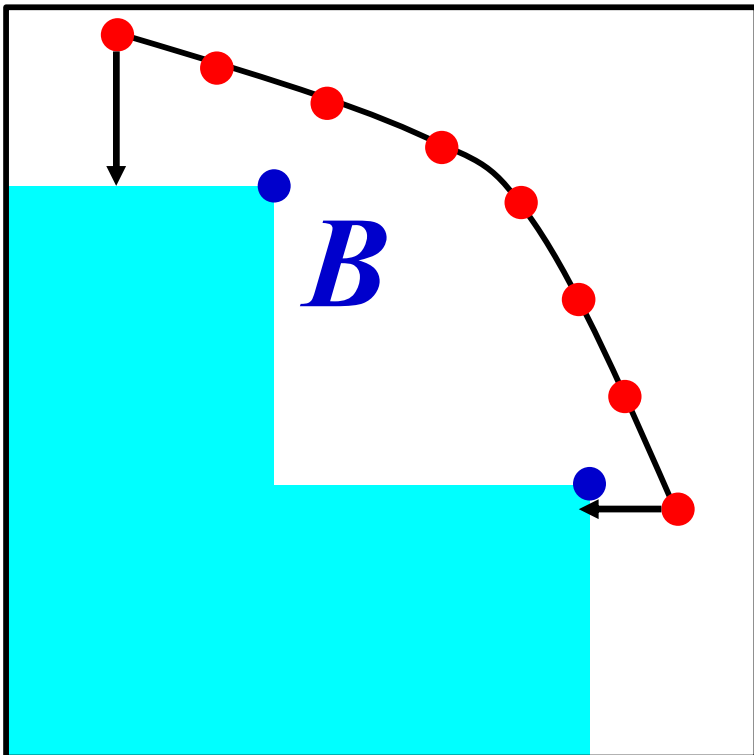
Pareto Compliance

$B \triangleleft A \Rightarrow IGD^+(B) > IGD^+(A)$ (This does not hold)

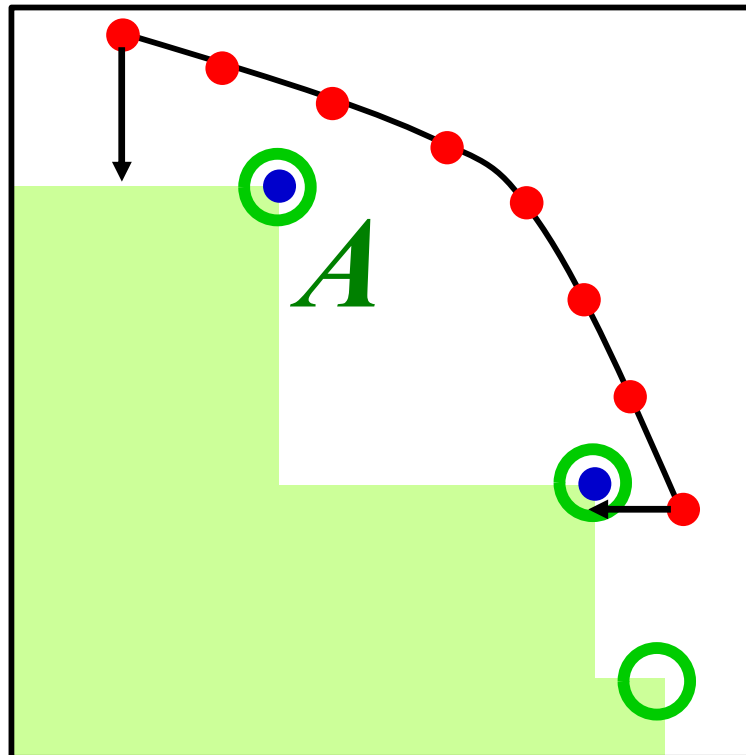
Weak Pareto Compliance.

$B \triangleleft A \Rightarrow IGD^+(B) \geq IGD^+(A)$ (This holds)

● Solution set B



○ Solution set A



$$IGD^+(A) = IGD^+(B)$$

Overall Performance Indicators

IGD (inverted generational distance) indicator

IGD⁺ (inverted generational distance plus) indicator

ϵ -indicator (Additive epsilon)

HV (hypervolume) indicator



ε -indicator (Additive epsilon)

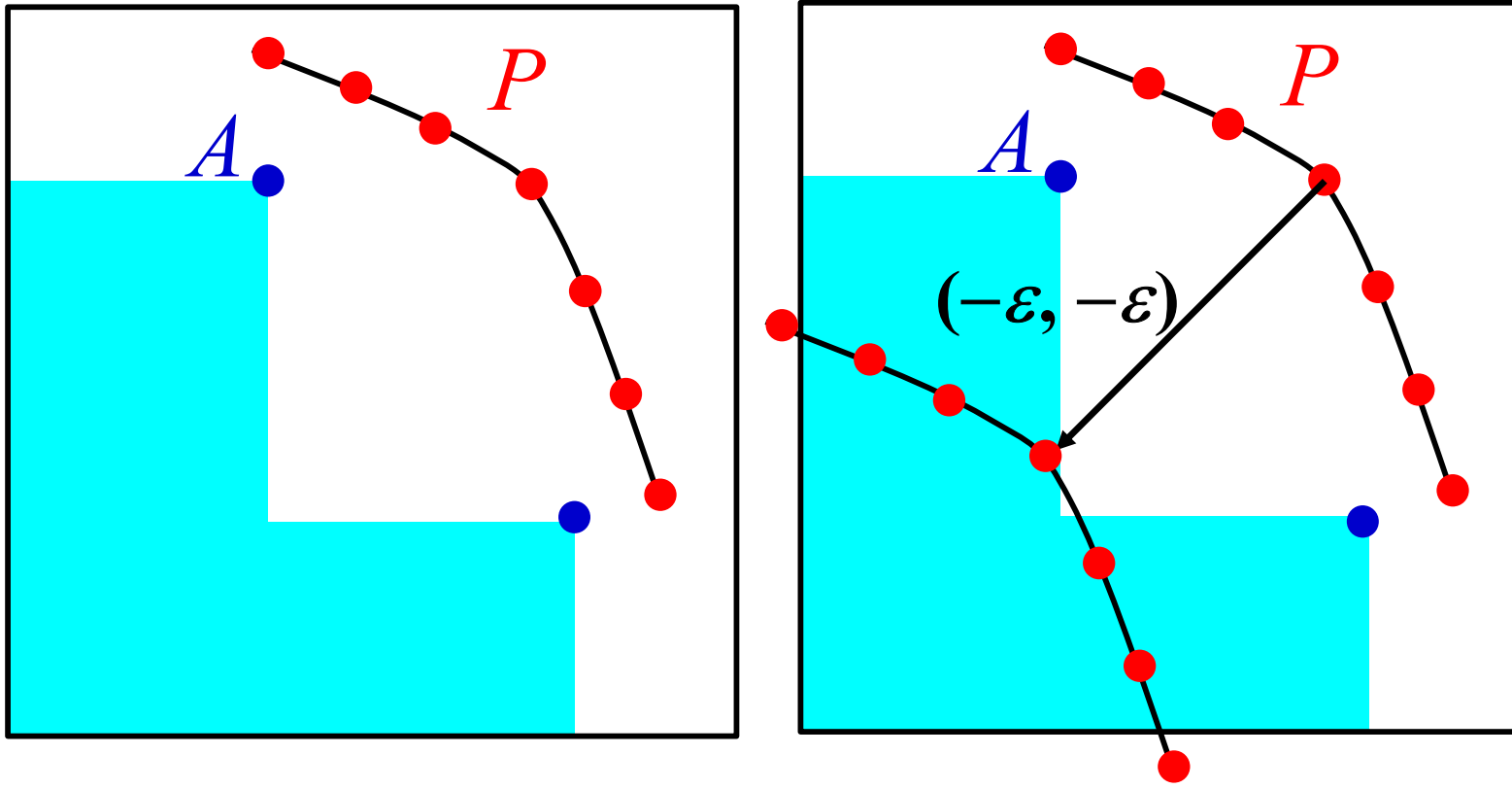
Solution Set $A = \{\mathbf{a}_1, \mathbf{a}_2, \dots\}$, $\mathbf{a}_i = (a_{i1}, \dots, a_{im})$ for $i = 1, 2, \dots$,

Reference Set $P = \{\mathbf{p}_1, \mathbf{p}_2, \dots\}$, $\mathbf{p}_i = (p_{i1}, \dots, p_{im})$ for $i = 1, 2, \dots$

$$I_{\varepsilon+}(A, P) = \min \{\varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A\}$$

$$P_{\varepsilon-} = \{\mathbf{p}_1 - \varepsilon, \mathbf{p}_2 - \varepsilon, \dots\}, \quad \mathbf{p}_i - \varepsilon = (p_{i1} - \varepsilon, \dots, p_{im} - \varepsilon)$$

for $i = 1, 2, \dots$

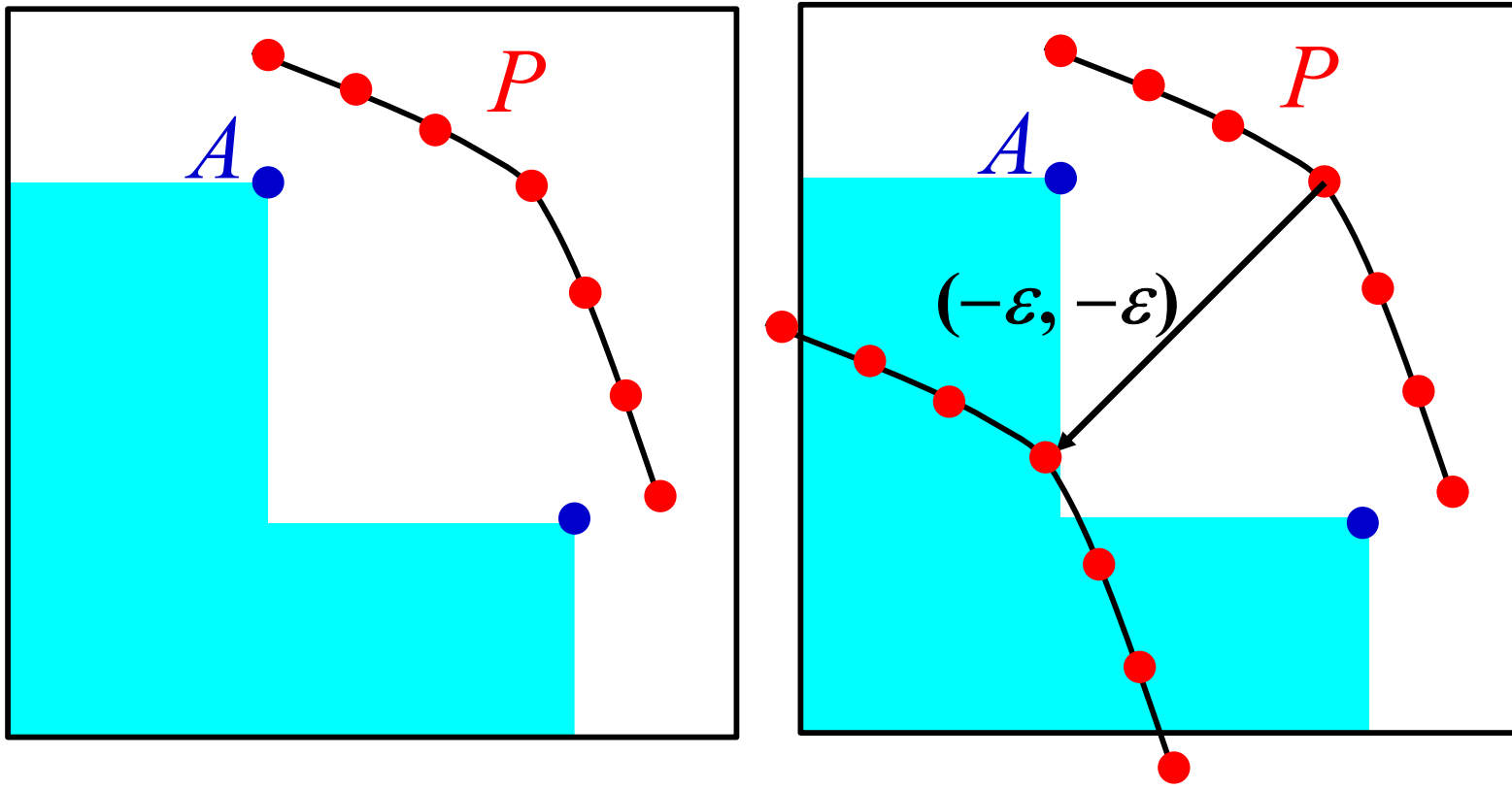


ε -indicator (Additive epsilon)

$$I_{\varepsilon+}(A, P) = \min \{ \varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A \} \quad P_{\varepsilon-} = \{ p_1 - \varepsilon, p_2 - \varepsilon, \dots \}$$

Basic Idea:

The minimum change (the minimum shift) of P so that the solution set A weakly dominates the reference set P .



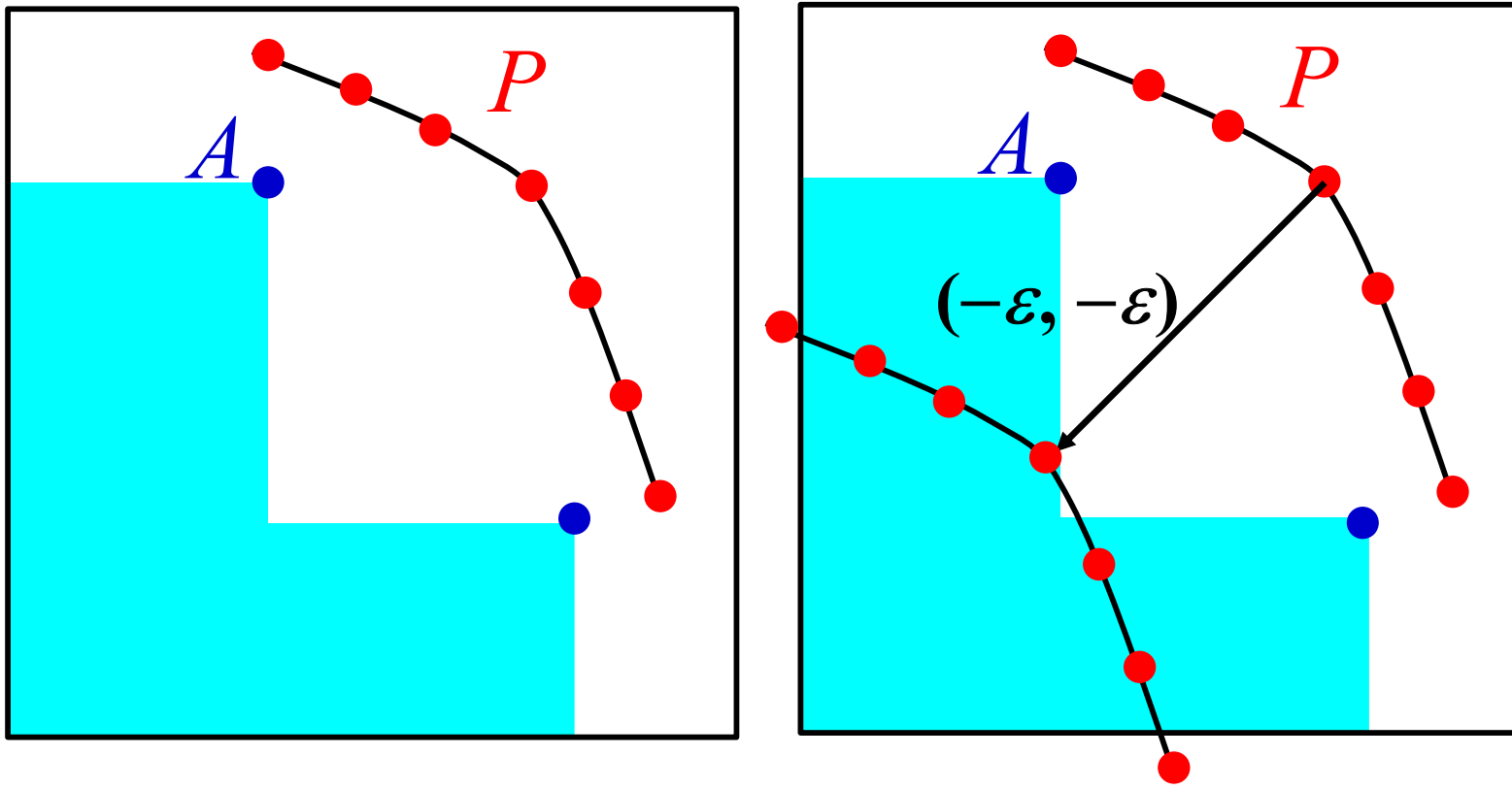
ε -indicator (Additive epsilon)

$$I_{\varepsilon+}(A, P) = \min \{ \varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A \} \quad P_{\varepsilon-} = \{ p_1 - \varepsilon, p_2 - \varepsilon, \dots \}$$

Basic Idea:

The minimum change (the minimum shift) of P so that the solution set A weakly dominates the reference set P .

All reference points in P should be moved into the dominated region by the solution set A .



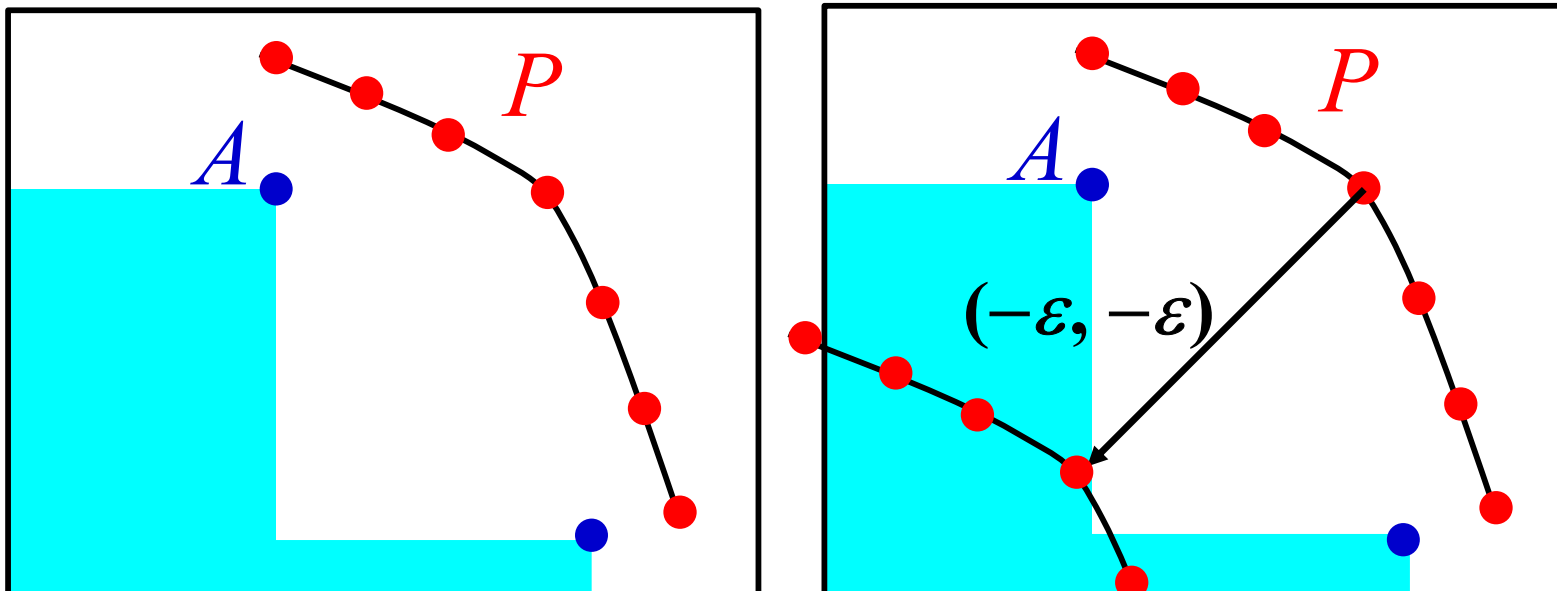
ε -indicator (Additive epsilon)

$$I_{\varepsilon+}(A, P) = \min \{ \varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A \} \quad P_{\varepsilon-} = \{ p_1 - \varepsilon, p_2 - \varepsilon, \dots \}$$

Basic Idea:

The minimum change (the minimum shift) of P so that the solution set A weakly dominates the reference set P .

All reference points in P should be moved into the dominated region by the solution set A .



Question: Is additive epsilon a Pareto compliant indicator ?
Your answer: _____.

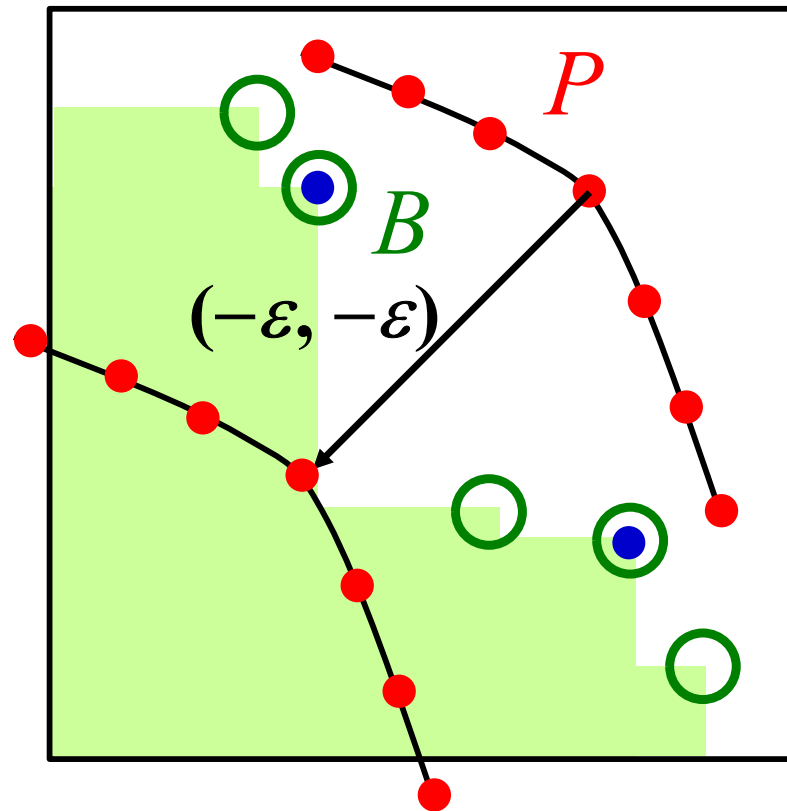
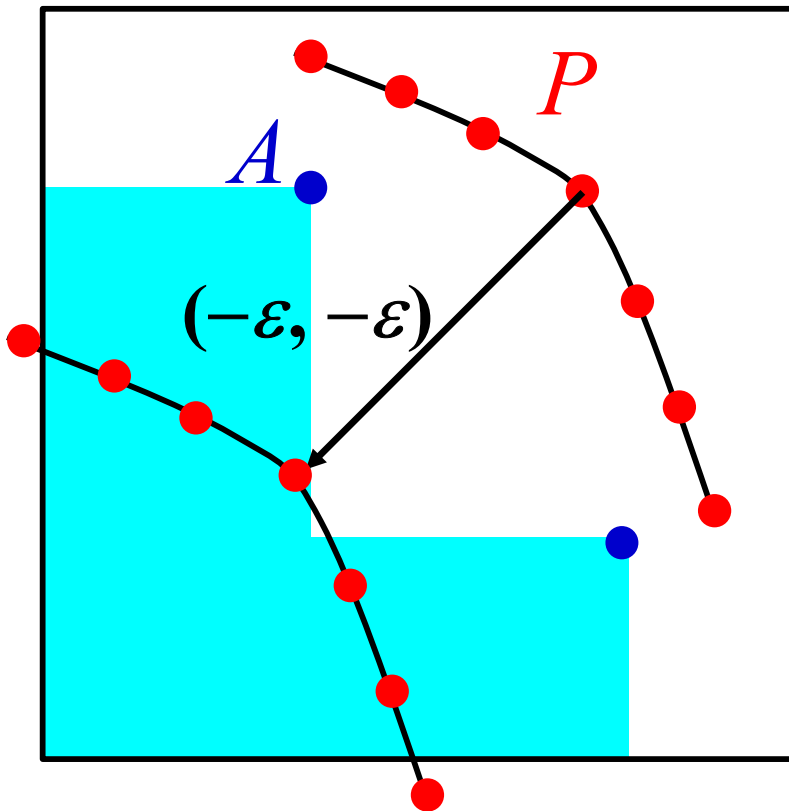
ε -indicator (Additive epsilon)

$$I_{\varepsilon+}(A, P) = \min \{ \varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A \} \quad P_{\varepsilon-} = \{ p_1 - \varepsilon, p_2 - \varepsilon, \dots \}$$

Basic Idea:

The minimum change (the minimum shift) of P so that the solution set A weakly dominates the reference set P .

All reference points in P should be moved into the dominated region by the solution set A . $\implies$ **Weak Pareto Compliant.**



ε -indicator (Additive epsilon)

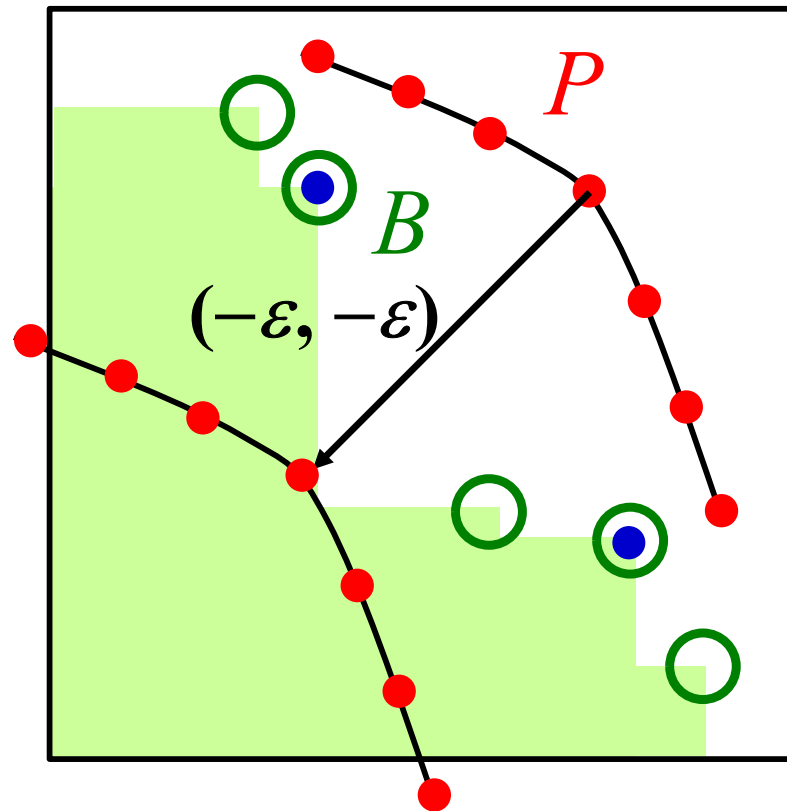
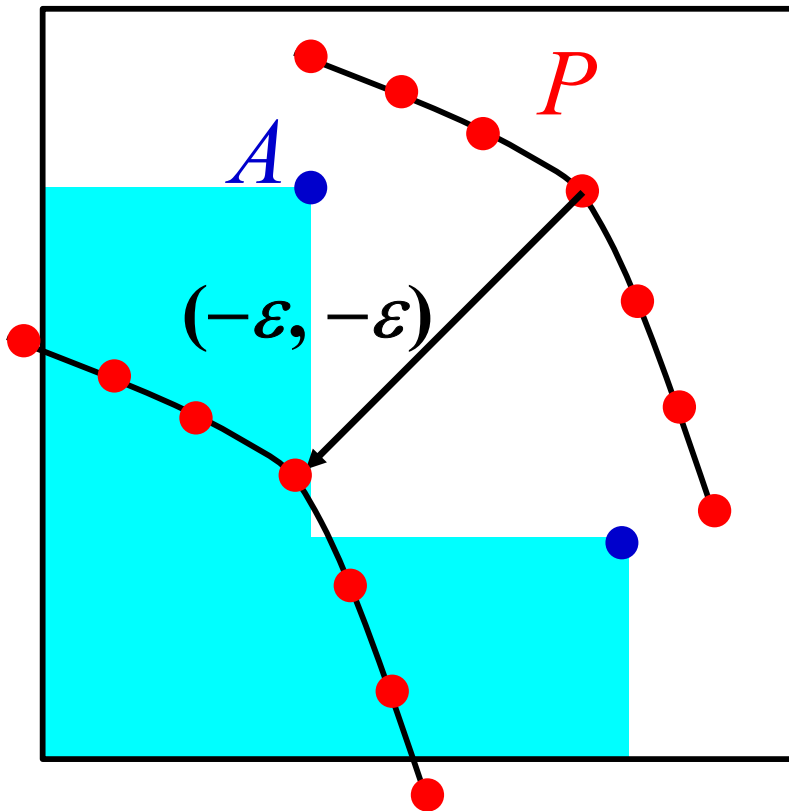
$$I_{\varepsilon+}(A, P) = \min \{ \varepsilon \in \mathbb{R} \mid P_{\varepsilon-} \preceq A \} \quad P_{\varepsilon-} = \{ \mathbf{p}_1 - \varepsilon, \mathbf{p}_2 - \varepsilon, \dots \}$$

Some difficulties:

- (i) $I_{\varepsilon+}$ depends on the reference set (as IGD and IGD⁺)
- (ii) Weak Pareto compliant (not Pareto compliant as IGD⁺)

==> Better solution sets can be evaluated as the same.

This is severe since the calculation is based on only a single point.



Computational Experiments

Step 1: Solution sets A and B ($A \triangleleft B$) are generated.

A and B are intermediate populations in a run of an EMO algorithm.

Step 2: It is examined which is hold among

- Pareto compliant: $I(A) > I(B)$,
- Weakly Pareto compliant: $I(A) = I(B)$,
- Inconsistent: $I(A) < I(B)$.

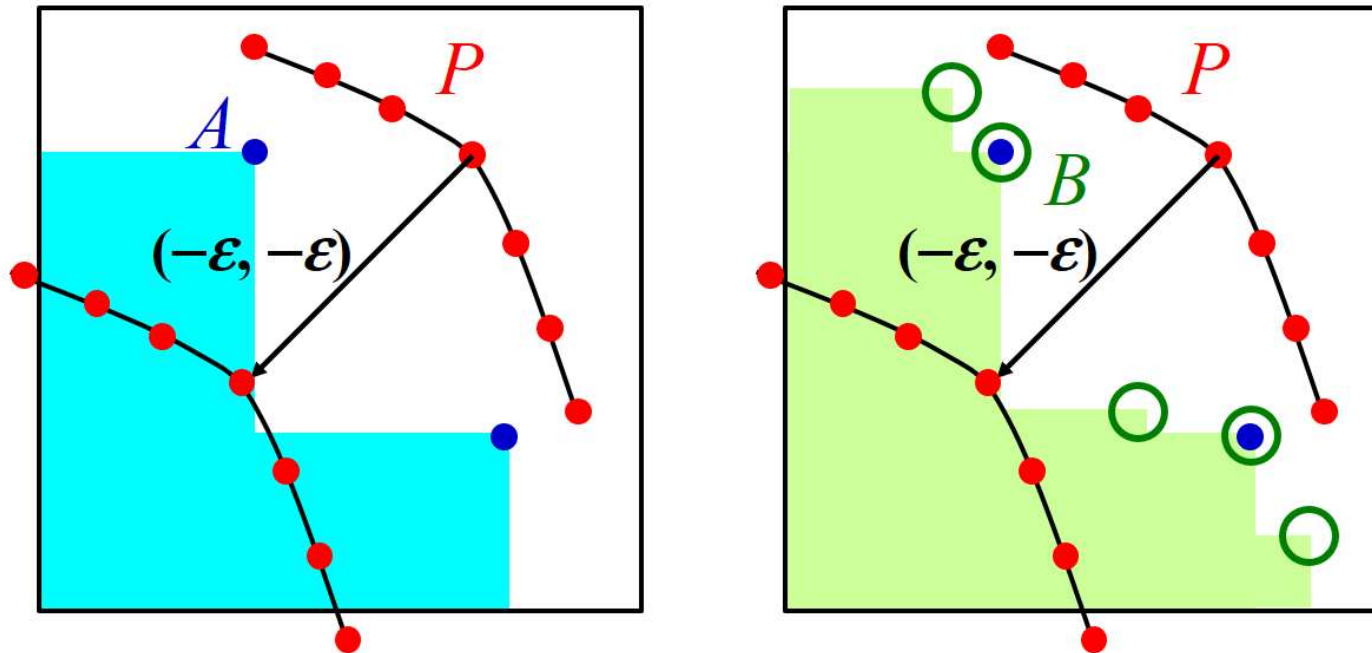


Table 1. Results on the 2193 pairs of ordered non-dominated solution sets of the **two-objective DTLZ2 problem.**

Indicator	Pareto Compliant $I(A) > I(B)$	Weakly Compliant $I(A) = I(B)$	Inconsistent $I(A) < I(B)$
GD	98.04%	0.00%	1.96%
IGD	97.95%	0.00%	2.05%
IGD ⁺	100.00%	0.00%	0.00%
Additive ε	91.24%	8.97%	0.00%

Table 2. Results on the 145 pairs of ordered non-dominated solution sets of the **four-objective DTLZ2 problem.**

Indicator	Pareto Compliant $I(A) > I(B)$	Weakly Compliant $I(A) = I(B)$	Inconsistent $I(A) < I(B)$
GD	97.93%	0.00%	2.07%
IGD	100.00%	0.00%	0.00%
IGD ⁺	100.00%	0.00%	0.00%
Additive ε	93.79%	6.21%	0.00%

Table 3. Results on the 47993 pairs of ordered non-dominated solution sets of **the two-objective 500-item knapsack problem.**

Indicator	Pareto Compliant $I(A) > I(B)$	Weakly Compliant $I(A) = I(B)$	Inconsistent $I(A) < I(B)$
GD	81.80%	0.00%	18.20%
IGD	77.22%	17.58%	5.21%
IGD ⁺	82.41%	17.59%	0.00%
Additive ε	14.83%	85.17%	0.00%

Table 4. Results on the 4704 pairs of ordered non-dominated solution sets of **the four-objective 500-item knapsack problem.**

Indicator	Pareto Compliant $I(A) > I(B)$	Weakly Compliant $I(A) = I(B)$	Inconsistent $I(A) < I(B)$
GD	98.15%	0.00%	1.85%
IGD	99.30%	0.60%	0.11%
IGD ⁺	99.36%	0.64%	0.00%
Additive ε	53.93%	46.07%	0.00%

Overall Performance Indicators

IGD (inverted generational distance) indicator

IGD⁺ (inverted generational distance plus) indicator

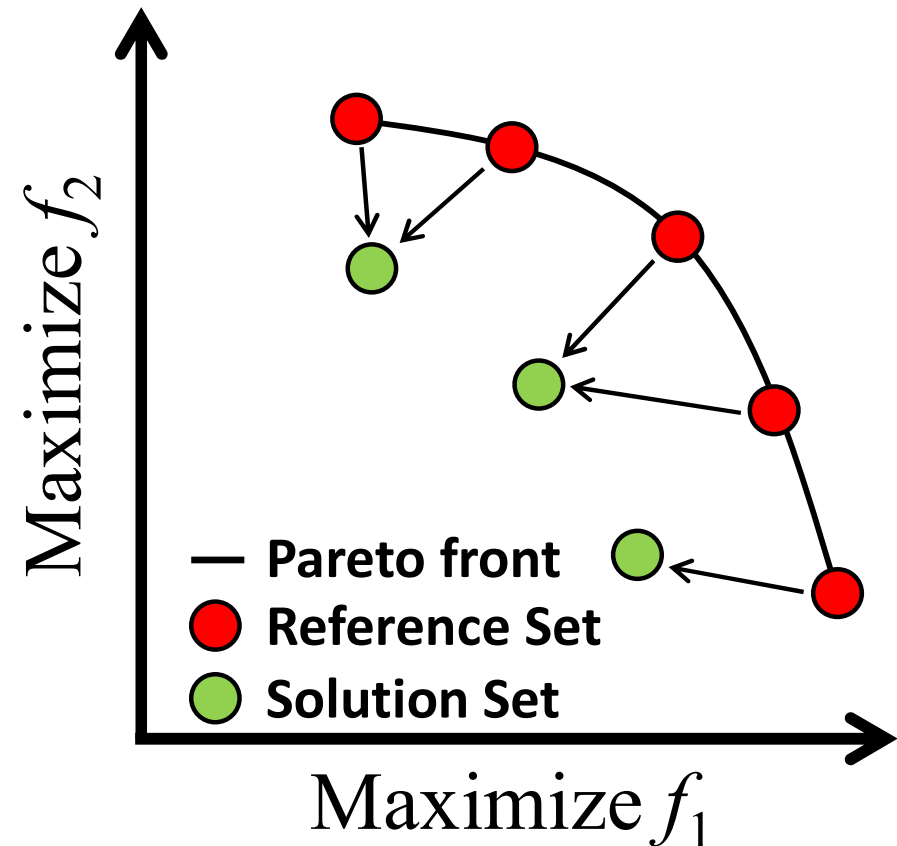
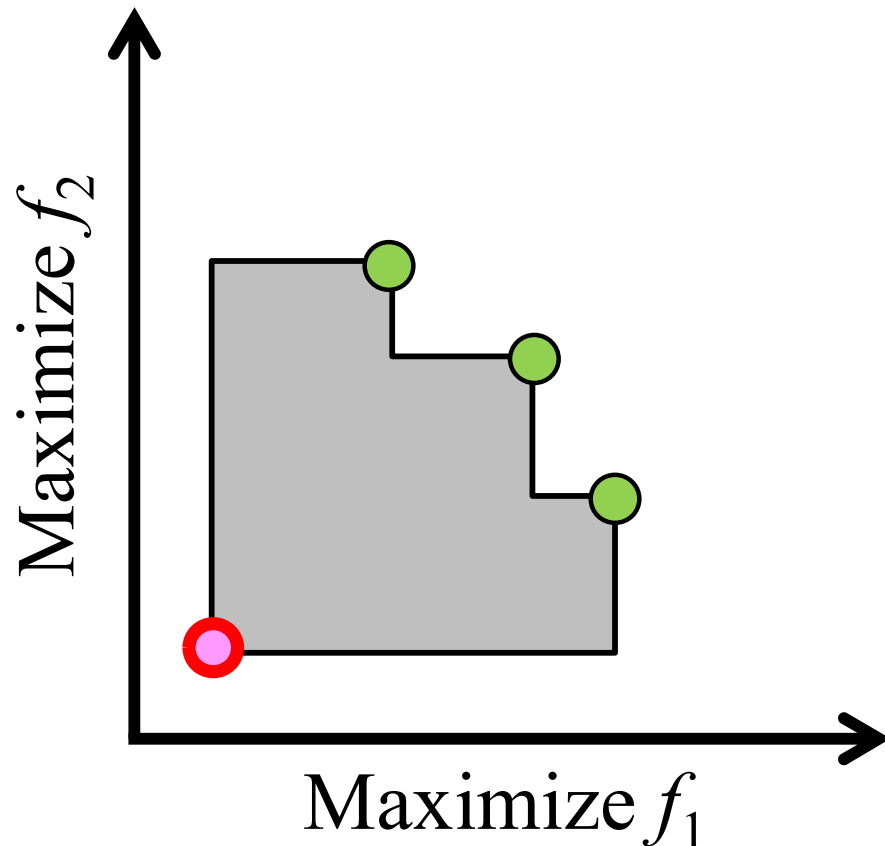
ε -indicator (Additive epsilon)

HV (hypervolume) indicator



Hypervolume

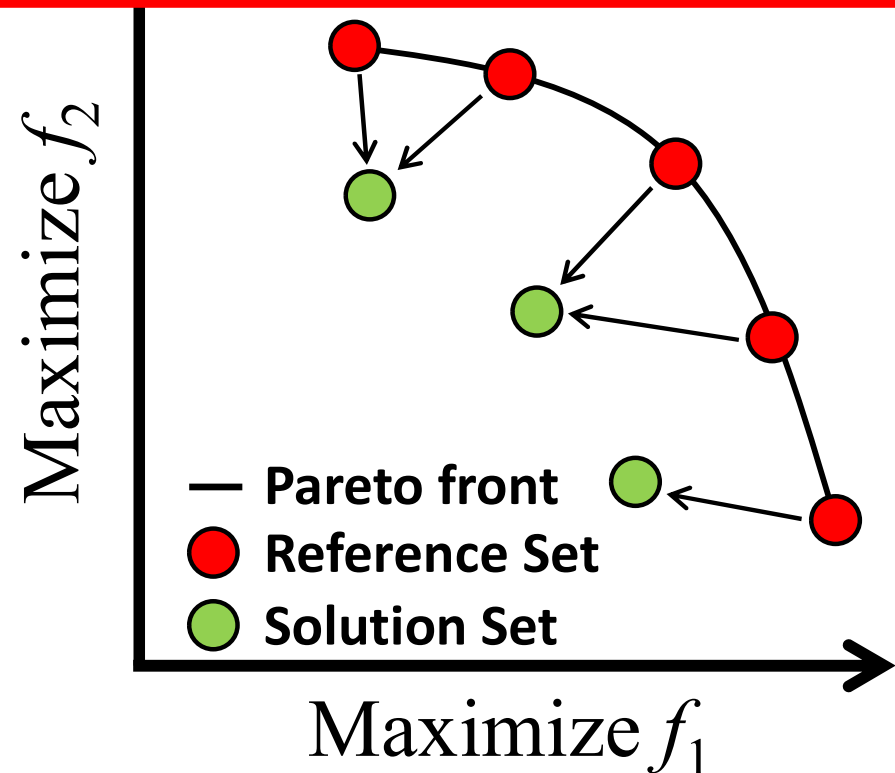
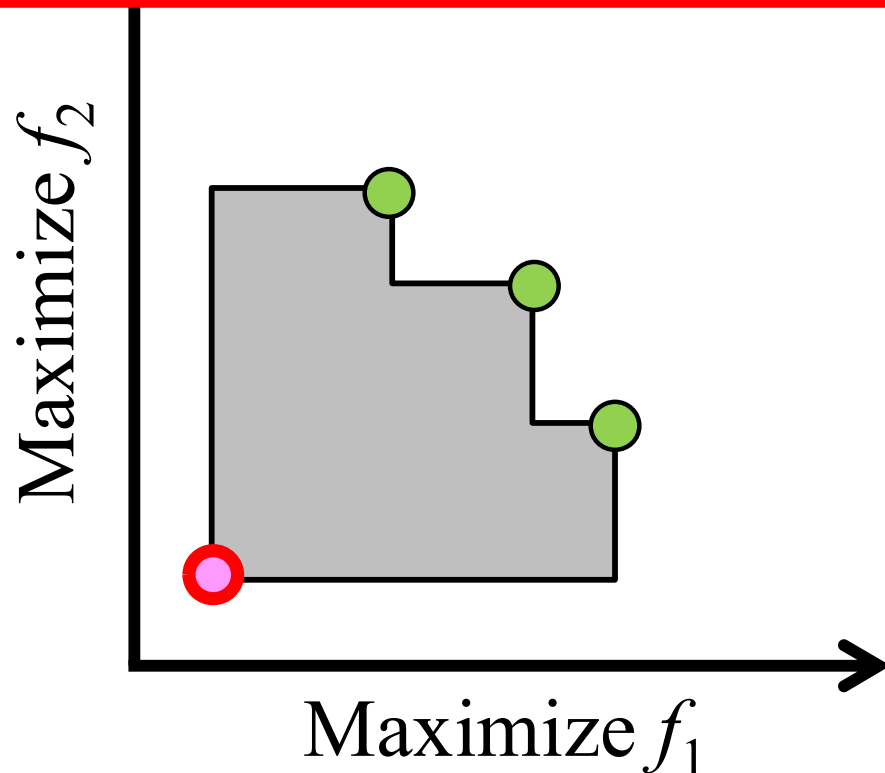
Hypervolume is the volume of the dominated region by the obtained solutions bound by the reference point.



Hypervolume

Hypervolume is the volume of the dominated region by the obtained solutions bound by the reference point.

Question: Is hypervolume a Pareto compliant indicator ?
Your answer: _____.

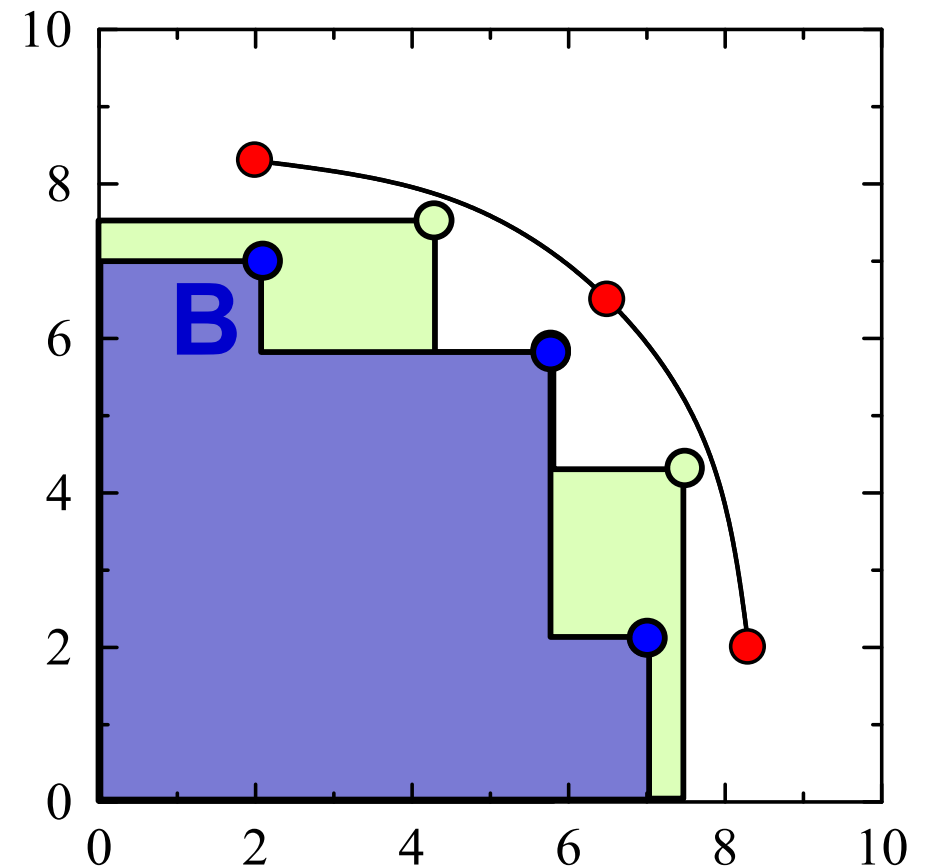
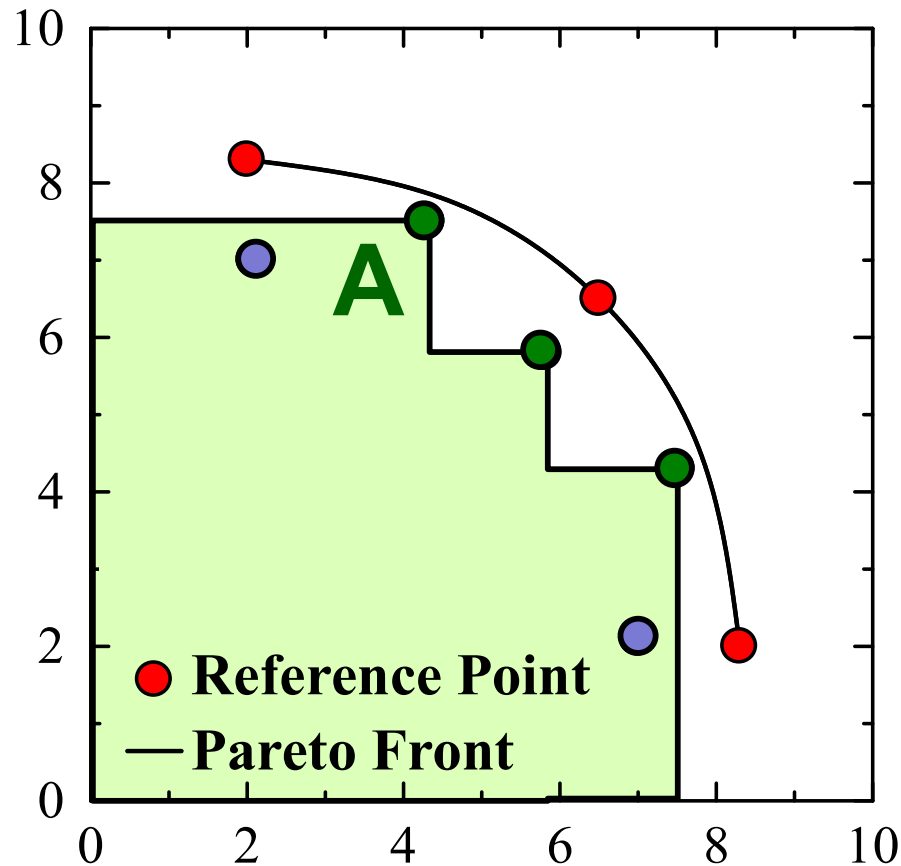


Theoretical Result

When solution set A is better than B

Hypervolume(A) > Hypervolume(B) always holds.

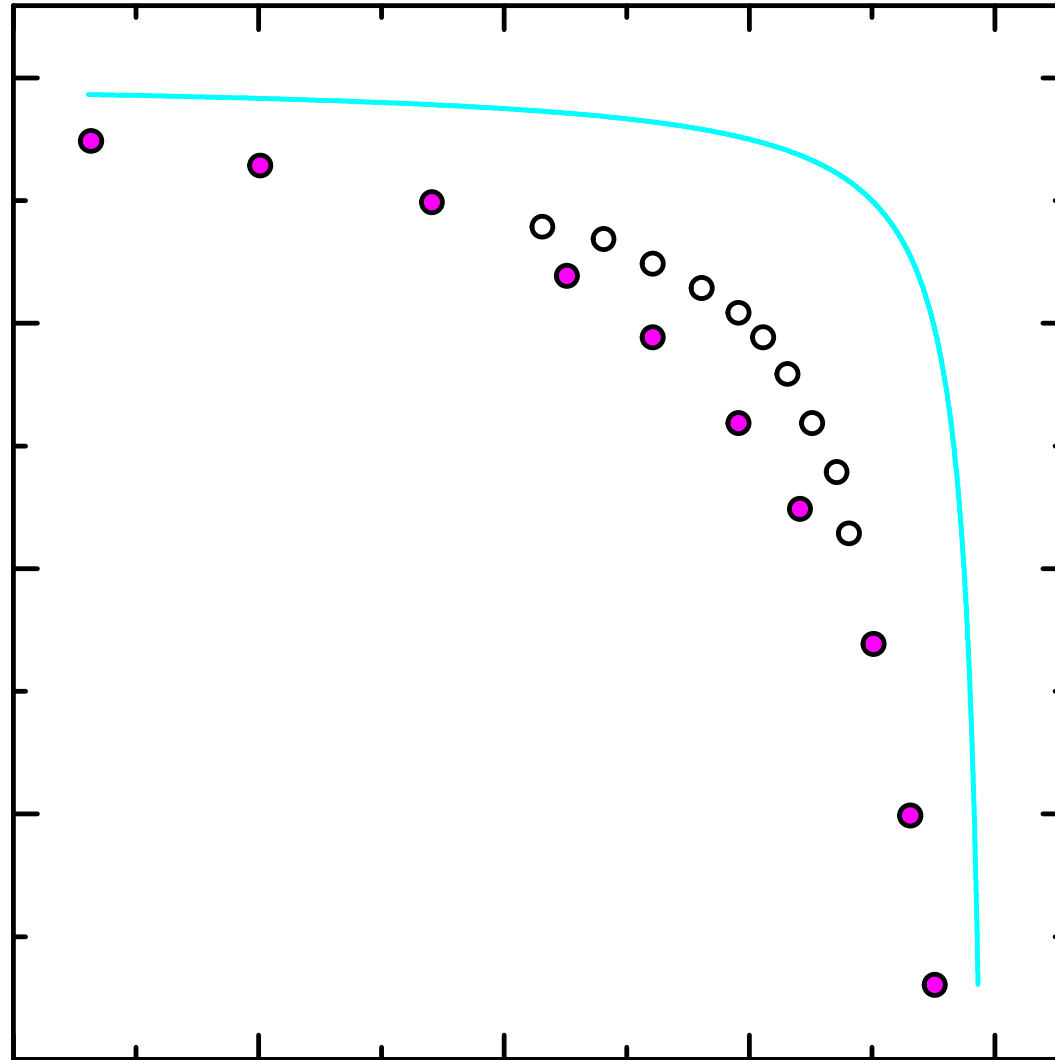
Well-Known Property: Hypervolume is Pareto compliant.



Hypervolume(A) > Hypervolume(B)

Two Solution Sets:

Which do you like?

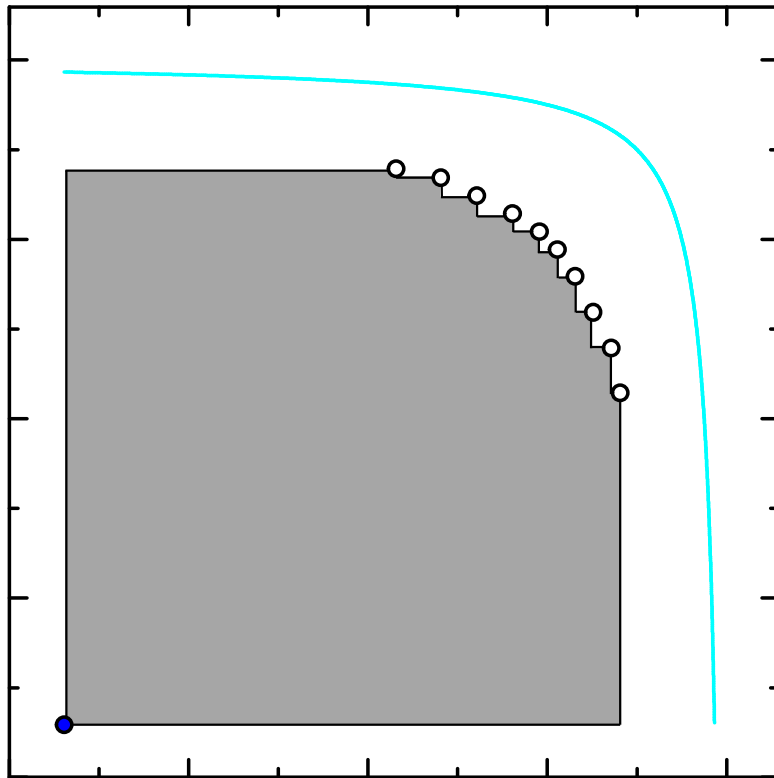


This is not an easy problem even for a two-objective case.

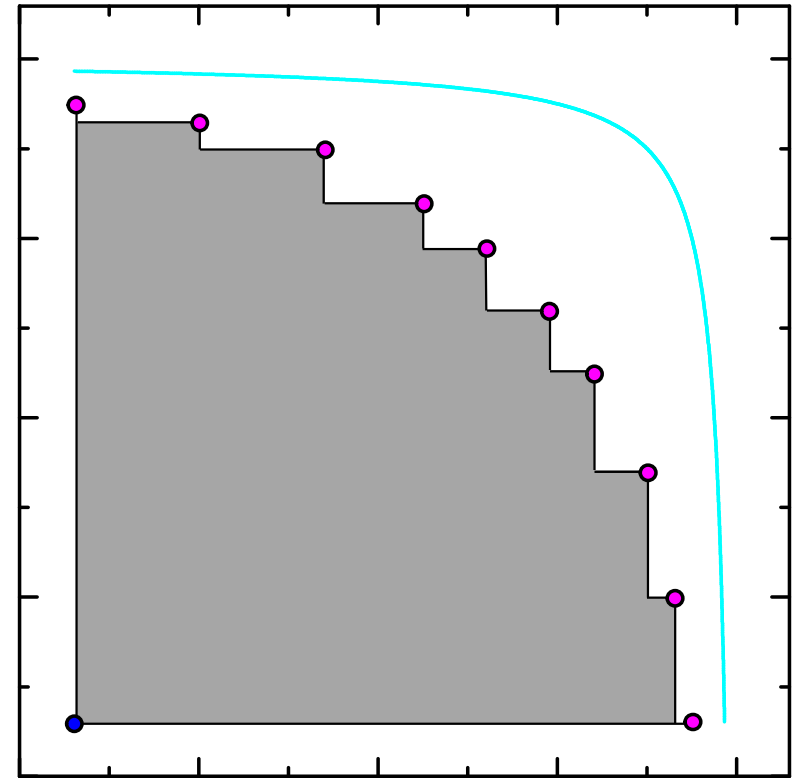
Hypervolume (HV)

Comparison results depend on the reference point

When the reference point is close to the Pareto front:



>

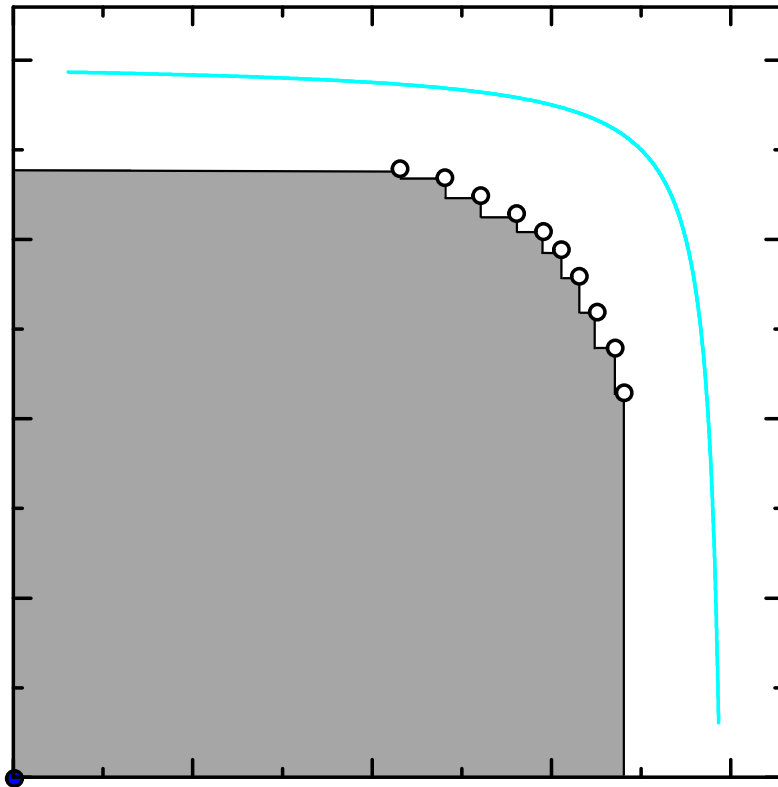


Better Solution Set

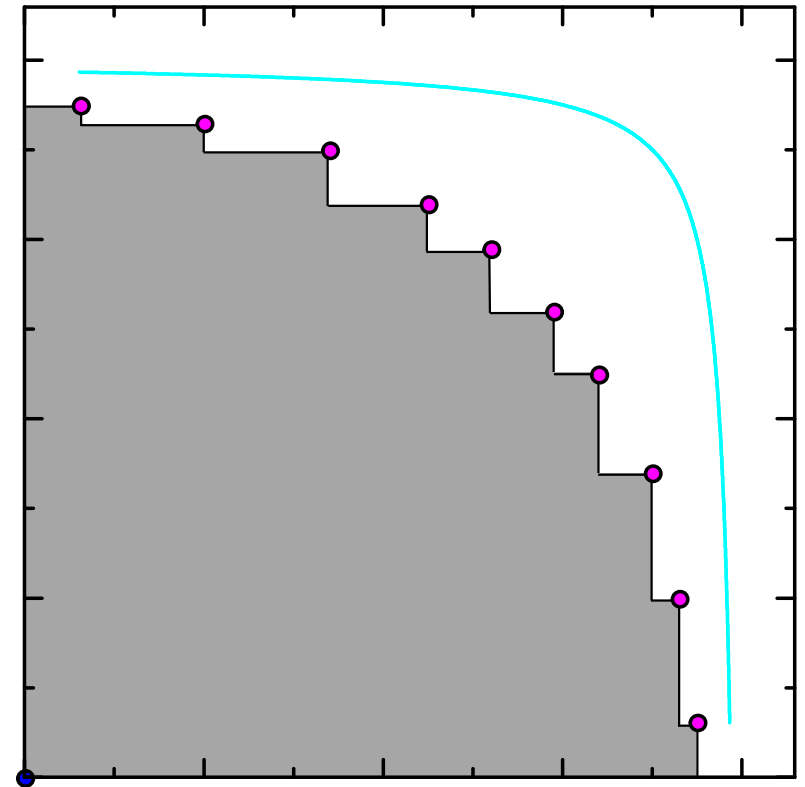
Hypervolume (HV)

Comparison results depend on the reference point

When the reference point is far from the Pareto front:

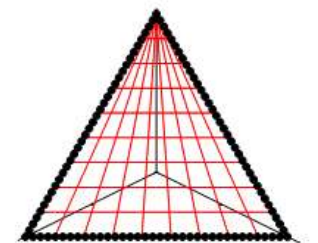
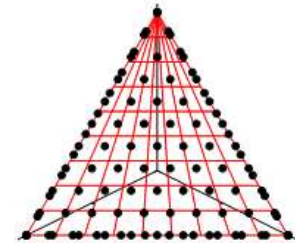
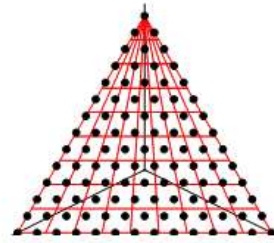
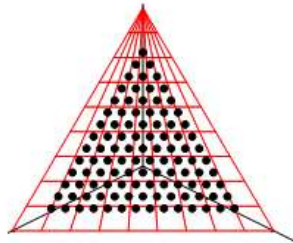
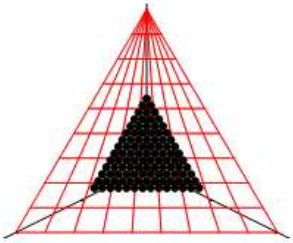


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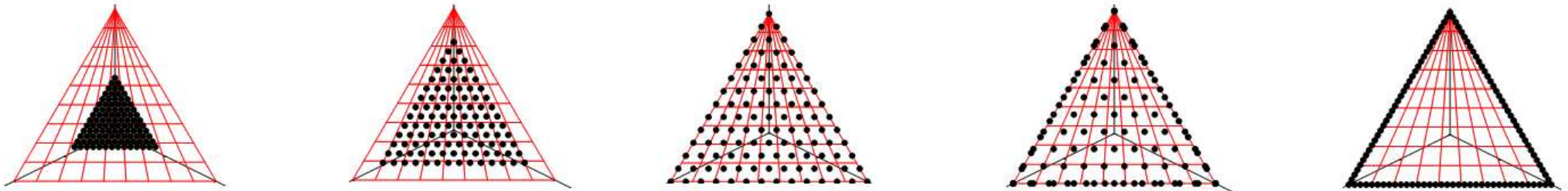
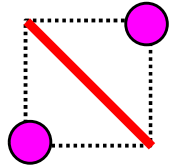
Better Solution Set

Which is the Best Solution Set by Hypervolume-Based Comparison?

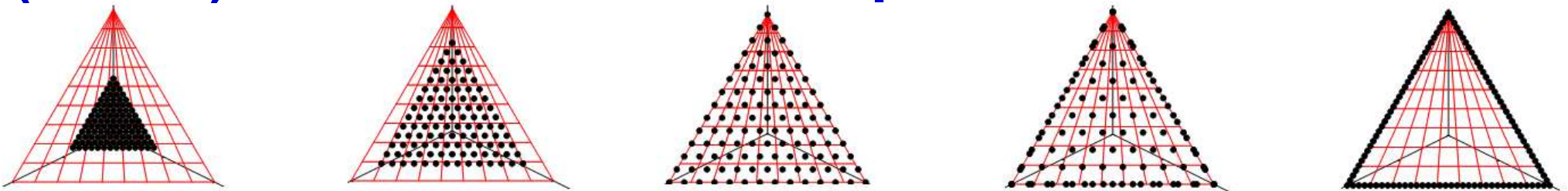


Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is the nadir point:
(0, 0, 0) for the maximization problem



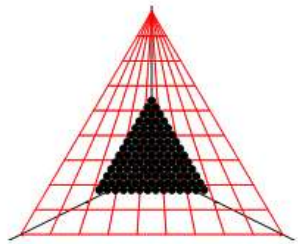
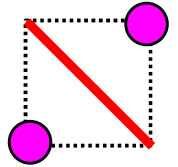
(1, 1, 1) for the minimization problem



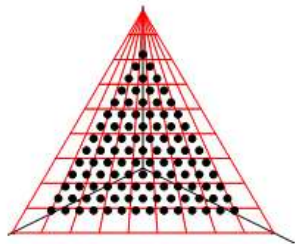
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is the nadir point:

$(0, 0, 0)$ for the maximization problem



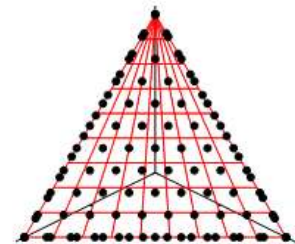
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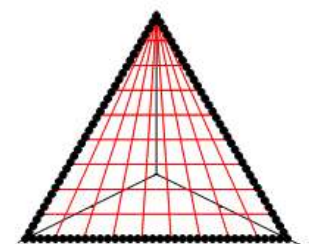
1



2

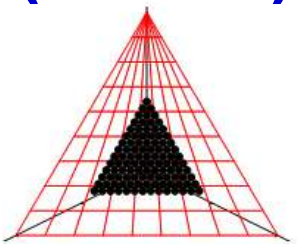


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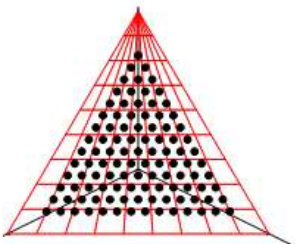


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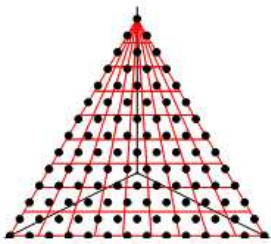
$(1, 1, 1)$ for the minimization problem



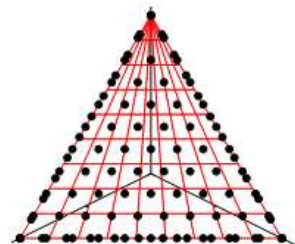
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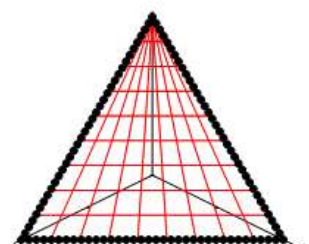
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1

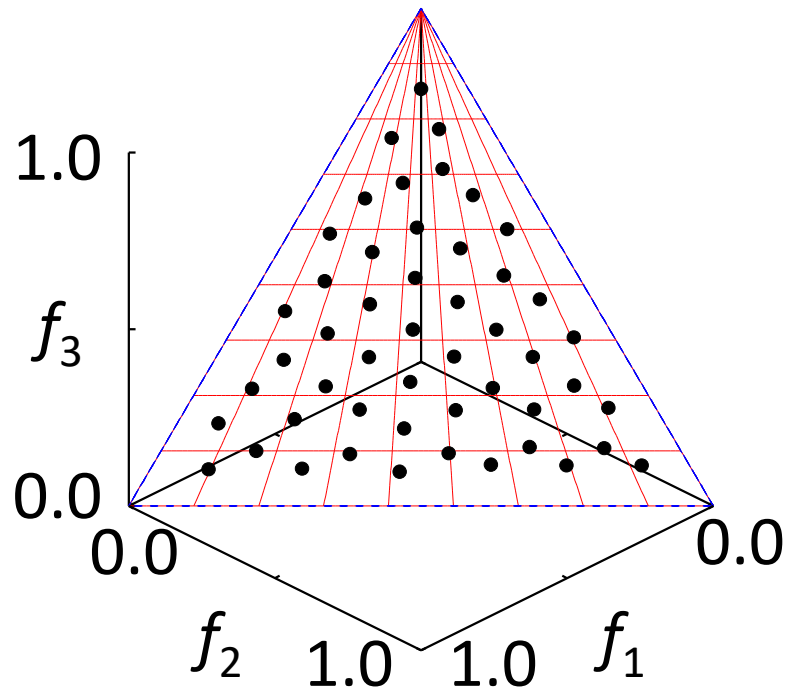


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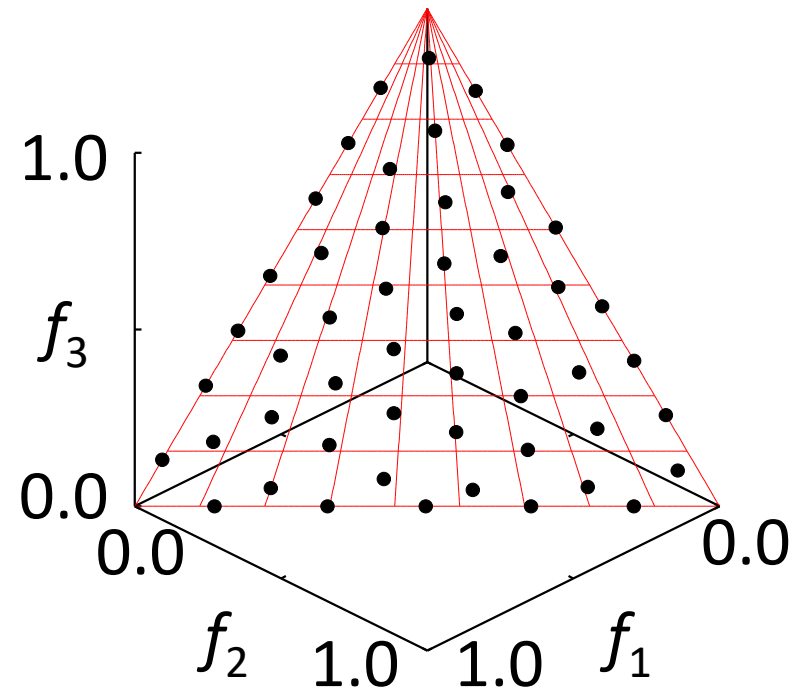


3

Near Optimal Distribution when the reference point is the nadir point



Maximization Problem

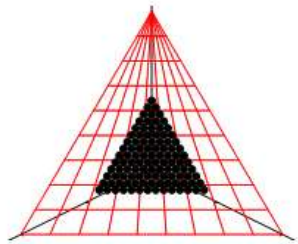
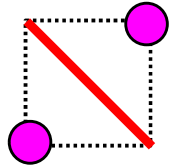


Minimization Problem

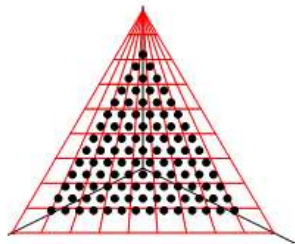
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is the nadir point:

$(0, 0, 0)$ for the maximization problem



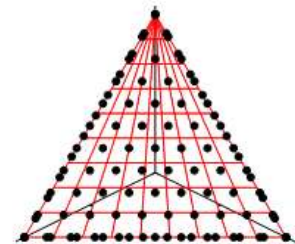
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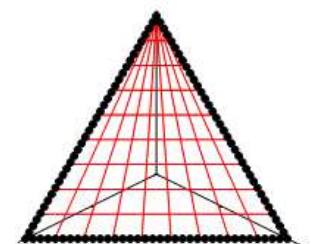
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2

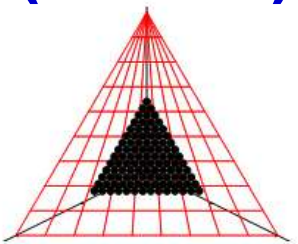


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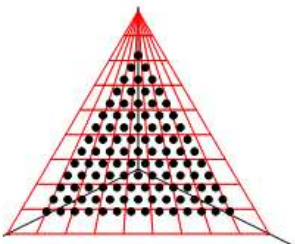


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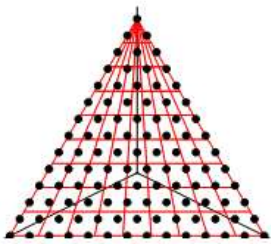
$(1, 1, 1)$ for the minimization problem



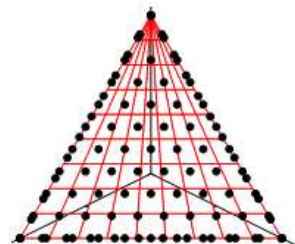
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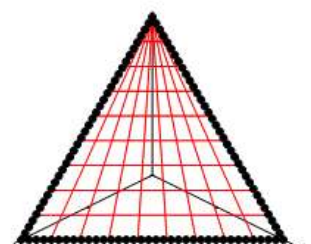
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1



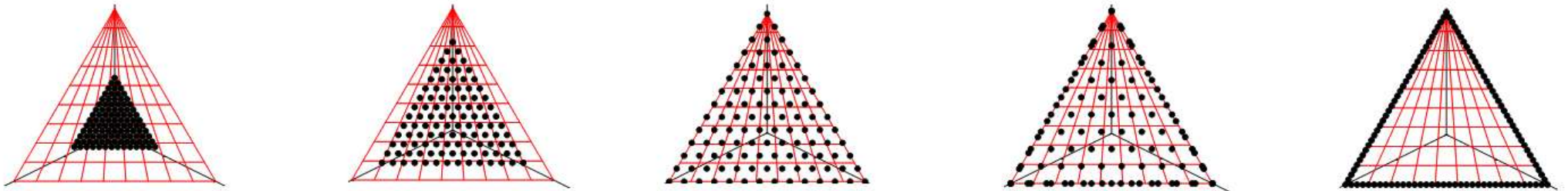
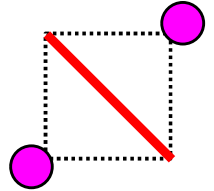
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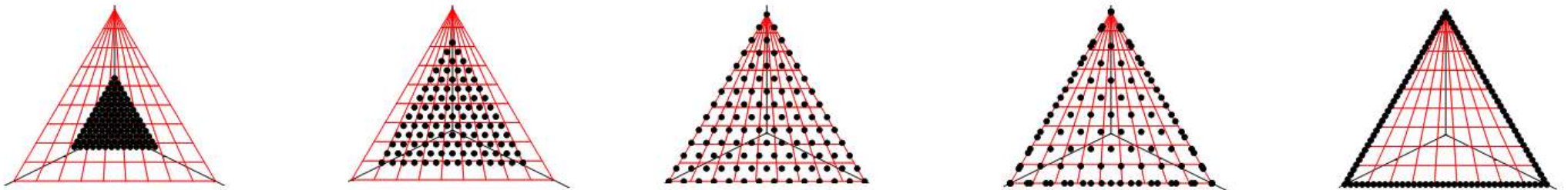
3

Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is a little bit far:
(-0.1, -0.1, -0.1) for the maximization problem

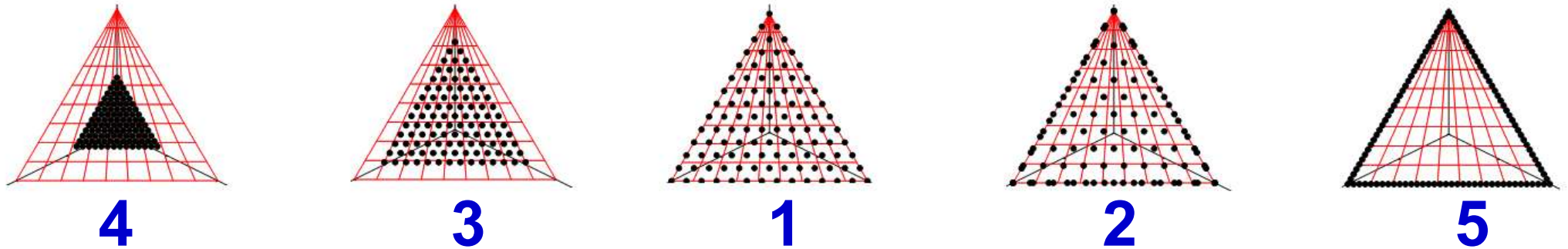
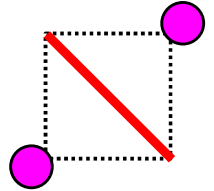


(1.1, 1.1, 1.1) for the minimization problem

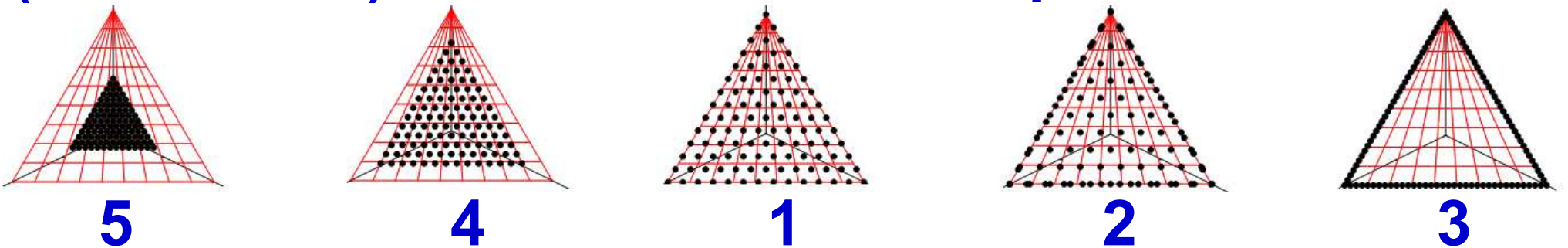


Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is a little bit far:
(-0.1, -0.1, -0.1) for the maximization problem



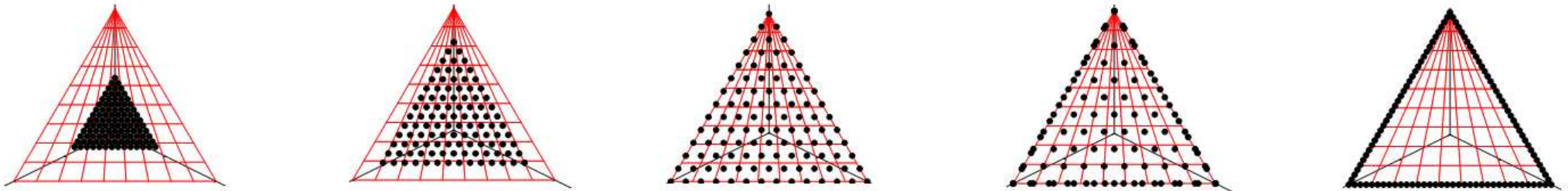
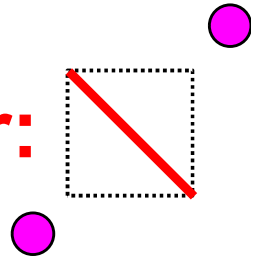
(1.1, 1.1, 1.1) for the minimization problem



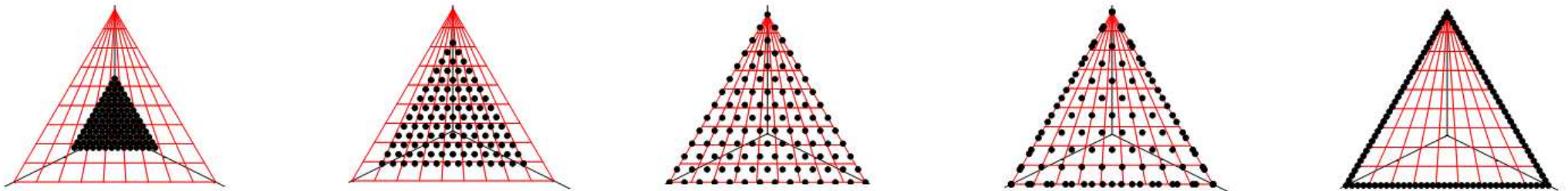
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is a little bit more far:

$(-0.5, -0.5, -0.5)$ for the maximization problem



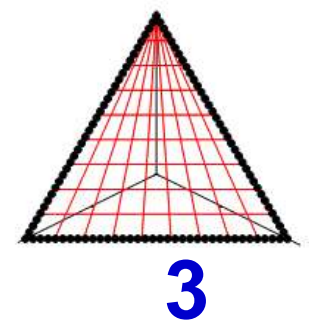
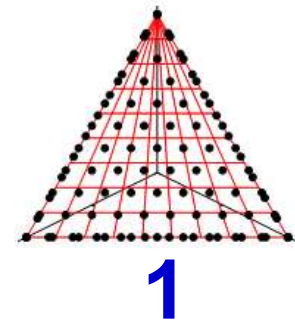
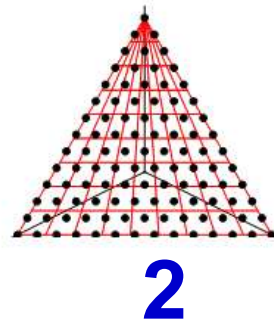
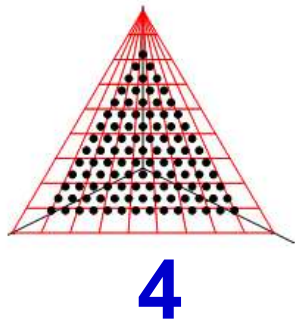
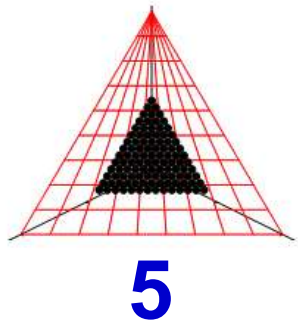
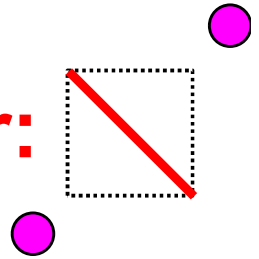
$(1.5, 1.5, 1.5)$ for the minimization problem



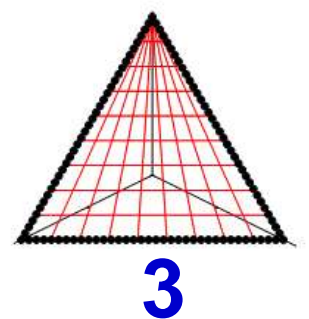
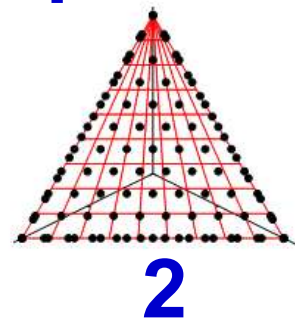
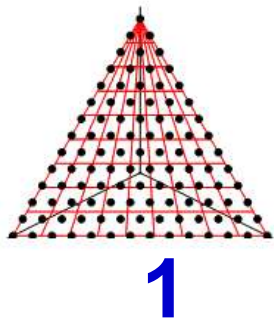
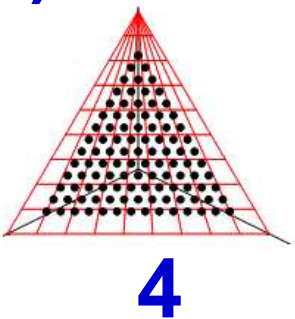
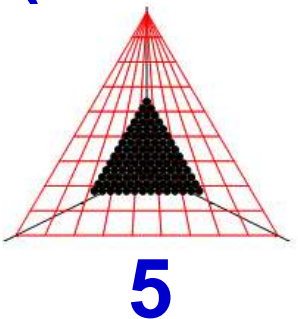
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is a little bit more far:

$(-0.5, -0.5, -0.5)$ for the maximization problem



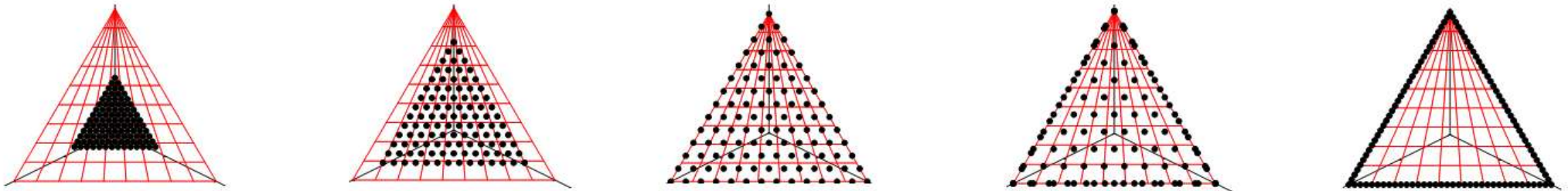
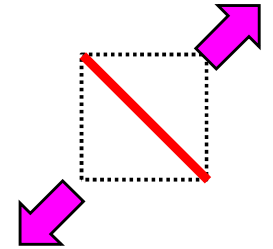
$(1.5, 1.5, 1.5)$ for the minimization problem



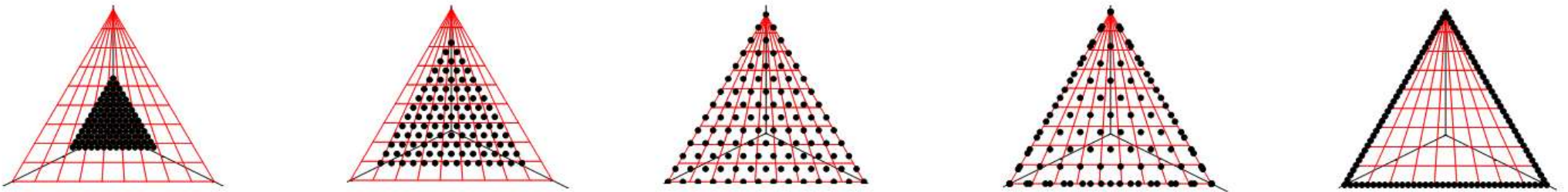
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is far away:

$(-20, -20, -20)$ for the maximization problem



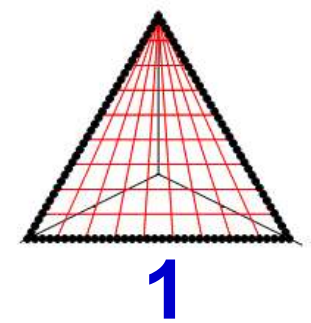
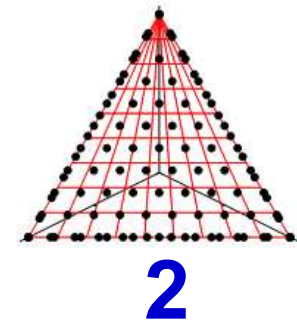
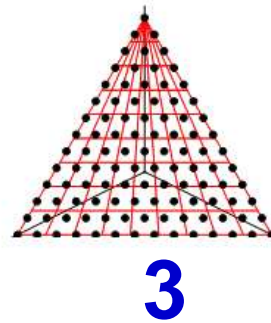
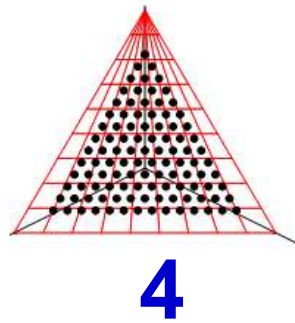
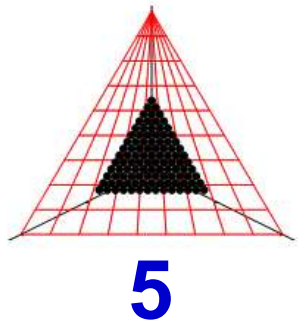
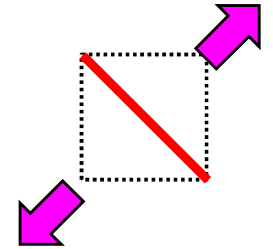
$(20, 20, 20)$ for the minimization problem



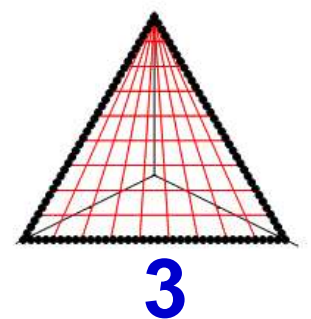
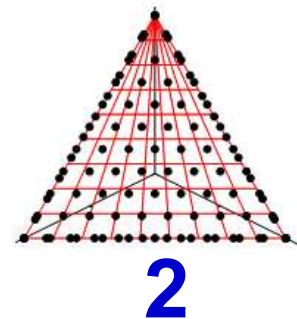
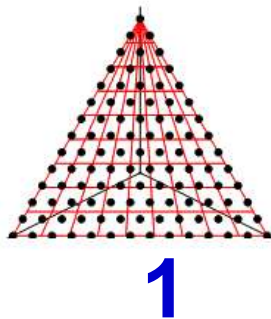
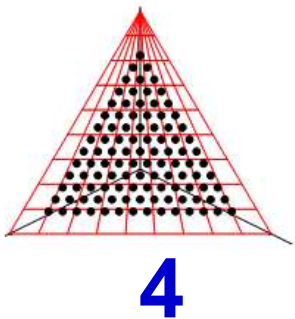
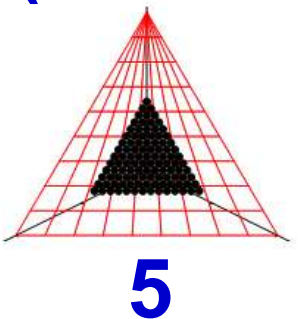
Which is the Best Solution Set by Hypervolume-Based Comparison?

When the reference point is far away:

$(-20, -20, -20)$ for the maximization problem



$(20, 20, 20)$ for the minimization problem



Dependency of the optimal distribution of solutions on the location of the reference point ($r = 1.0$: Nadir Point)

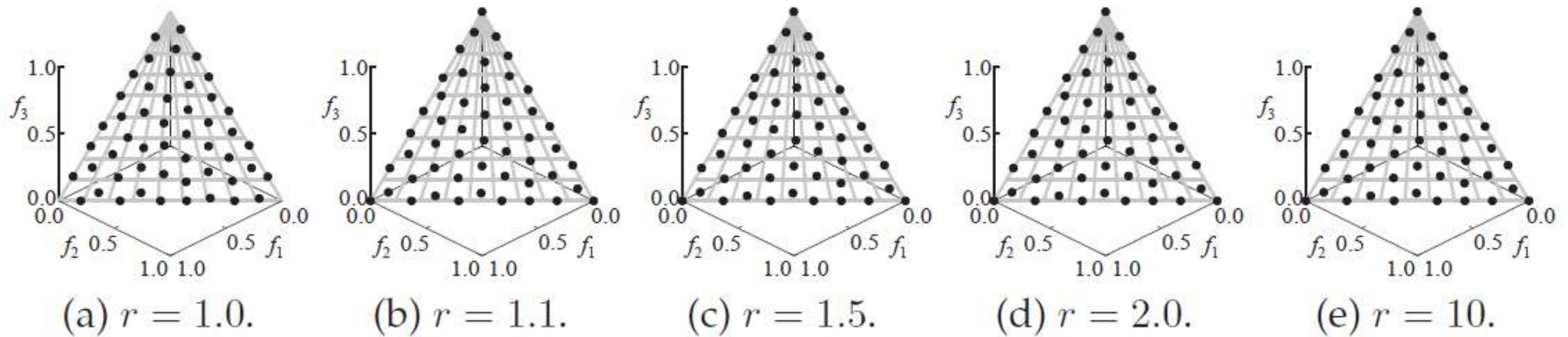


Figure 1: Obtained solution sets for the three-objective normalized DTLZ1.

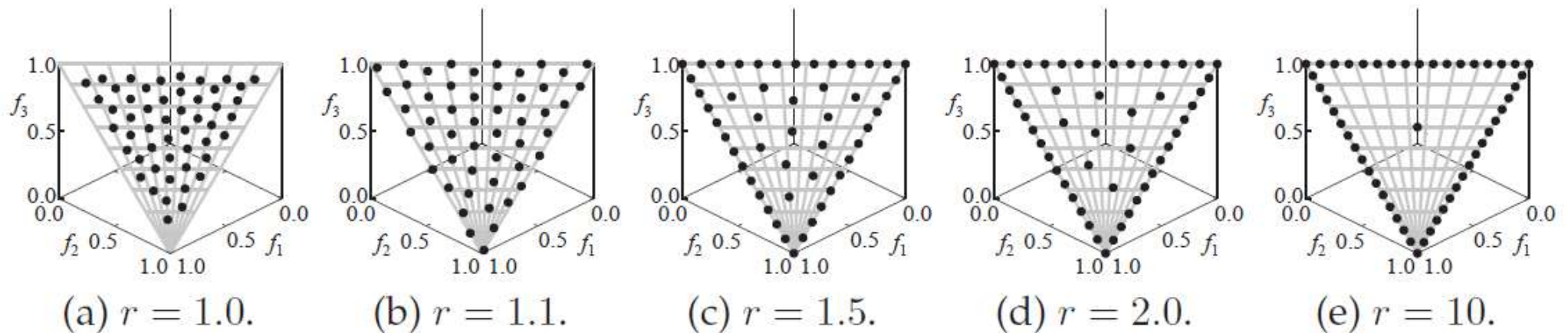


Figure 2: Obtained solution sets for the three-objective normalized Minus-DTLZ1.

H. Ishibuchi et al., How to specify a reference point in hypervolume calculation for fair performance comparison," *Evolutionary Computation* (2018).

Lab Session

Maximize $f_1(\mathbf{x}) = x_1$, $f_2(\mathbf{x}) = x_2$,
subject to $x_1 + x_2 \leq 1$, $0 \leq x_1$, $0 \leq x_2$.

Pareto front is a line segment between $(0, 1)$ and $(1, 0)$.

Task 1: When the reference point is $(0, 0)$, show a single solution which maximizes the hypervolume indicator.

Task 2: When the reference point is $(0, -1)$, show a single solution which maximizes the hypervolume indicator.

Task 3: When the reference point is $(0, 0)$, show a set of two solutions which maximizes the hypervolume indicator.

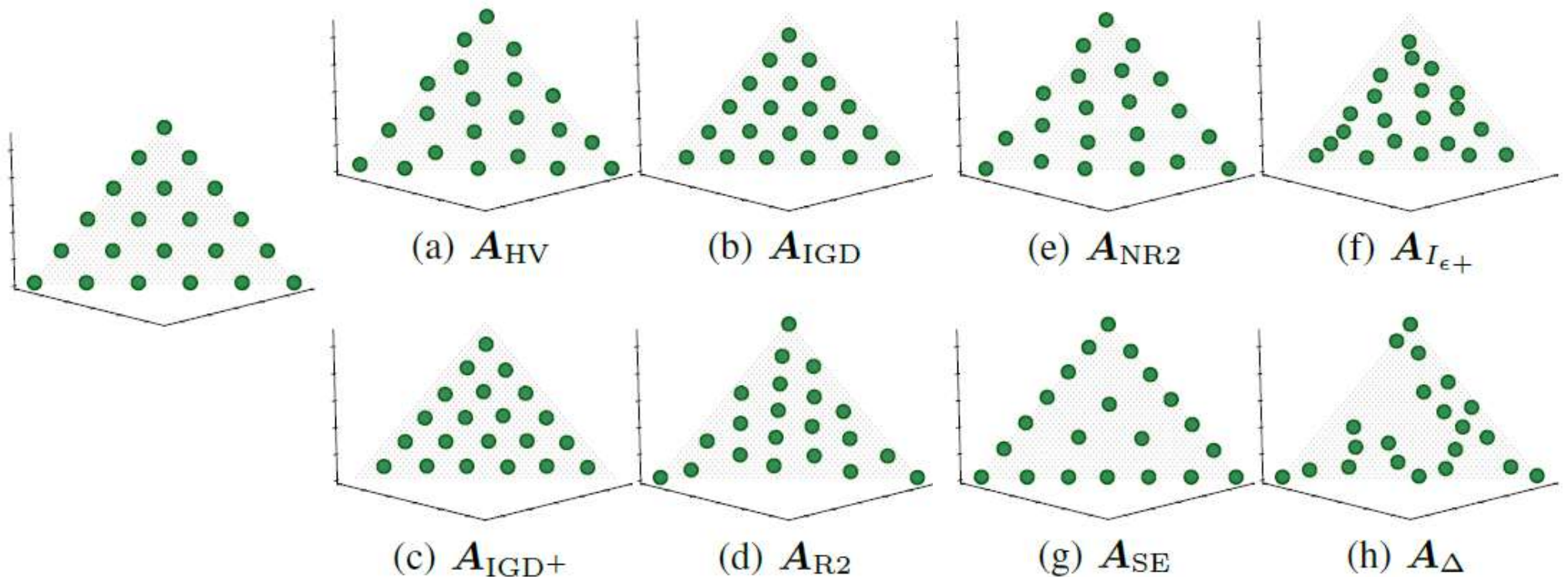
Task 4: When the reference point is $(-2, -2)$, show a set of two solutions which maximizes the hypervolume indicator.

Task 5: When the reference point is $(-2, -2)$, show a set of five solutions which maximizes the hypervolume indicator.

Task 6: When an infinitely large number of reference points are uniformly given over the entire Pareto front, show a set of five solutions which minimize the IGD indicator.

One Additional Note on Performance Indicators

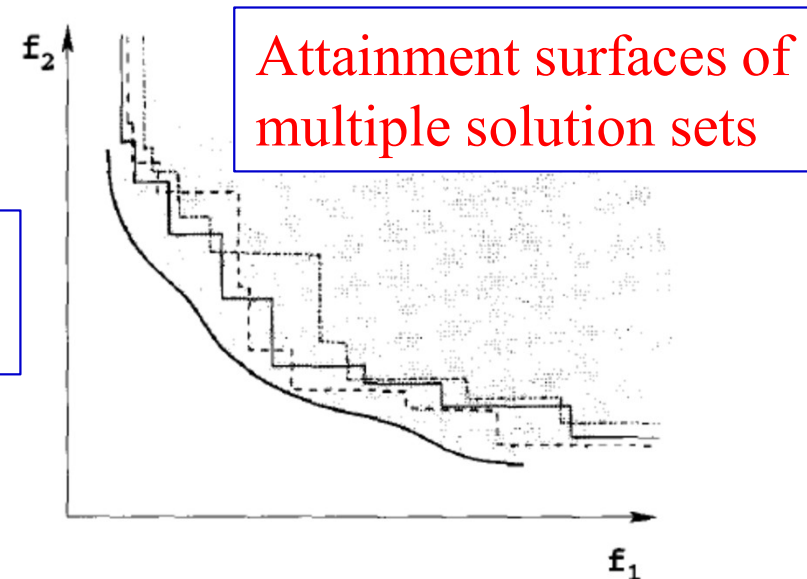
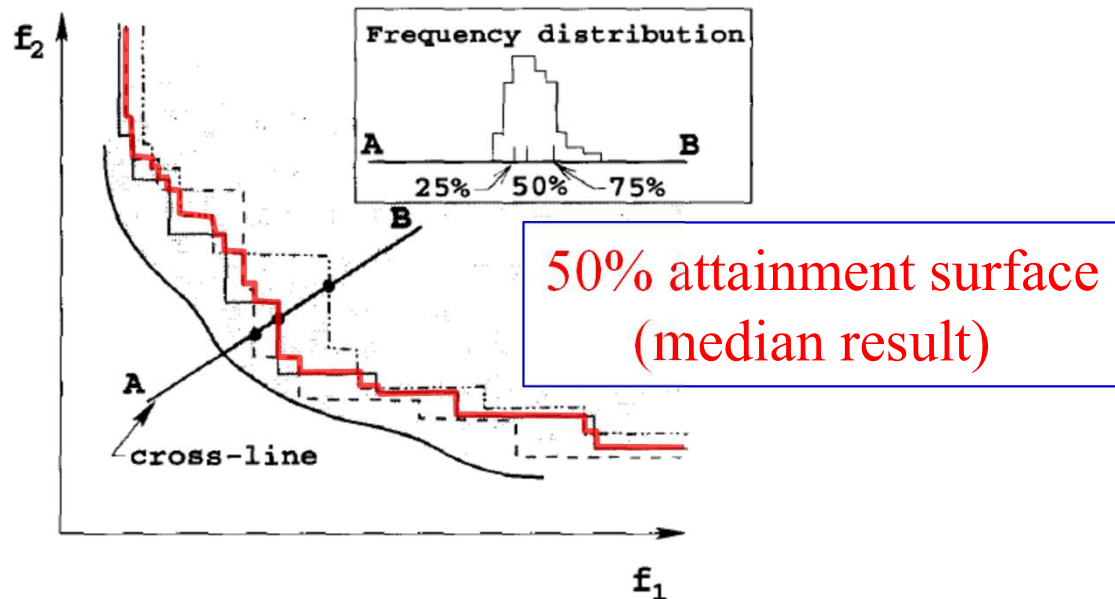
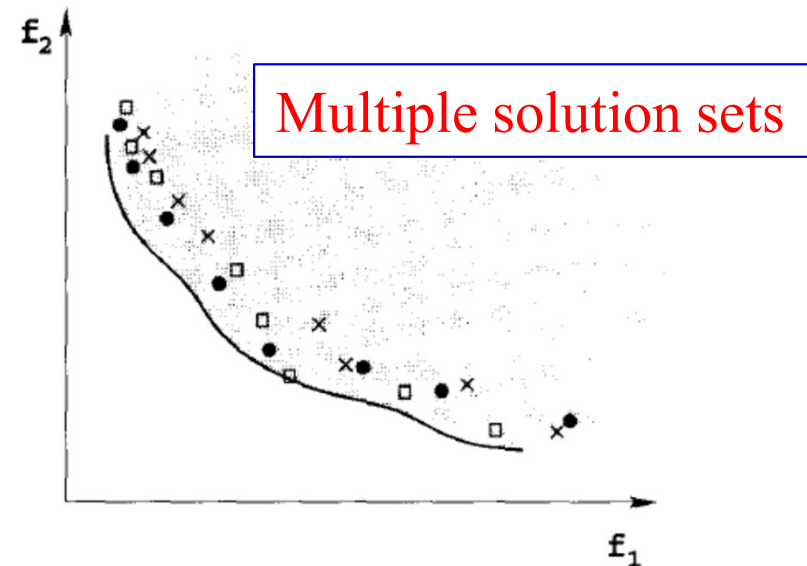
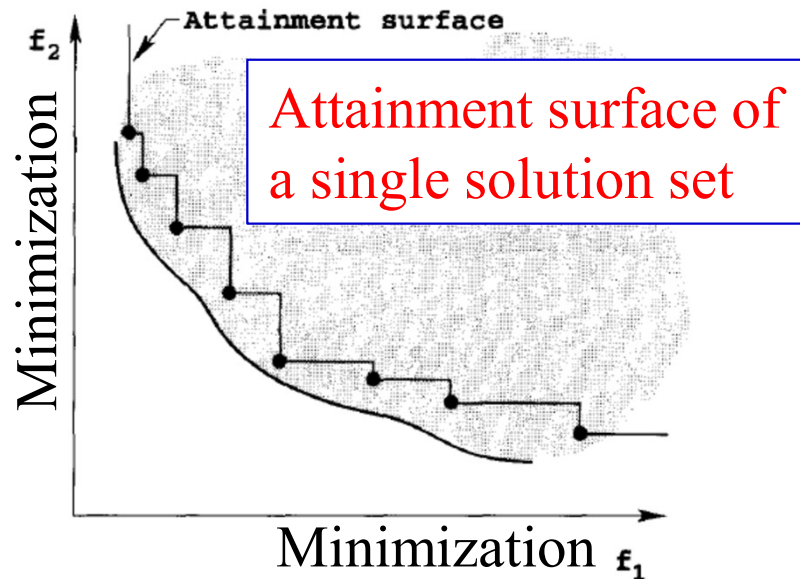
Uniformly-distributed solutions over the entire Pareto front (including the boundary) are usually not optimal for any performance indicators even for linear Pareto fronts.



R. Tanabe and H. Ishibuchi, **“An analysis of quality indicators using approximated optimal distributions in a three-dimensional objective space,”** *IEEE Trans. on Evolutionary Computation* (Early Access).

Other Performance Evaluation Method (Visual Evaluation)

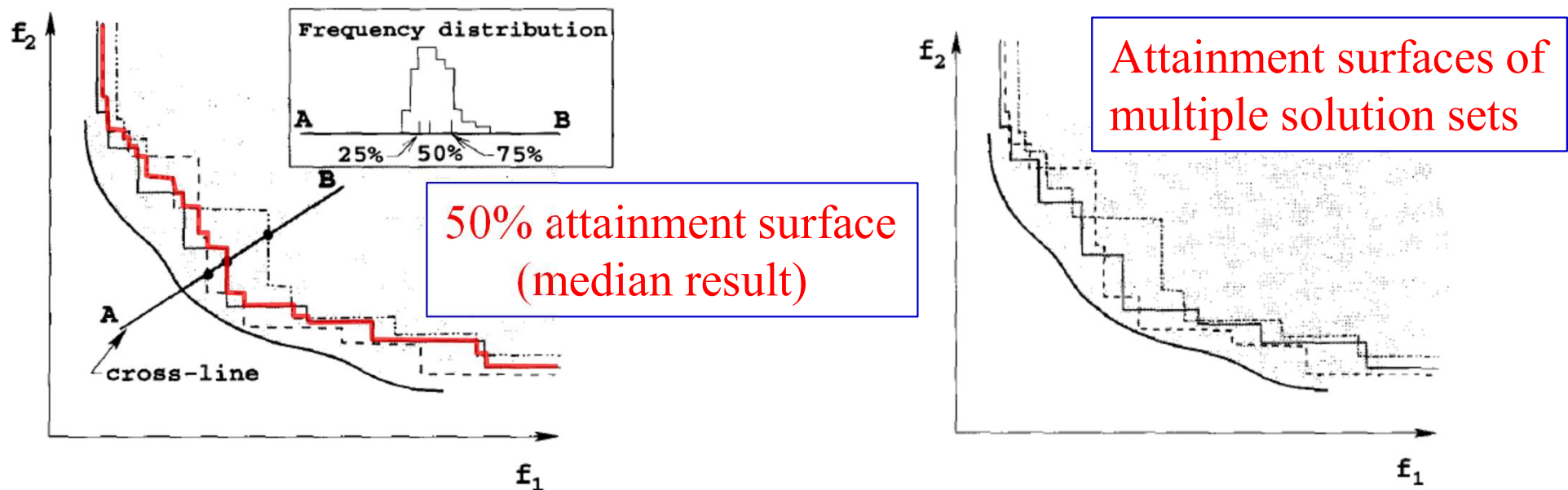
(1) Attainment Surface for two-objective problems



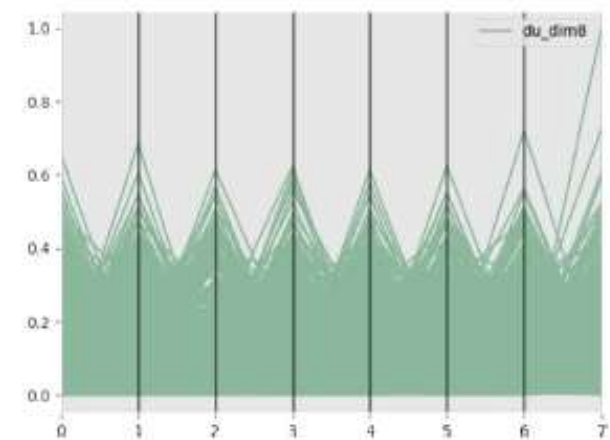
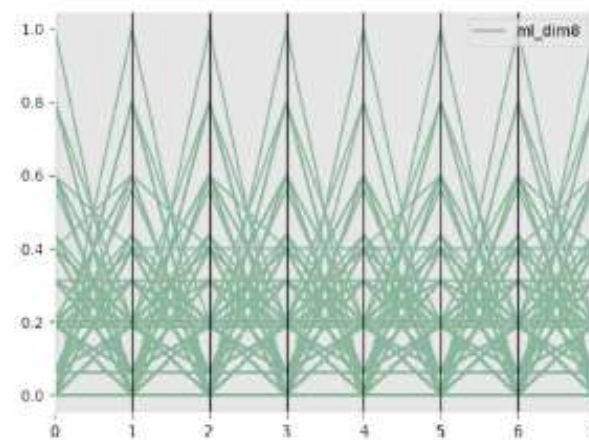
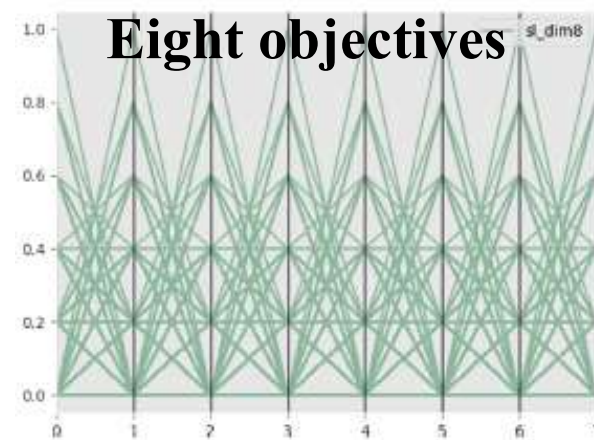
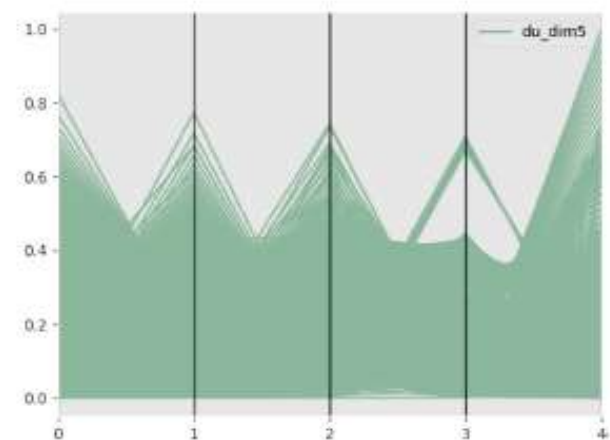
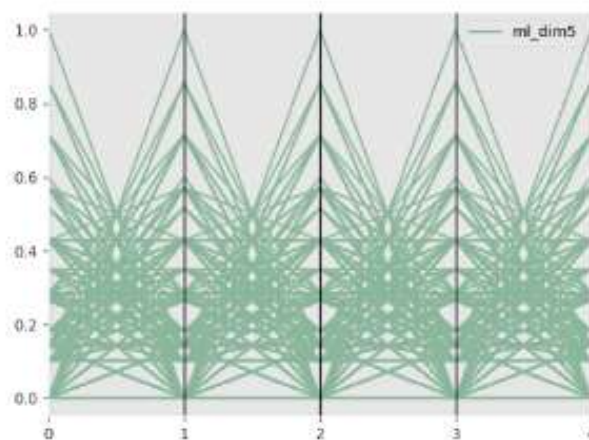
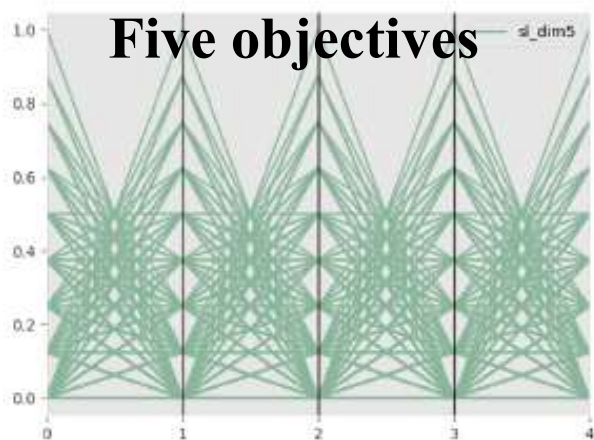
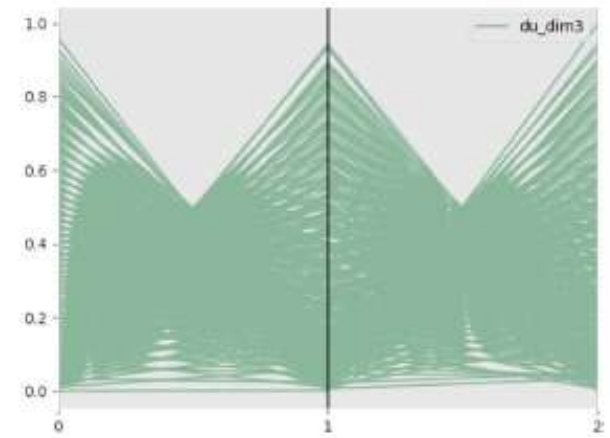
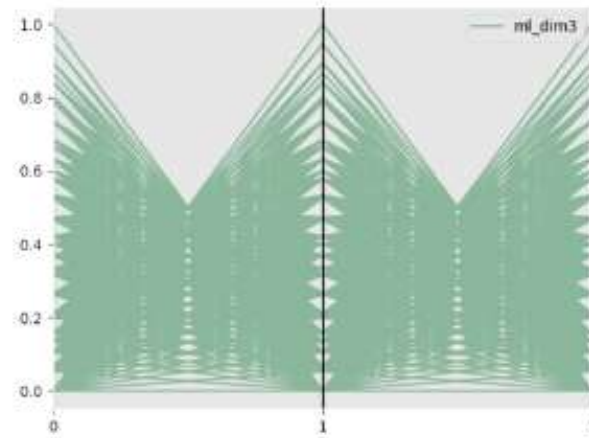
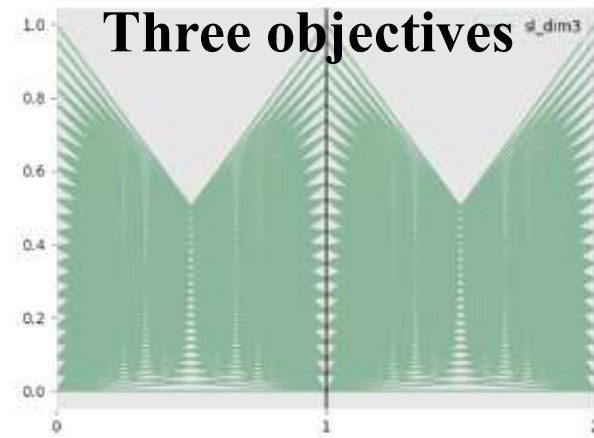
Other Performance Evaluation Method (Visual Evaluation)

(1) Attainment Surface for two-objective problems

For a two-objective problem, it is often recommended (by reviewers for a journal paper, by the session chair for a conference paper) to use the 50% attainment surface for visual performance comparison of the proposed algorithm with other existing algorithms. By using the 50% attainment surface, you can visually compare experimental results of different algorithms in a somewhat statistically solid manner.



(2) Parallel coordinates plots for many-objective problems

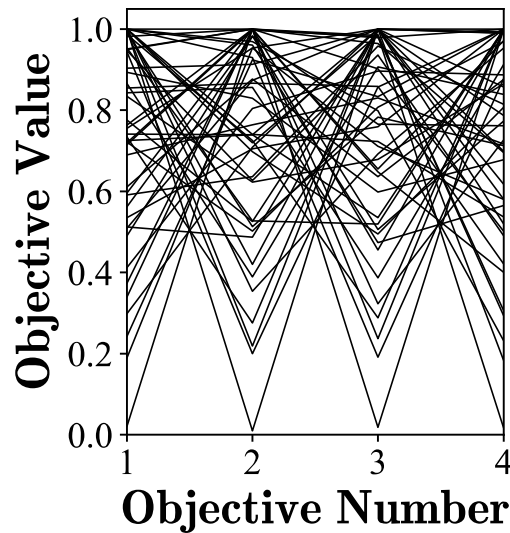


References for Visualization of Solution Sets

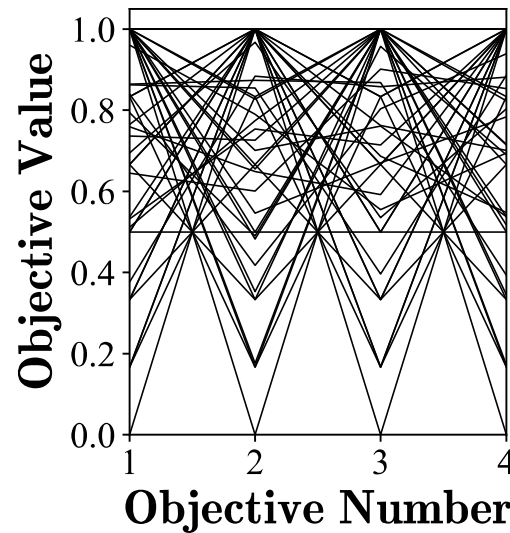
- [1] T. Tušar and B. Filipič, “**Visualization of Pareto front approximations in evolutionary multiobjective optimization**: A critical review and the projection method” *IEEE Transactions on Evolutionary Computation*, 2015.
- [2] M. Li, L. Zhen, and X. Yao, “**How to read many-objective solution sets in parallel coordinates**,” *IEEE Computational Intelligence Magazine*, 2017.

Visualization is not easy for the case of four or more objectives

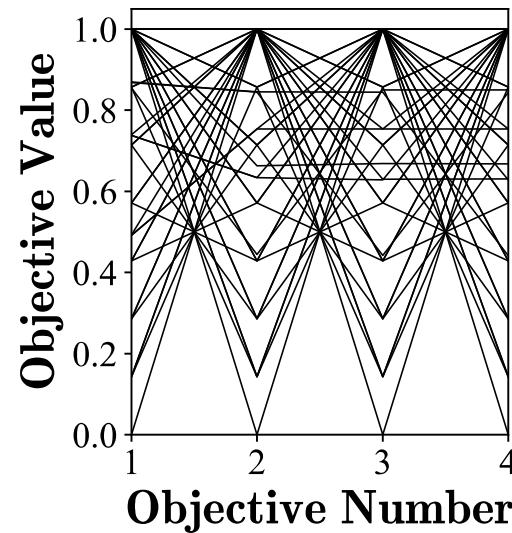
- Can you explain the difference among the four solution sets?
- Which do you think the best solution set?



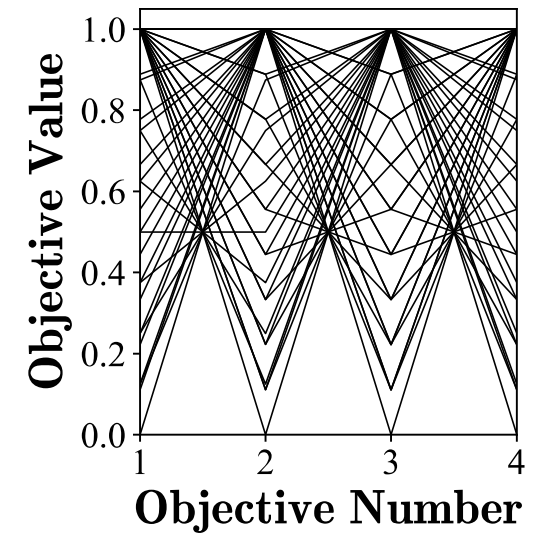
(a) $r = 1.2$.



(b) $r = 1.4$.



(c) $r = 2.0$.

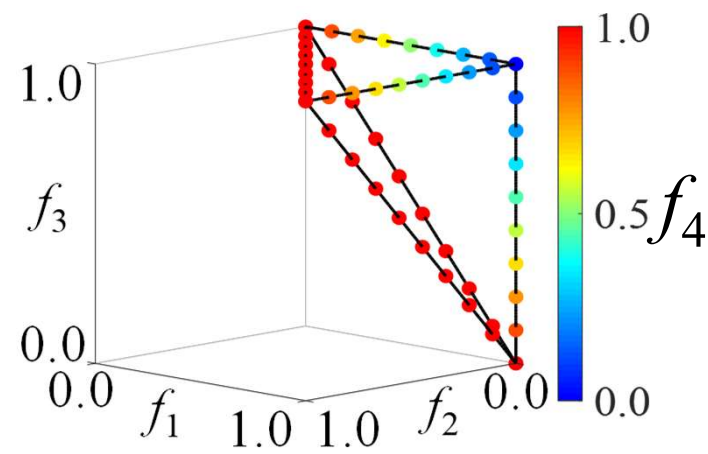
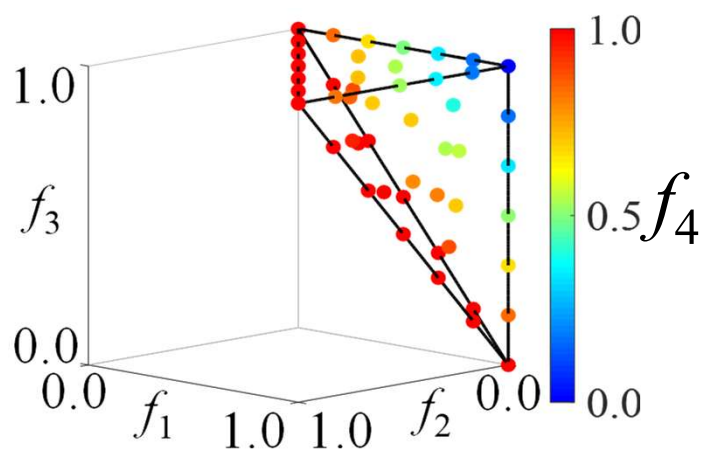
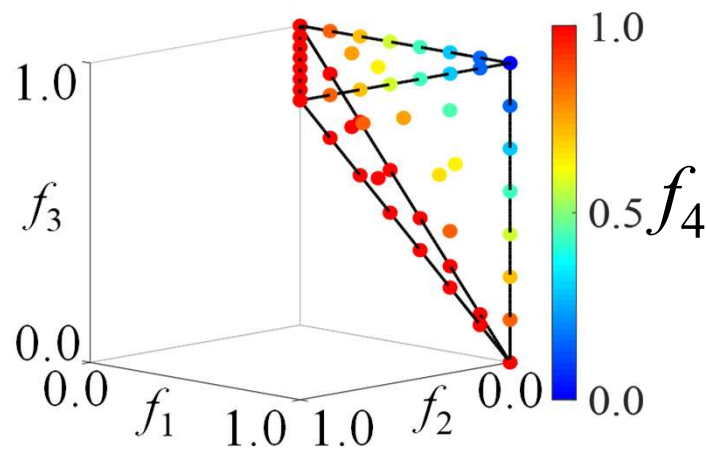
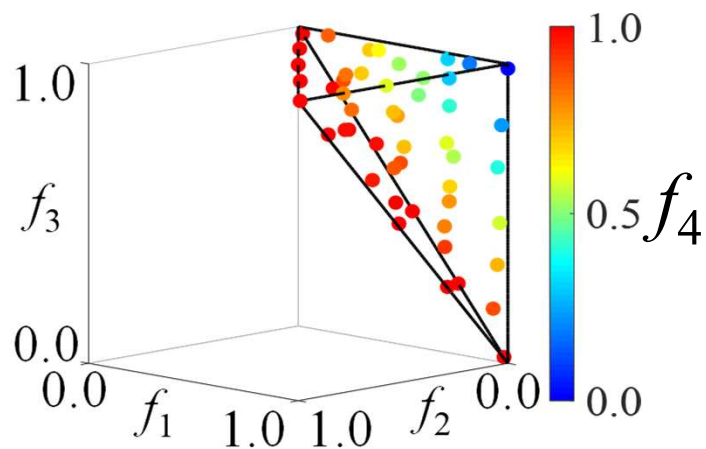
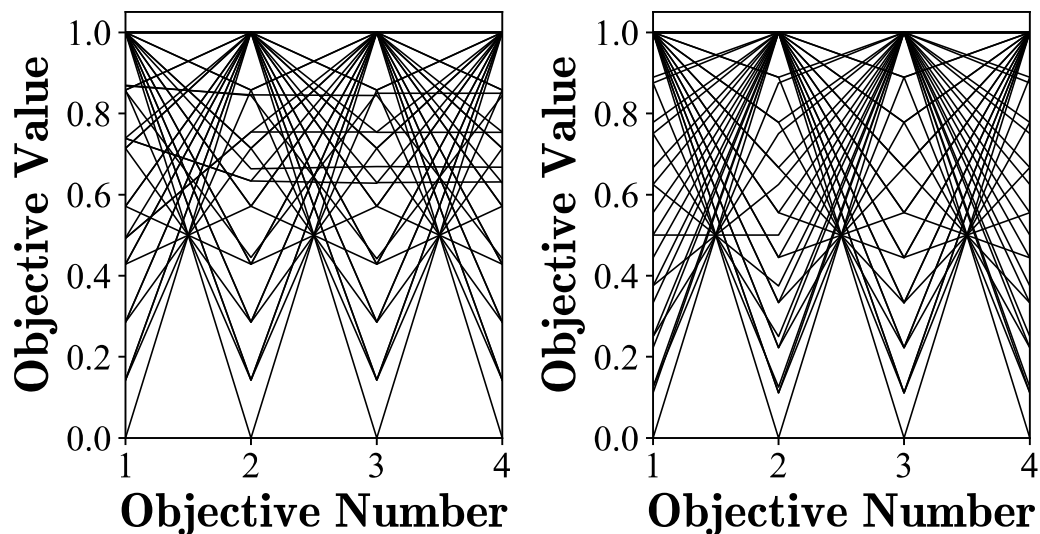
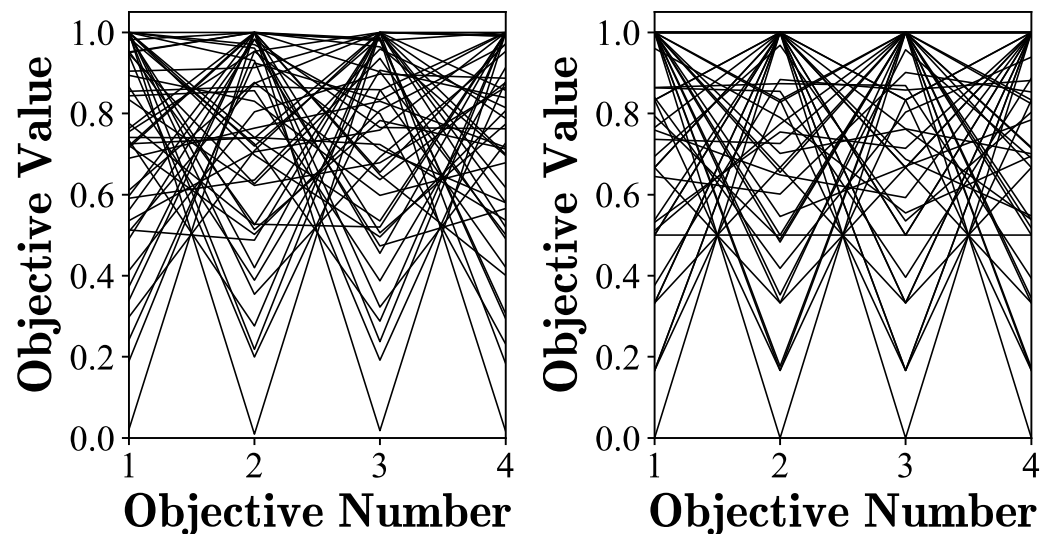


(d) $r = 100$

Optimal distributions of solutions of the four-objective minus-DTLZ1 problem for HV maximization with respect to different specifications of the reference point $\mathbf{r} = (r, r, r, r)$ in the normalized objective space.

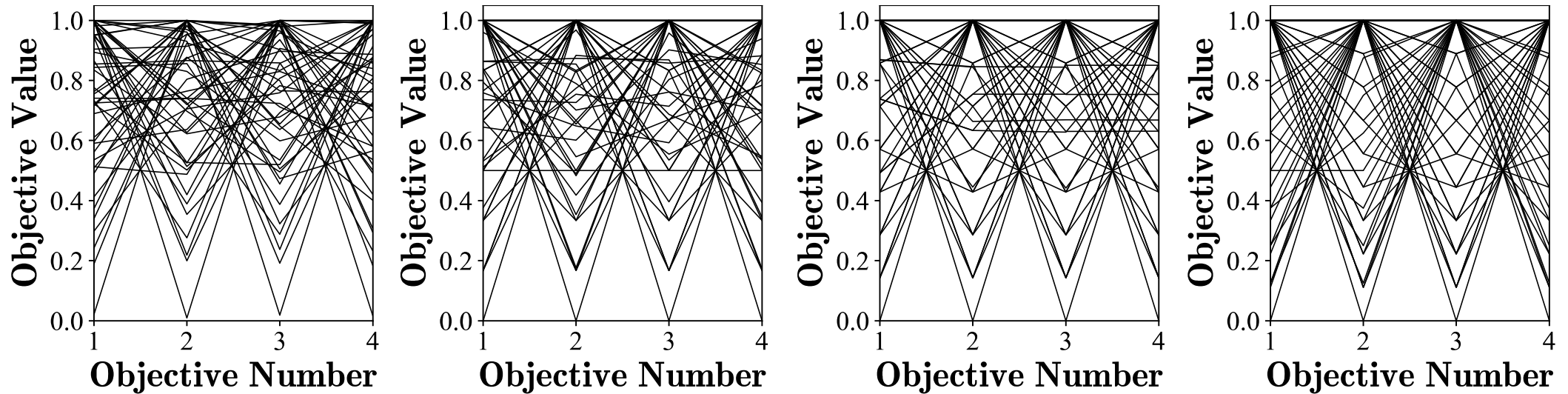
H. Ishibuchi, T. Matsumoto, N. Masuyama, and Y. Nojima, "Optimal distributions of solutions for hypervolume maximization on triangular and inverted triangular Pareto fronts of four-objective Problems," IEEE SSCI 2019.

Visualization is not easy for the case of four or more objectives

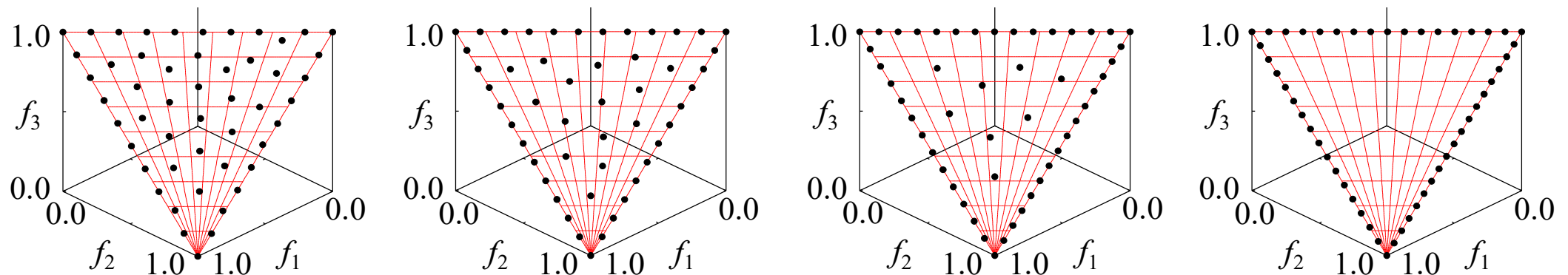


Visualization is not easy for the case of four or more objectives

Results on Four-Objective Minus-DTLZ1

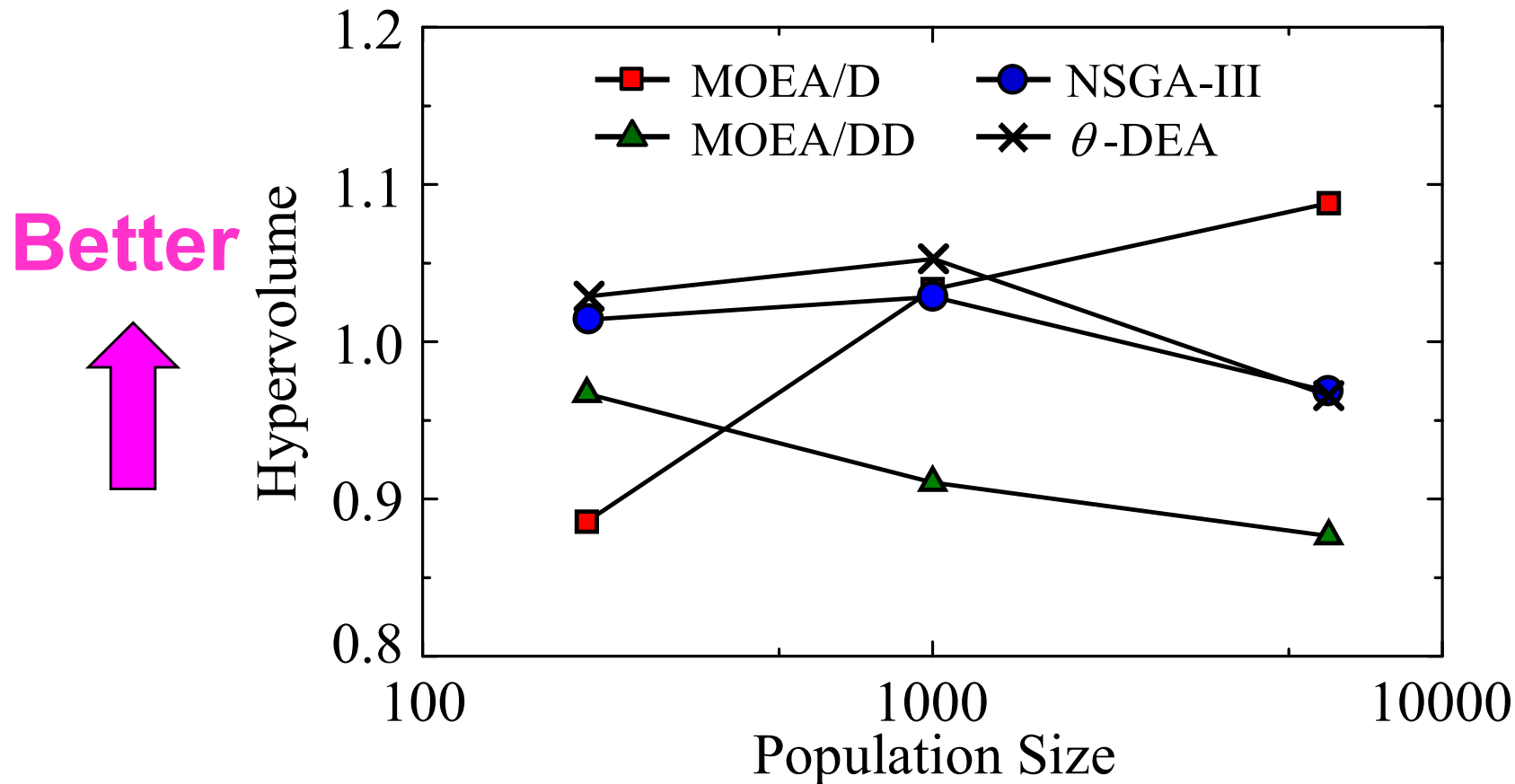


Results on Three-Objective Minus-DTLZ1



Another Difficulty related to Performance Evaluation

Performance comparison results strongly depend on the specification of the population size.



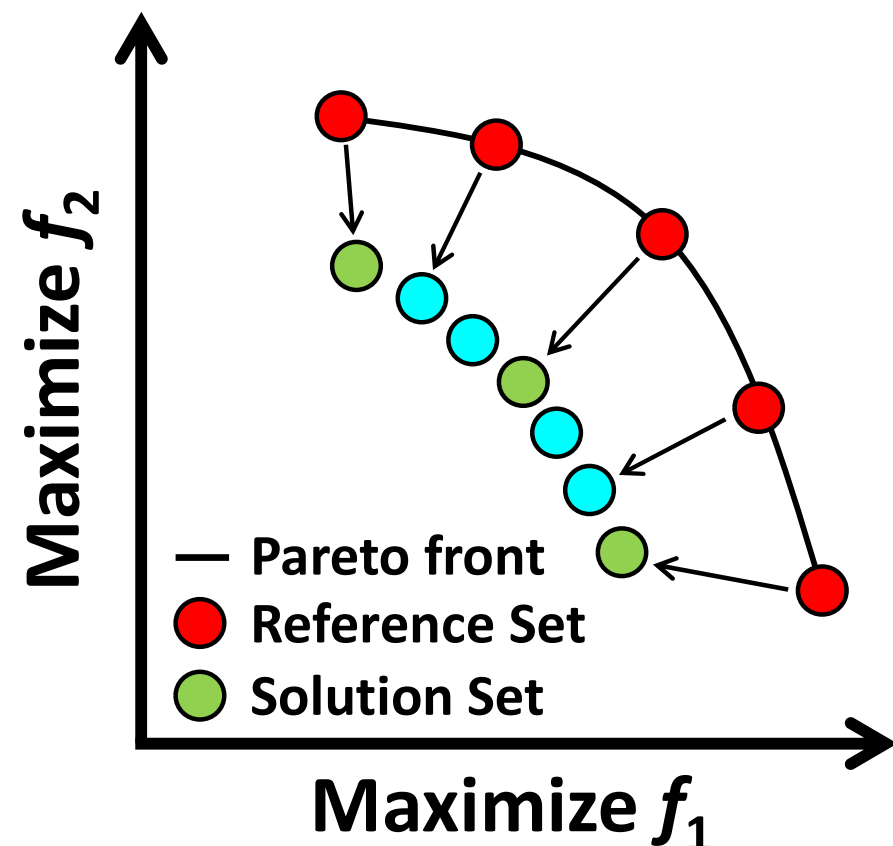
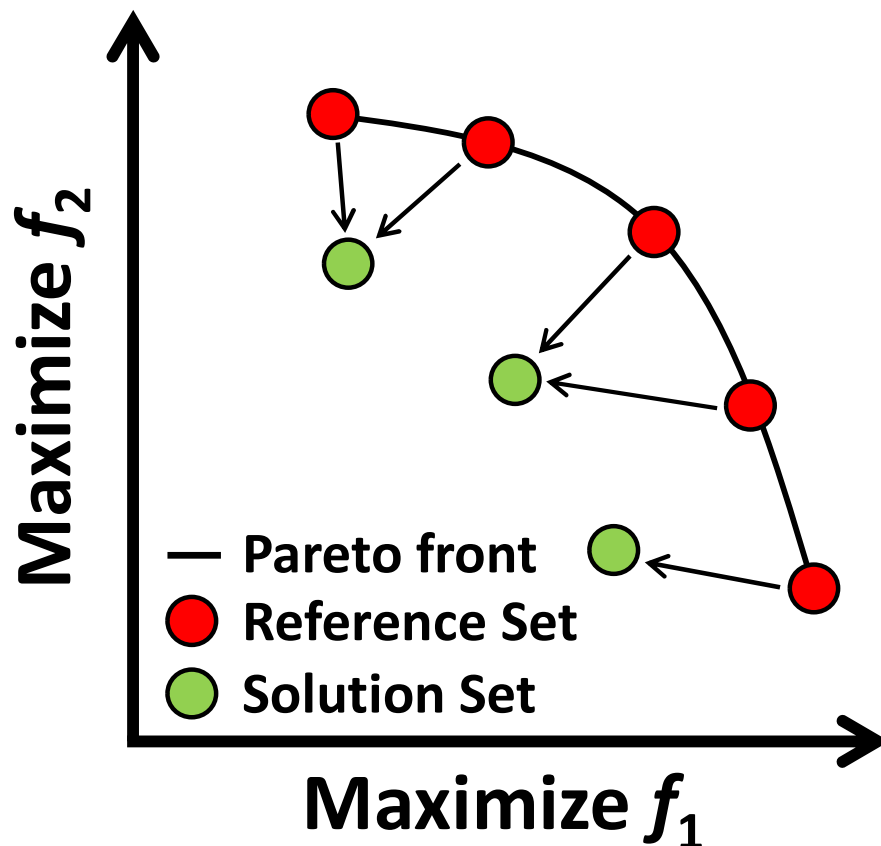
Performance of the final population with different population size
(Comparison results on the five-objective WFG3)
Ishibuchi et al. (IEEE CEC 2016)

Another Difficulty related to Performance Evaluation

Performance comparison results strongly depend on the specification of the population size.

The same size solution sets are needed for fair comparison.

Overall performance indicator values are usually improved by increasing the number of solutions in the examined solution set.

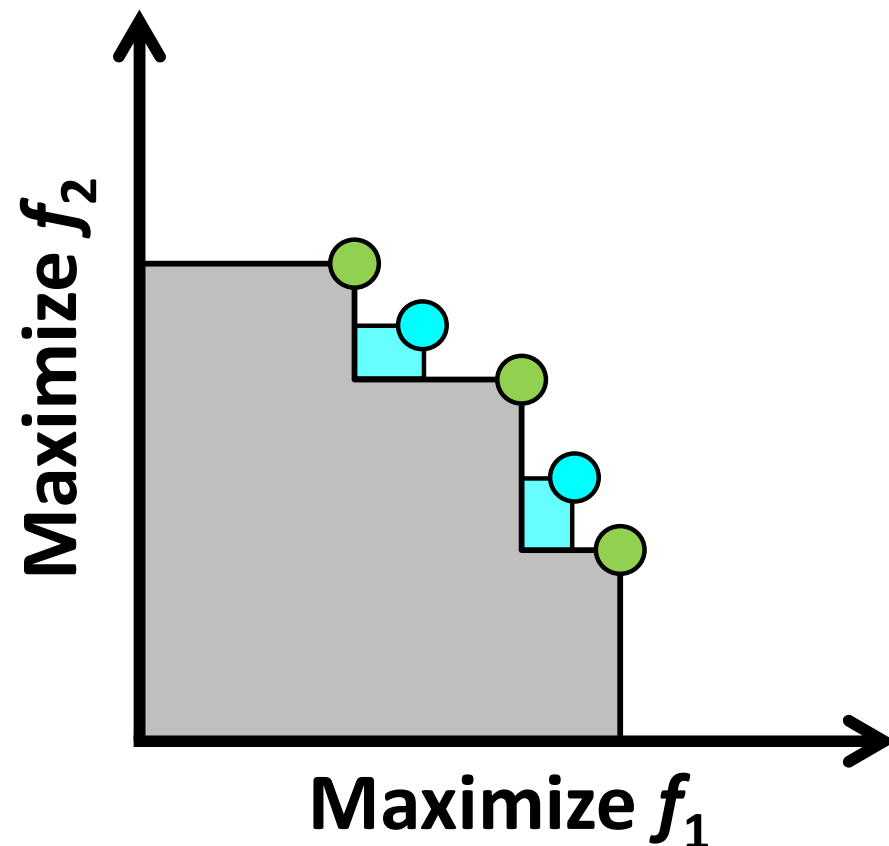
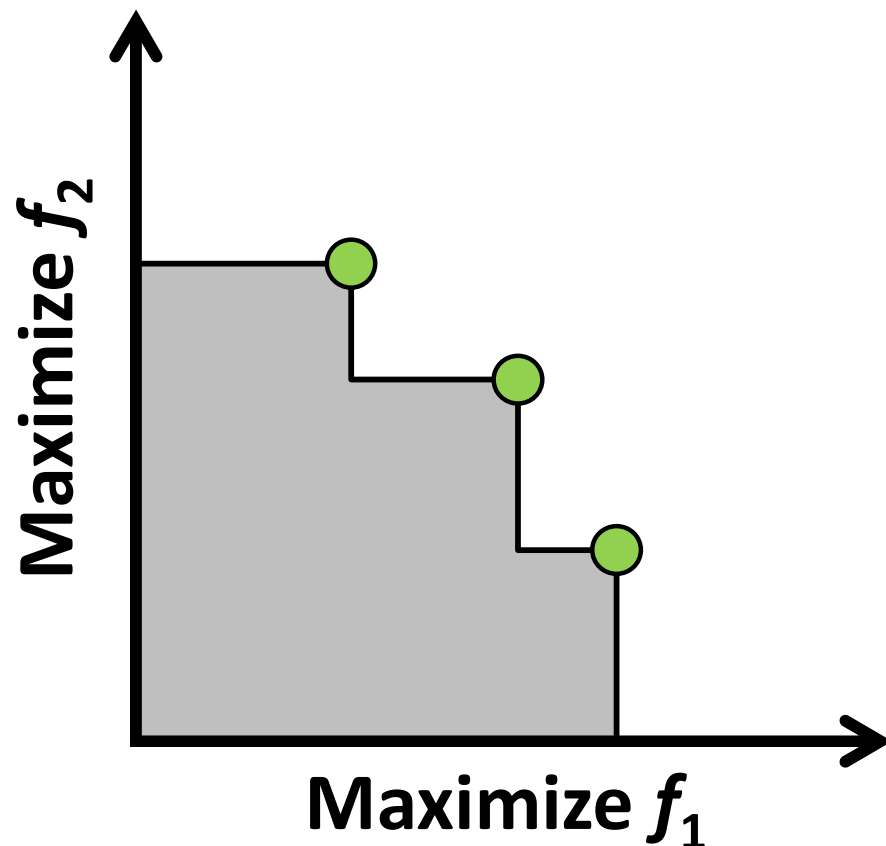


Another Difficulty related to Performance Evaluation

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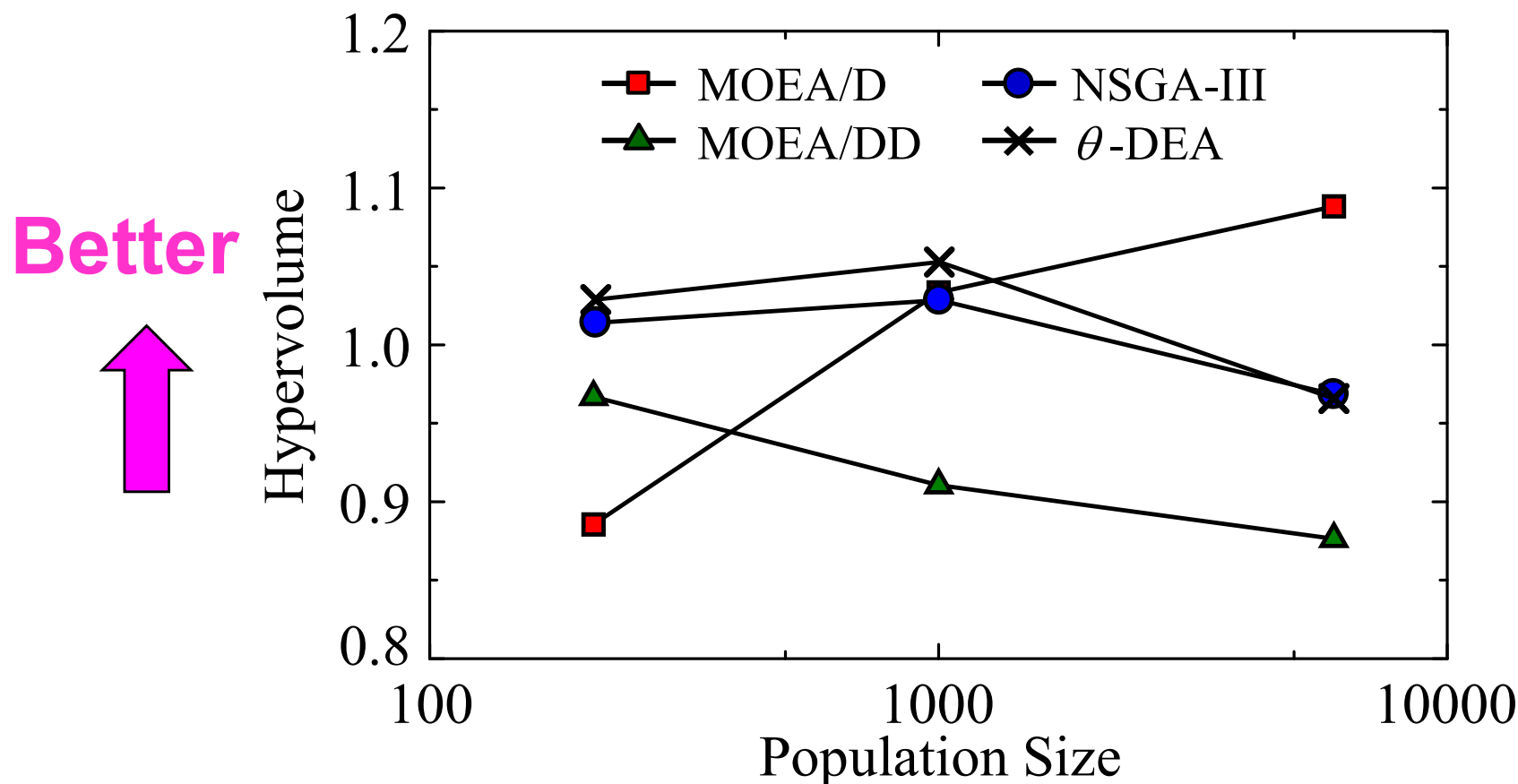


Another Difficulty related to Performance Evaluation

Performance comparison results strongly depend on the specification of the population size.

The same size solution sets are needed for fair comparison.

Q: How to choose the same population size for all algorithms.



Another Difficulty related to Performance Evaluation

Performance comparison results strongly depend on the specification of the population size.

The same size solution sets are needed for fair comparison.

Recent Trend: Selection of a pre-specified number of solutions from all the examined solutions. Population size can be different as long as the same number of solutions are selected.

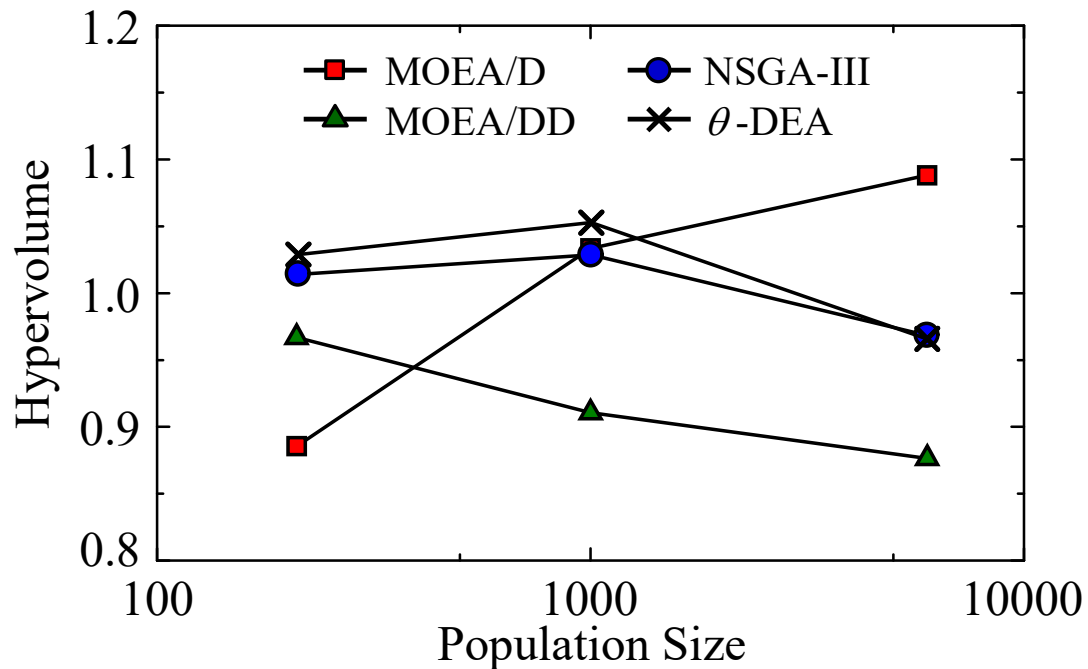
H. Ishibuchi, Y. Setoguchi, H. Masuda, and Y. Nojima, **“How to compare many-objective algorithms under different settings of population and archive sizes,”** *Proc. of 2016 IEEE Congress on Evolutionary Computation*, pp. 1149-1156, Vancouver, Canada, July 24-29, 2016.

R. Tanabe, H. Ishibuchi, and A. Oyama, **“Benchmarking multi- and many-objective evolutionary algorithms under two optimization scenarios,”** *IEEE Access*, vol. 5, pp. 19597-19619, December 2017.

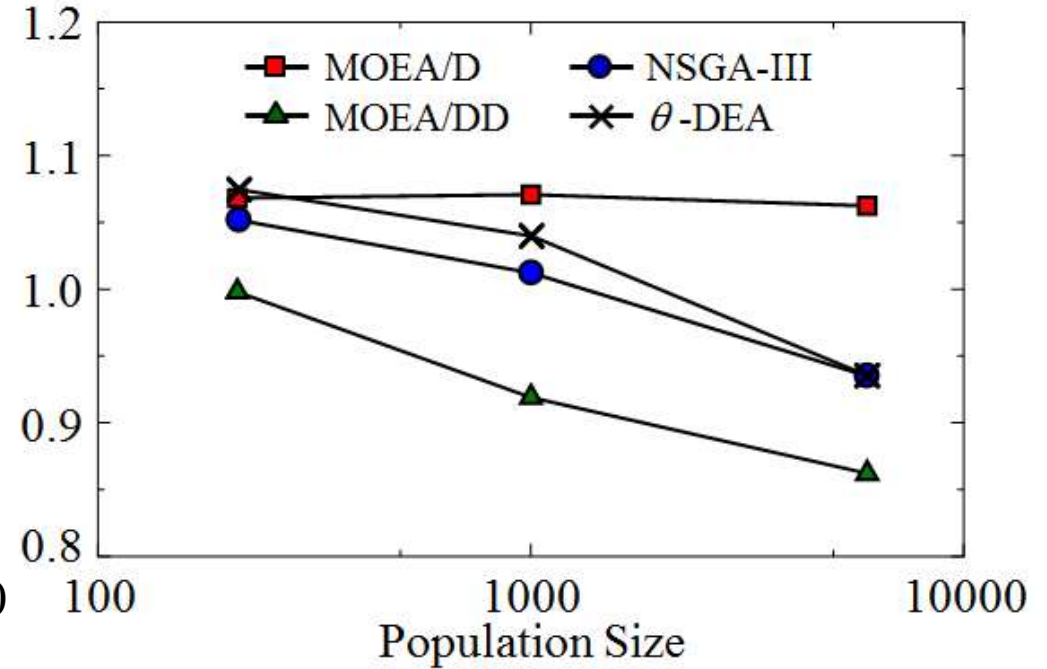
H. Ishibuchi, L. M. Pang, and K. Shang, **“A new framework of evolutionary multi-objective algorithms with an unbounded external archive,”** *Proc. of 24th European Conference on Artificial Intelligence (ECAI 2020)*, Santiago, Spain, August 29 - September 02, 2020 (Accepted).

Final Population vs. Selected Solutions

(50 solutions are selected)



Final Population Results



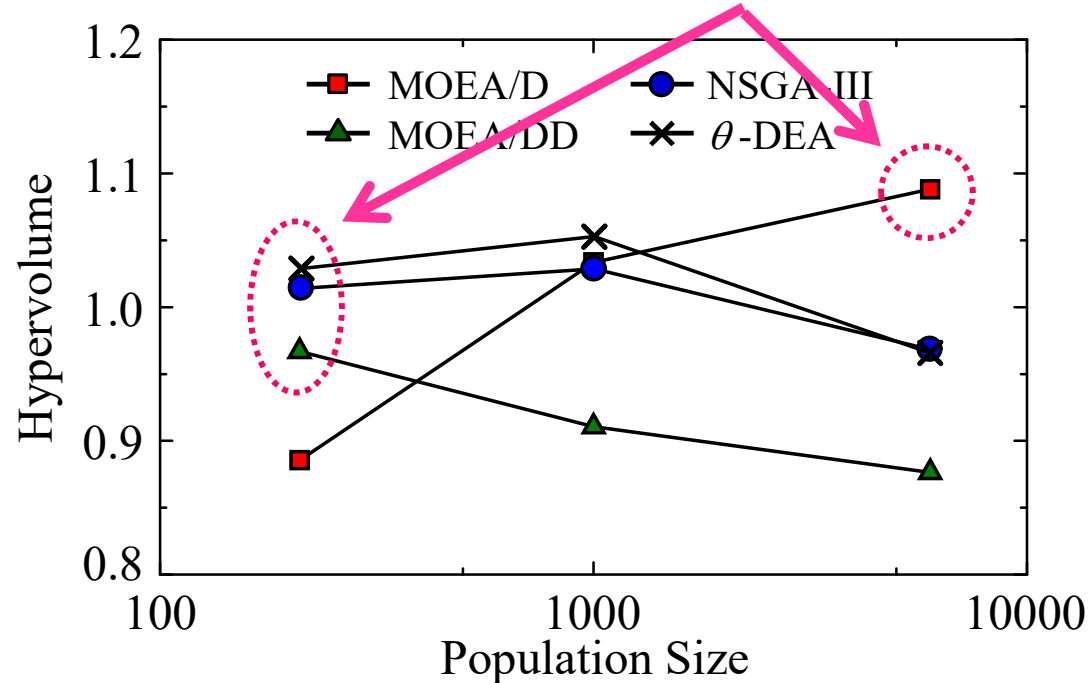
Selected Solutions Results

When the four algorithms are compared using the final population with the standard population size (about 200), MOEA/D is the worst. However, when they are compared by the selected solutions, MOEA/D is the best.

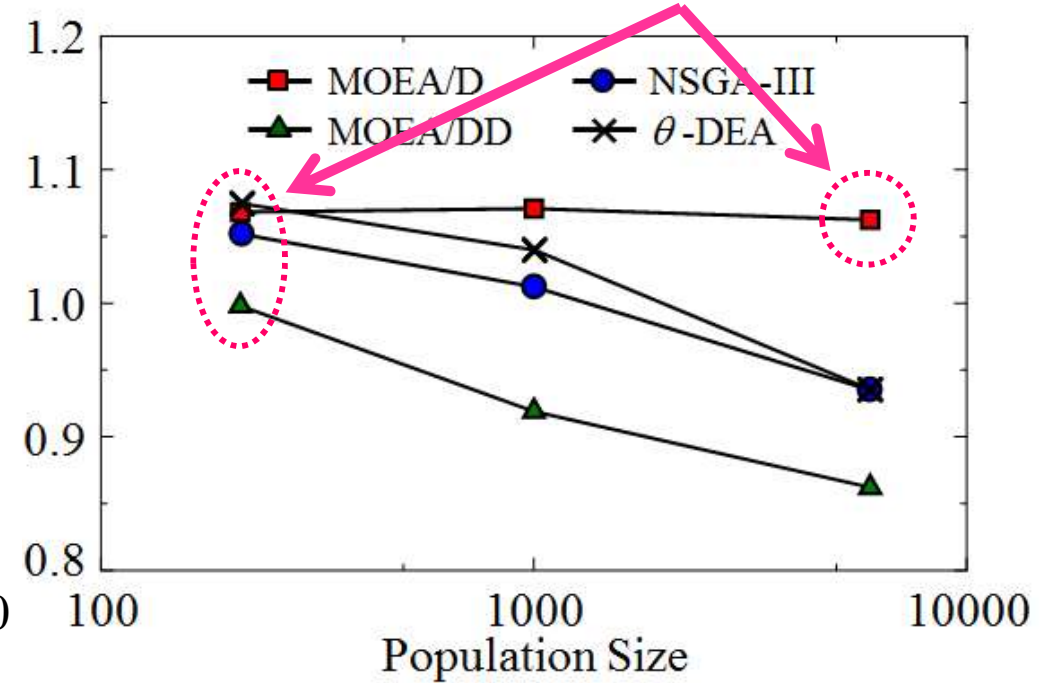
Final Population vs. Selected Solutions

(50 solutions are selected)

We cannot compare these results but we can compare these results.



Final Population Results



Selected Solutions Results

When the four algorithms are compared using the final population with the standard population size (about 200), MOEA/D is the worst. However, when they are compared by the selected solutions, MOEA/D is the best.